Verifying Imported Python Modules via PSDL

Integrating Lean-Style Expressivity with
Cubical Duck-Type Analysis

The deppy Project[1]

[1]Source: deppy repository. This treatise is generated alongside the source tree at `docs/psdl_treatise/`.

Abstract

This monograph develops a verification methodology for *imported* Python libraries. The user names a library — e.g. `import sympy.geometry` — and supplies a sidecar specification file (`.deppy`) carrying typed property annotations and proof scripts. The deppy pipeline introspects the live module, parses the sidecar, elaborates each verification block into a *proof term* in deppy's kernel, discharges arithmetic obligations to Z3, runs counterexample search on the negation, and emits a self-contained Lean 4 certificate that round-trips through Lean's elaborator.

The proof scripts are written in PSDL, the *Python-Semantic Deppy Language*: a Python-syntax tactic language whose vocabulary includes both CIC-flavoured primitives (induction, transport, function extensionality, path composition) *and* primitives intrinsic to reasoning about cubical-duck-typed effectful Python (`isinstance` fibrations, exception fibres under the $\Diamond/\Box$ modalities, monkey-patch paths, Kan fillers, Σ-projection through `__getattr__`). Each PSDL statement compiles in lock-step to a deppy `ProofTerm` that the kernel checks and a Lean tactic that Lean's kernel checks.

We work out the design in detail, present the formal model, document every component of the pipeline against the source tree, and walk through the worked example of verifying properties of `sympy.geometry.point.Point.distance` end-to-end. We are explicit about the trust surface: foundations admit, opaque `axiom`s admit, `Structural` leaves admit; the certificate is honest about which subgoals it has actually discharged and which it has merely passed through.

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Chapter 1

Introduction

1.1 The verification gap for imported modules

Most theorem provers ask the user to *rewrite* their program in the prover's language: Coq's `Inductive`, Lean's `def`, Agda's `data`. The mature ecosystems (de Moura and Ullrich, 2021; The mathlib Community, 2020) demonstrate that this works, but it imposes a barrier: a working Python library — say, `sympy.geometry` or `torch.nn.functional` — consists of tens of thousands of lines of mature, peer-reviewed Python that *nobody is going to retype* into Lean syntax. The user wants to verify properties of the library *as it stands*, with the library's source bytes left untouched on disk.

This is the *imported-module verification gap*. The library is a third-party object: dynamically dispatched, exception-throwing, duck-typed, monkey-patchable, and comfortable with `eval` and metaclasses. The user wants to verify a property such as "`Point.distance(p, q) >= 0`" *about the live* SymPy*'s actual* `Point.distance` *implementation*, not about a Lean shadow.

deppy resolves this gap by treating each Python module as an *external object* and each property as a *sidecar specification*. The flow is:

1. The user names a module: `import sympy.geometry`.
2. The user supplies a sidecar file: `sympy_geometry.deppy`.
3. deppy introspects the live module via `importlib` and `inspect`, parses the sidecar, elaborates each verification block to a kernel `ProofTerm`, runs each through `ProofKernel.verify`, discharges arithmetic foundations to Z3, runs counterexample search on negations, and emits a Lean 4 certificate that — ideally — round-trips through the Lean toolchain with exit code zero.

The library's own bytes are never modified. Verification artefacts live exclusively in the sidecar plus the generated certificate; the working programmer can delete both at any time without breaking the library.

1.2 Goals and non-goals

Goals

- **Library-agnostic.** The pipeline must work on any Python module. Nothing is hard-coded for SymPy, NumPy, or PyTorch; the sidecar file alone determines vocabulary.
- **Honest trust accounting.** Every claim ends up at one of seven trust levels (Chapter 8). The certificate counts exactly how many proofs went through `Lean`'s kernel, how many were admitted, how many bottomed at `Z3`, and how many leaked `Structural` pass-throughs into the trust chain.
- **`Lean`-style expressivity.** Anything provable in CIC-flavoured systems should be expressible in the proof language: Π/Σ, induction, transport, function extensionality, path composition, equivalences. The user should not feel that they have been confined to a baby tactic language.
- **Cubical / duck-type primitives where Python demands them.** `isinstance` is a fibration, exceptions live in a $\Diamond$-modal exception fibre, monkey-patches are paths between class versions, `__getattr__` is a Σ-projection. The proof language should make these *native vocabulary*, not afterthoughts behind escape hatches.
- **Real `Lean` certificates.** When the pipeline says it verified something, it must mean `Lean` actually accepted the certificate. "We emitted a tautology because `Sidecar_X_prop` is opaque" is a failure mode, not a victory.

Non-goals

- **We are not a literal CIC implementation.** The `deppy` kernel is a *distinct* type theory: a multi-trust-level proof checker that admits proof terms with explicit cubical, effect, and duck primitives. Chapter 4 formalises `deppy`'s metatheory; Chapter 6 explains the impedance mismatch between `deppy`'s language and CIC. We are explicit where the synthesis admits, approximates, or trusts.
- **We do not verify Python's runtime semantics from scratch.** The kernel does not model CPython's bytecode. We work from `inspect`-recovered source plus `ast.parse`; we trust that CPython runs that source faithfully (otherwise a verified Lean proof would be of an idealisation that does not match what Python actually does).
- **We do not require Mathlib.** The generated certificate is a self-contained `.lean` file using only `Lean` core. Foundation axioms involving real numbers (`sqrt`) admit at "`axiom`" level rather than reaching for `Mathlib.Analysis.SpecialFunctions.Pow.Real`.

1.3 The two traditions we synthesise

`deppy` sits at the intersection of two traditions, each with its own vocabulary, idioms, and proof culture.

Tradition I: Lean-style expressivity

The first is the proof-theoretic tradition descending from Per Martin-Löf (Martin-Löf, 1975; Nordström et al., 1990), the Calculus of (Inductive) Constructions (Coquand and Huet, 1988),

and modern CIC-flavoured systems including Lean 4 (de Moura and Ullrich, 2021), Coq, and Agda. This tradition treats proofs as *terms* in a dependent type theory. Inductive types come for free; equality is propositional; tactics elaborate to terms; the kernel checks terms.

Lean 4 specifically gives us:

- Universe-polymorphic dependent types.
- Tactic-mode and term-mode proofs interoperating.
- Type-class inference.
- A small kernel that any user can read and (in principle) audit.
- A generic-elaborator infrastructure: notation, macros, elaborators.

We borrow this tradition's *expressivity*: PSDL accepts $\forall$, $\exists$, induction, Σ-projections, function extensionality, path induction (Martin-Löf's J), congruence, substitution, transport, and equivalence. The compiled Lean target uses real Lean tactics: `intros`, `exact`, `rfl`, `rw`, `induction`, `simp`, `omega`. When a PSDL proof closes, *Lean's elaborator* closes the corresponding tactic.

Tradition II: Cubical / duck-type analysis of Python

The second tradition is deppy's own. deppy treats Python through the lens of cubical homotopy type theory (The Univalent Foundations Program, 2013; Cohen et al., 2015) adapted to a dynamically-typed, exception-throwing language. The philosophical move is that Python's apparent informality is in fact *unusually amenable* to higher-categorical analysis:

Types are spaces. A Python class is a topological space whose points are its instances and whose paths are method identifications.

`isinstance` is a fibration. A subclass relation $C \leq D$ is the fibration whose total space is the disjoint union of C's and $D \setminus C$'s instances; the Python check `isinstance(x, C)` is the projection map's pre-image test. This is encoded by the kernel's `Fiber` proof term and surfaces in PSDL as the `if isinstance(x, T): ...` idiom.

Duck typing is path equivalence. Two classes that supply the same method signatures are *path-equivalent* in the universe of types — a univalent identification. The kernel records this via the `DuckPath` term.

Monkey-patching is a path. Replacing a method $C.m \mapsto C.m'$ at runtime is a 1-cell in the path space of classes. Whether the monkey-patch *preserves* a previously verified property is encoded by `Patch` which records the *behavioural-contract* preservation proof.

Exceptions live in a $\Diamond$-modality. Pure functions inhabit $\Box A$; functions that may raise inhabit $\Diamond A$. Exception handlers are eliminations of the exception-fibre face of $\Diamond$. The kernel's `EffectWitness` and `ConditionalEffectWitness` witness this layered modal structure.

`__getattr__` is a Σ-projection. Dynamic attribute lookup against a proxy is the first projection of a Σ-type whose second component is the resolved value; conditional duck paths record the predicate under which the projection inhabits the protocol fibre.

This metatheory is *cubical* in that it treats paths — not quotienting — as the primitive notion of equality, with explicit Kan-fillers, ech-glue, and cohomology classes for control-flow and inter-procedural analysis.

The synthesis

PSDL bridges the two. Each PSDL statement compiles in lock-step to:

1. a *deppy `ProofTerm`* that the deppy kernel type-checks;
2. a *Lean tactic line* that Lean's elaborator type-checks.

Both checks must succeed for a verify-block to be accepted. If the deppy kernel sees a `Refl(Var("_lem_body"))` that closes a side condition but Lean's elaborator rejects the corresponding `rfl`, the certificate marks the verify-block REJECTED.

We are explicit that deppy's metatheory is *not* literally CIC. We do not claim that every PSDL proof faithfully encodes a CIC derivation, nor that deppy's effect modalities are the same formal object as Lean's monads. What we claim is that PSDL is *rich enough*, in practice, to express every property a user hand-wrote against SymPy in our test suite, and that the generated Lean certificates close.

1.4 What this monograph contains

Chapter 2 reviews dependent type theory at the level needed to read the rest of the book: Π/Σ, identity types, induction principles, propositional vs. judgemental equality, HoTT's univalence axiom, and the cubical interpretation (The Univalent Foundations Program, 2013; Cohen et al., 2015; Angiuli et al., 2019).

Chapter 3 reviews the Python language constructs that the verifier actually has to deal with: dynamic dispatch and method resolution order, duck typing and structural subtyping (van Rossum et al., 2014; Levkivskyi et al., 2017), the exception protocol, descriptor machinery, monkey-patching, metaclasses. We do not assume the reader is a Python expert; the relevant idioms are introduced as we need them.

Chapter 4 formalises deppy's cubical-duck-type metatheory: the universe of `SynTypes`, the `ProofTerm` grammar, the seven `TrustLevels`, the kernel's verification rules. This is the longest single chapter; everything that follows refers back to it.

Chapter 5 documents the `.deppy` sidecar file format: `module`, `foundation`, `spec`, `verify`, `lemma`, `code_types`, `lean_imports`, `lean_preamble`.

Chapter 6 is a complete reference for PSDL: every statement shape, the `ProofTerm` it compiles to, the Lean tactic it emits, soundness arguments, and worked examples.

Chapter 7 walks through the `run_sidecar` entry point step by step: module-name resolution, sidecar parsing, the several rendering passes (`derive_all`, `audit_all`, `mechanize`, `link_all`, `verify_all`, `discharge_foundations`, `run_counterexample_search`), Lean round-trip via `check_lean_module_source`.

Chapter 8 is the Lean target: how the certificate is laid out, how the trust chain bottoms at concrete Lean `Prop`s where possible and at admitted axioms where not, how foundations are mechanised, how body translation interacts with the preamble.

Chapter 9 is the Z3 backend: which formulas it discharges, how counterexample search works, how the trust level promotes from `AXIOM_TRUSTED` to `Z3_VERIFIED`.

Chapter 10 covers the cohomology layer: intra-procedural cubical analysis, inter-procedural simplicial cohomology, the bicomplex bridge, and Kan-fill obligations. We are honest that this layer is mostly *advisory* — it surfaces obstructions and provides structural metadata — and is not, in the current implementation, the source of trust for any verify-block.

Chapter 11 works `sympy_geometry.deppy` end-to-end: the foundations, the spec for `Point`, the `verify dist_nonneg_psdl` block, the emitted certificate, and the Lean round-trip. We compare this against the `distance_pythagoras_proof` (chained derivation) and against the deeper `euler_line` theorems which we admit honestly.

Chapter 12 is the soundness chapter: what is verified, what is admitted, where the trust surface lives, where soundness gates fire, what `certify_or_die` actually checks.

Chapter 13 surveys related work and concludes.

1.5 Reading guide

A reader who already knows dependent type theory and Lean can skip Chapter 2. A reader who knows Python well can skim Chapter 3. Everyone should read Chapter 4: it is where deppy's metatheory is fixed. The shortest path to seeing the system in action is Chapter 4 → Chapter 6 → Chapter 11.

The cross-references are extensive and intentional. Whenever we quote a code excerpt we cite the file and an approximate line range; whenever we name a kernel proof term we cite its data class in `deppy/core/kernel.py`; whenever we mention a tactic we cite the corresponding PSDL compiler line in `deppy/proofs/psdl.py`.

1.6 Notational conventions

We adopt:

- **Bold sans-serif** for system names (deppy, PSDL, Lean, Z3).
- `teletype` for code, identifiers, and source paths.
- Italic for emphasis and for the (rare) defined terms.
- Mathematical italic (T, σ, φ) for type-theoretic metavariables.
- $[\![e]\!]$ for the *interpretation* (semantic value) of a syntactic expression e.
- Throughout, $A \simeq B$ denotes type equivalence (a triple of forward, backward, and witnesses); $A = B$ denotes propositional equality of types (only meaningful in a univalent setting); $a =_A b$ denotes propositional equality of terms in type A.

When we give code excerpts of more than two lines we use `listings` blocks; when we give a one-liner we inline it in `teletype` font. Excerpts are reproduced verbatim from the source tree at the time of writing; we cite the file path explicitly so the reader can verify with `git log -p`.

1.7 A first taste

To anchor the discussion we end this introduction with a single worked example: the `verify dist_nonneg_psdl` block from `sympy_geometry.deppy`. We will return to this throughout the book.

Listing 1.1: The `dist_nonneg_psdl` block, taken verbatim from `sympy_geometry.deppy` (top of repository).

```
verify dist_nonneg_psdl:
  function: sympy.geometry.point.Point.distance
  property: "Point.distance(p, q) >= 0"
  via: foundation Real_sqrt_nonneg
  binders: p: Point, q: Point
  psdl_proof: |
    if isinstance(other, Point):
        # value-fibre: body returns sqrt(sum_zip_sub_sq(self, other)).
        inline(Point.distance, sqrt(sum_zip_sub_sq(self, other)))
        apply(Real_sqrt_nonneg)
        assert sum_zip_sub_sq(self, other) >= 0, "z3"
    else:
        # exception-fibre: TypeError raised before return.
        raise NotImplementedError
```

That is a PSDL proof. Reading it left-to-right:

- `if isinstance(other, Point):` compiles to a `Fiber` proof term (file `deppy/core/kernel.py`, dataclass `Fiber`) with two type branches; in Lean this becomes `by_cases h : isinstance other Point`.
- `inline(Point.distance, sqrt(sum_zip_sub_sq(...)))` emits an `Unfold` that asserts the body translation against the explicit RHS.
- `apply(Real_sqrt_nonneg)` emits an `AxiomInvocation` that cites the foundation declared at the top of the same `.deppy` file.
- `assert ..., "z3"` emits a `Cut` whose lemma proof is a `Z3Proof` on the formula `sum_zip_sub_sq(self, other) >= 0`.
- In the `else`-branch, `raise NotImplementedError` is PSDL's *vacuous* marker, gated by the kernel's `Structural` term.

The pipeline then:

1. resolves `sympy.geometry.point.Point.distance` via `importlib`;
2. reads the live source via `inspect`;
3. parses the PSDL block via `compile_psdl` (file `deppy/proofs/psdl.py`);
4. type-checks the resulting `ProofTerm` tree against the property's `RefinementType`;

5. emits a Lean certificate including a concrete `Verified_dist_nonneg_psdl_property : Prop` (built from the binders) and a tactic-mode `theorem`;
6. runs the certificate through `lean_runner.check_lean_module_source`;
7. reports `CERTIFIED` or surfaces the Lean rejection verbatim.

The last two steps are what distinguish "we ran a kernel and it returned True" from "we emitted a real Lean certificate that Lean accepted." We emphasise the latter at every opportunity in this monograph; the former is a necessary but not sufficient condition.

1.8 Honest limits

Before diving in, we summarise the trust surface a careful reader should carry through the rest of the book. These are revisited in detail in Chapter 12; we list them here as fair warning.

- Foundations admit. The user lists arithmetic facts as `foundation` declarations; the pipeline tries Z3 on each (Chapter 9); whichever Z3 cannot discharge stay admitted. In `sympy_geometry.deppy`, `Real_sqrt_nonneg` stays admitted because Z3 over `Int` cannot model `sqrt`.
- `axiom` declarations admit. Anything stated as `axiom` in the sidecar is taken on the user's authority; the certificate emits it as a Lean `axiom` declaration and exposes its statement so a reviewer can reject it.
- `Structural` leaves leak. When a `ProofTerm` bottoms out at `Structural("...")`, the kernel accepts at `STRUCTURAL_CHAIN` trust. The soundness gate (Chapter 12) flags chains containing such leaves and `certify_or_die` will refuse to certify if any verify block leaks one.
- Body translation is best-effort. Some Python features (`eval`, dynamic class creation, C-extension fall-throughs) are *not* translated; the body translator emits `sorry` and the certificate flags it.
- Lean round-trip is conservative. We invoke Lean with a 60-second timeout (configurable); if it does not return in time we flag the certificate as not round-tripped. Round-tripped is *required* for `CERTIFIED`.
- We trust Lean's kernel. At the very bottom of the trust chain is Lean itself. We do not duplicate Lean's kernel; when Lean accepts, we accept.

Chapter 2

Background: Dependent Types, Lean, HoTT, Cubical Type Theory

This chapter reviews the proof-theoretic background needed to read the rest of the monograph. Readers comfortable with Lean 4 (de Moura and Ullrich, 2021) and the homotopy-type-theory book (The Univalent Foundations Program, 2013) can treat this as a refresher and skim.

We are deliberately concise: this is not a textbook in dependent type theory. Our aim is to fix vocabulary and notation, point out the constructions that recur in PSDL and the kernel, and flag the places where deppy's metatheory *diverges* from canonical treatments.

2.1 Dependent types in one breath

A type theory is a system of *judgments* about *terms*. The two primitive judgments are

$$\Gamma \vdash A : \mathsf{Type} \qquad \text{“}A \text{ is a (well-formed) type in context } \Gamma\text{”}$$
$$\Gamma \vdash a : A \qquad \text{“}a \text{ is a term of type } A \text{ in context } \Gamma\text{”}$$

where Γ is a list of typed bindings $x_1 : A_1, \ldots, x_n : A_n$.

A theory is *dependent* when types may mention previously introduced terms. The two canonical dependent connectives are:

Π-types (dependent function space). Given a type A and a family $B : A \to \mathsf{Type}$, the type

$$\Pi(x : A).\, B(x)$$

is the type of functions f such that $f(a) : B(a)$ for every $a : A$. When B does not depend on x, this reduces to the ordinary non-dependent function type $A \to B$.

In Lean 4 syntax: `(x : A) → B x`. In deppy's `deppy.core.types`: `PiType(param_name, param_type, body_type`

Σ-types (dependent pair). The type

$$\Sigma(x : A).\, B(x)$$

contains pairs $\langle a, b\rangle$ where $a : A$ and $b : B(a)$.

In Lean 4: `(x : A) 𝔼 B x` or `Sigma A B`. In deppy: `SigmaType`.

Identity / path types. Given a type A and two terms $a, b : A$, the type

$$a =_A b$$

classifies *proofs that a and b are propositionally equal.* The canonical inhabitant is $\mathrm{Refl}_a : a =_A a$ for any a. The elimination principle is *path induction* (Martin-Löf's J): to prove a property $P(x, y, p)$ for arbitrary $x, y : A$ and $p : x =_A y$, it suffices to prove $P(a, a, \mathrm{Refl}_a)$ for arbitrary a.

In Lean 4: `a = b`; the eliminator is `@Eq.rec`. In deppy: `PathType` as the type, `Refl` / `Sym` / `Trans` / `TransportProof` as proof-term constructors.

Universes. Lean 4 has a polymorphic universe hierarchy $\mathsf{Type}\,0 : \mathsf{Type}\,1 : \mathsf{Type}\,2 : \ldots$ along with an impredicative universe Prop for propositions. deppy encodes universes via `UniverseType(level=...)` but does not currently exploit polymorphism beyond a fixed level—the generated certificates target `Type 0` / `Prop` only. This is a deliberate simplification we acknowledge openly: see Chapter 12 §12.7.

2.2 The Calculus of Inductive Constructions

The Calculus of Constructions (CoC) (Coquand and Huet, 1988) extends Π-types with impredicative quantification at Prop. The *Calculus of Inductive Constructions* (CIC) adds inductive type families ('`Inductive Nat : ...`', '`Inductive List ...`'), automatically generating constructors and eliminators that satisfy the appropriate η- and β-rules.

Lean 4 is built on a CIC-style core (with definitional proof-irrelevance and quotient types). The kernel checks *terms* against types; tactics elaborate to terms; the user mostly does not see the kernel because the elaborator is a thick intermediate layer.

2.2.1 Terms vs. tactics

Lean 4 supports two styles:

Listing 2.1: Term-mode and tactic-mode proofs in Lean 4 are interchangeable.

```
-- Term mode:
def two_plus_two : 2 + 2 = 4 := rfl

-- Tactic mode:
theorem two_plus_two' : 2 + 2 = 4 := by
  rfl
```

PSDL compiles to tactic mode. Each PSDL statement emits one or more lines of tactic-mode Lean which the elaborator then expands to a term checked by Lean's kernel. See Chapter 6 for the mapping table.

2.2.2 Inductive types and induction principles

A Lean 4 inductive type:

```
inductive List ( : Type) : Type where
  | nil : List
  | cons : List List
```

generates an automatically-elaborated eliminator

$$\mathsf{List.rec} : \forall (P : \mathsf{List}\,\alpha \to \mathsf{Type}),\ P\,\mathsf{nil} \to (\forall x\,xs,\ P\,xs \to P\,(x :: xs)) \to \forall l,\ P\,l.$$

and similarly for any user inductive. The eliminator is the engine behind both `induction` and `cases` tactics.

deppy mirrors this for the two builtin shapes `NatInduction` and `ListInduction` (kernel terms in `deppy/core/kernel.py`). PSDL surfaces them as the Python loops:

```
for x in xs: # ListInduction
    <step body>

while n: # NatInduction
    <step body>
```

These compile to Lean's `induction xs with` / `induction n with` respectively.

2.2.3 Type-class inference

Lean 4 supports type-class inference via the `class`/`instance` mechanism (de Moura et al., 2024): `Add`, `Mul`, `Decidable`, etc. deppy does *not* encode arbitrary Lean type classes. Generated certificates use a fixed vocabulary of operators (`Int` arithmetic via `omega`, `Eq`-based reasoning, `by_cases` for decidable propositions) and admit anything that requires class inference beyond that vocabulary.

2.3 The HoTT view of identity

The HoTT book (The Univalent Foundations Program, 2013) reframes Martin-Löf's identity types as *path spaces*. In this view:

- Types are spaces (more precisely, ∞-groupoids).
- Terms are points.
- Identity proofs $p : a =_A b$ are paths.
- Identity proofs of identity proofs ($p =_{a=_A b} q$) are homotopies.
- This continues to all dimensions.

The headline axiom is *univalence* (Voevodsky):

$$(A \simeq B) \simeq (A =_{\mathsf{Type}} B).$$

Equivalent types are propositionally equal. This makes algebraic isomorphisms transport-able: if $G \simeq H$ as groups, every property of G is a property of H.

In deppy, univalence shows up in two places:

1. `Univalence(equiv_proof, from_type, to_type)` as a kernel `ProofTerm` (file `deppy/core/kernel.py`, line around 838).
2. Practically, in the form of `DuckPath`: if classes A and B implement the same protocol, they are structurally-equivalent in the universe, and a property proved of A over the protocol-relevant operations transports to B.

We do *not* export univalence to Lean: the generated certificate cannot use `ua` as a primitive because Lean's core is not univalent. Where deppy's kernel uses `Univalence`, the Lean side admits the resulting identification.

2.3.1 Cubical interpretations

The cubical type theories (Cohen et al., 2015; Angiuli et al., 2019) give *computational* interpretations of univalence. Instead of postulating `ua`, they introduce an interval type $\mathbb{I}$ and define identity types as functions out of $\mathbb{I}$ (roughly), with Kan-fillings ensuring the necessary closure properties.

The cubical apparatus in deppy mirrors this without claiming to *be* cubical type theory. Specifically:

- `IntervalType` is the deppy analogue of $\mathbb{I}$.
- `PathAbs(ivar, body)` is path abstraction $\lambda i.\, t$.
- `PathApp(p, r)` is path application $p\, r$.
- `Transport(path, src, tgt)` is the type-level transport.
- `KanFill(dimension, faces, ...)` is the cubical Kan filler.
- `Comp(...)` is Kan composition.

These data classes live in `deppy/core/types.py`. Their kernel verifiers live in `deppy/core/path_engine.py`.

We re-emphasise: this is a *model* suited to the analysis of Python's structural shapes (fibrations on `isinstance`, homotopies between control-flow paths, glue patches across method implementations). It is *not* a `lean-port` of cubical-type-theory. The relationship is one of *inspiration* and *shared vocabulary*, not formal equivalence.

2.4 Lean 4 specifics that we use

We close this chapter with a quick reference for the Lean 4 features the generated certificate exercises.

2.4.1 `def` vs. `abbrev` vs. `axiom` vs. `opaque`

The certificate uses each of these:

`def f : T := body` declares a *definitional unfolding*; tactics can `unfold f` to expose `body`. Used for translated bodies and concrete foundation `Props`.

`abbrev T : Type := ...` is a special "reducible" `def` that the elaborator unfolds eagerly. Used for "`abbrev Point : Type := Int`" bridges where the generated certificate identifies a Python class with the underlying `Int` carrier. This is a deliberate simplification (see Chapter 8).

`axiom A : T` declares `A : T` on the user's authority. No `unfold` possible; nothing computes. Used for foundations Z3 cannot discharge.

`opaque A : T` is an `axiom` variant the elaborator treats as definitionally opaque. Used historically for "`Sidecar_X_prop : Prop`"; the `CONCRETE-PROPS` pass replaces these with concrete `def`s where annotations permit.

The trust ramification is straightforward: a `def` contributes no trust surface; an `axiom` or `opaque` adds one admission to the chain. The certificate counts these (see Chapter 12).

2.4.2 Tactics we emit

The most-frequent Lean tactics emitted by PSDL and the certificate are:

- `rfl` — reflexivity, used when the body translates directly to the property's RHS.
- `exact e` — term-mode discharge, used for foundation citations (`exact Foundation_Real_sqrt_nonneg`) and for direct proof terms.
- `intros / intro` — Π-introduction.
- `by_cases h : P` — decidable case split, used for `isinstance` fibration descents.
- `cases h with` — case analysis on inductives.
- `induction n with` — induction on `Nat` / `List`.
- `unfold f` — expose the body of `def f`.
- `rw [L]` — rewrite using a propositional equality `L`.
- `simp [L1, L2, ...]` — normalise modulo a list of lemmas.
- `omega` — the linear-integer-arithmetic decision procedure; used when PSDL's `assert P, "z3"` can be discharged in linear arithmetic.
- `sorry` — placeholder; emitted when body translation or PSDL compilation cannot close a goal. The certificate counts `sorry`s and refuses certification when too many appear in any verify block (Chapter 12).

2.4.3 The kernel of Lean 4

Lean's kernel is a small, separately-implemented core that type-checks λ-terms. It does not run the elaborator, tactics, or macros; it just sees the terms the elaborator produced and checks them. When we say *Lean's kernel accepted*, we mean exactly that: the elaborator wrote a term, handed it to the kernel, and the kernel returned "OK".

The contractual meaning of an *verified* verify block is that its compiled Lean tactic, when elaborated against the concrete `Prop`, produces a term the kernel accepts. This is the gold

standard deppy aims for. Where the certificate falls short of it (admitting a foundation, leaking a `Structural`, emitting a `sorry`), the gap is recorded and counted.

2.5 What we will and will not formalise

Chapter 4 formalises deppy's metatheory. We will:

- give a syntax for `SynType` and `SynTerm`;
- give the typing judgments and the kernel's verification rules;
- enumerate the proof terms and document each rule;
- state soundness as a meta-theorem and discuss the gaps between the meta-theorem and the implementation.

We will *not*:

- give a fully-formal denotational semantics for Python;
- prove that deppy's kernel reductions are confluent or strongly-normalising as a CIC-style theory;
- claim that an admitted foundation is anything other than an admission.

The reason is honesty. deppy is engineered for utility on real Python code, not for rigorous metamathematical foundations. We say what we have, where the gaps are, and where the user should apply additional scrutiny.

2.6 Summary

The proof-theoretic vocabulary we will assume from now on:

- Π-types, Σ-types, identity types.
- Propositional vs. judgmental equality.
- Term-mode and tactic-mode proofs in Lean 4.
- Inductive types and their eliminators.
- The HoTT picture: types as spaces, paths as identity proofs.
- Univalence as "equivalent types are equal".
- Cubical type theory's interval $\mathbb{I}$, Kan fillers, Kan composition.
- Lean's distinction between `def`, `abbrev`, `axiom`, `opaque`.

The non-trivial point is that deppy re-uses these notions *operationally* but is not *itself* a HoTT or cubical type theory. The next chapter switches sides: it reviews the *Python* constructs the verifier has to actually deal with.

Chapter 3

Background: Python's Semantics

This chapter reviews the Python language constructs that `deppy` has to interpret. It is shorter than the dependent-type-theory chapter because most readers are at least familiar with Python; we focus on the corners that the verifier has to model carefully.

We use "Python" to mean CPython 3.10+ unless we specifically need to discuss earlier versions or alternative interpreters. The sidecar pipeline is implemented against Python 3.10 minimum (the `match`/`case` statement is a frequent dependency).

3.1 Duck typing and structural subtyping

The Python community has long used *duck typing*: "if it walks like a duck and quacks like a duck . . . " Operationally, a function does not insist that its argument inhabit a particular nominal type — it just calls methods on the argument and *expects them to work*.

Listing 3.1: Duck typing in idiomatic Python.

```python
def total_lengths(items):
    return sum(len(x) for x in items)

# Works on lists, tuples, generators, dicts, strings, custom objects:
total_lengths([[1, 2], (3, 4, 5), "abc"]) # 8
```

`total_lengths` works for any iterable whose elements respond to `len()`. Standard *nominal* subtyping ("`List` extends `Iterable`") would require every such carrier to declare itself a subclass of some abstract iterable. Duck typing skips this: the argument just has to *look right* at call time.

PEP 544 (Levkivskyi et al., 2017) formalises this with *Protocols*: structurally-typed records of method signatures. A class implements `Sized` if it has `__len__` with the right signature, full stop — no `class C(Sized)` declaration needed.

3.1.1 Why this matters for verification

The implication for verification is significant. A property such as

> "`total_lengths` returns a non-negative `int`."

is true *for every input that the function does not reject.* The set of acceptable inputs is open — any future class that implements `__len__` returning `int` ≥ 0 qualifies. A nominal-types-only verifier cannot say anything about this property without enumerating all candidate types.

The cubical-duck-type view makes this explicit: a class C that implements a protocol P *inhabits the same fibre* of the protocol-fibration as every other class that implements P. The proof of a protocol-level property descends through the fibration. We discuss the formal model in Chapter 4; the kernel term is `DuckPath`.

3.2 Method resolution order (MRO)

When you call `x.method()`, Python looks up `method` on `type(x)`, then on `type(x)`'s bases, and so on, in a linearised MRO order computed by C3 linearisation.

```
class A: pass
class B(A): pass
class C(A): pass
class D(B, C): pass
D.__mro__
# (D, B, C, A, object)
```

The MRO is *the* static piece of information `deppy` uses to resolve dispatched methods. When the verifier sees `x.distance(y)` on `x : Point`, it consults `Point.__mro__` to find the resolved `distance`, reads its source via `inspect.getsource`, and translates that source.

Multiple inheritance complicates this. When `distance` is defined in two ancestors, the C3 linearisation determines which wins. `deppy` respects the linearisation but does not, in the current implementation, prove anything about the fact that C3 was used (it is taken on the user's authority that CPython implements C3 correctly).

3.3 Exceptions and the $\Diamond$-modality

Python exceptions are first-class control-flow. Every function either *returns a value* (the *value fibre*) or *raises an exception* (the *exception fibre*). These two fibres are disjoint — a function call's outcome is exactly one of them.

The cubical-duck-type model formalises this as a $\Diamond$-modality on types:

- $\Box A$ — the type of guaranteed-pure expressions of type A. No exception, no I/O, no mutation.
- $\Diamond A$ — the type of *maybe-effectful* expressions whose value, when one is produced, is of type A.

A `try / except` block is a $\Diamond$-eliminator that breaks into the two fibres:

Listing 3.2: `try / except` as $\Diamond$-elimination.

```
try: # branch
    result = risky_operation(x)
except ValueError: # exception fibre
    result = fallback(x)
```

In deppy's kernel: `EffectWitness(effect="value", proof=`P_v`)` plus `ConditionalEffectWitness(preconditi`
`ValueError", effect="exception", proof=`P_e`)`.

3.3.1 Why the modality matters

A property like "`Point.distance(p, q) >= 0`" is meaningful *only on the value fibre*. On the exception fibre, the function does not return a value, so the property is vacuously not violated. The PSDL `else`-branch typically discharges the exception fibre with a `raise NotImplementedError` that the kernel records as a `Structural` vacuous claim.

The soundness gate (Chapter 12) flags `Structural` leaves; in the current implementation, vacuity claims on the exception fibre are *permitted* but counted, and `certify_or_die` accepts them only when no counterexample-search outcome contradicts them.

3.4 Monkey-patching

Python permits runtime mutation of classes:

```
import sympy.geometry as G
G.Point.distance = my_faster_distance # monkey-patch
```

After this assignment every `Point` instance, including ones created before the patch, dispatches `distance` to `my_faster_distance`. Any property previously verified about `G.Point.distance` now claims something about `my_faster_distance`.

deppy models monkey-patching as a *path* between class versions:

$$\mathrm{C} =_{\mathsf{PyClass}} \mathrm{C}[\mathrm{m} := \mathrm{v}]$$

which holds *only if* the *behavioural contract* of `C.m` survives the patch. Kernel term: `Patch`.

The honest position is that we do not verify monkey-patches at the moment. The kernel *has* the term, but the pipeline does not detect monkey-patching at run time — the user is implicitly responsible for not patching a class while a verified certificate is in scope. A future iteration of the pipeline could compare the `inspect.getsource` of `C.m` against a recorded SHA-256 (the `expects_sha256:` field on `verify` blocks already exists, see Chapter 5); the gate currently only checks the hash, not the patch.

3.5 Descriptors, properties, and `__getattr__`

Python has *three* attribute-lookup protocols, in priority order:

1. *Data descriptors* on `type(x)` (e.g. `@property`).
2. Instance dictionary entries.
3. Non-data descriptors and class attributes.
4. `__getattr__` on `type(x)` as a fallback.

The `@property` decorator turns a method into a data descriptor: `circle.radius` calls the getter even though it looks like attribute access. The `verify segment_length_code` block in `sympy_geometry.deppy` verifies a property of `Segment.length` which is a `@property`; the verifier has to call `method.fget` explicitly to get the underlying function. See `sidecar_verify._process_one`, lines around 190.

The `__getattr__` fallback is more subtle. It is only called when normal lookup fails, so a class that defines `__getattr__` effectively has an *open* method set. In deppy, `__getattr__`-based proxies are modelled as fibered lookups (kernel term `FiberedLookup`): the resolved attribute is the second projection of a Σ-type $\Sigma\,(t : \mathit{Target}).\,\mathsf{methods}(t)$, and the kernel records the predicate under which the projection inhabits the protocol fibre.

3.6 Metaclasses

A metaclass is a class whose instances are themselves classes. `type` is the default metaclass; many libraries (including SymPy) define custom metaclasses for class-registration tricks.

deppy's kernel records metaclass-defined hierarchies as fibrations:

```
Fiber.from_metaclass(MetaC)
```

returns a `Fiber` proof term whose `type_branches` enumerate registered subclasses (`deppy/core/kernel.py`). This is again *advisory*; the verify-block pipeline does not currently consult metaclass-derived fibrations except when the user explicitly requests fibration descent in the `fibration:` field of a `verify` block.

3.7 The Python AST

The verifier ingests Python source as an `ast.Module` — the same AST the CPython compiler walks. Every `verify` block's target is resolved to a `FunctionDef` whose body is then recursively translated to either:

- a Lean expression, via `deppy.lean.body_translation`;
- a kernel `ProofTerm`, via `deppy.proofs.psdl` applied to a `psdl_proof` block.

The AST nodes that PSDL maps directly to proof-term constructors are listed in Chapter 6; the body-translation mapping (relevant when a verify block does not have an explicit `psdl_proof` but does want its body translated to Lean) is in Chapter 8.

We use `ast.parse(src, mode="exec")` for module-level parsing and `ast.parse(src, mode="eval")` for property expressions. The `ast.unparse` inverse is used heavily in the emitter to produce Lean-side text.

3.8 Notes on dynamic features

The verifier does not, in general, model:

- `exec` / `eval`.

- `globals()` / `setattr` modulo runtime conditionals.
- Dynamically-generated classes via `type(name, bases, dict)`.
- C-extension boundaries (SymPy's `gmpy2` backend, etc.).
- Threads, asyncio scheduling order, GIL semantics.

When the body translator hits one of these (most commonly a C-extension fall-through where `inspect.getsource` fails), it returns `sorry` with a note. The verify block records this in its outcome (`VerifyOutcome.translation_sorry_count > 0`) and the certificate flags it.

3.9 Summary

The Python features the verifier explicitly models:

- Duck typing, structurally typed via PEP 544 protocols (Levkivskyi et al., 2017), encoded as protocol fibrations and `DuckPath`s.
- C3-linearised MRO, taken on faith from CPython, used to resolve dispatched method bodies.
- Exceptions as the exception fibre of a $\Diamond$-modality; `try/except` as $\Diamond$-elimination.
- Monkey-patching as a `Patch` path; not currently auto-detected at runtime.
- Descriptors, especially `@property`, handled by `method.fget` dereferencing.
- `__getattr__` proxies via `FiberedLookup`.
- Metaclass-derived fibrations via `Fiber.from_metaclass`.
- AST-level body translation via `deppy.lean.body_translation`.

The dynamic features outside this set fall into the trust surface described in Chapter 12. The next chapter formalises the cubical-duck-type metatheory that ties together the type theory of Chapter 2 with the Python features of this chapter.

Chapter 4

The Cubical-Duck-Type Metatheory

This chapter formalises the metatheory of `deppy` as it exists in the source tree. The core data lives in:

- `deppy/core/types.py` — types and terms;
- `deppy/core/kernel.py` — judgments, proof terms, `ProofKernel.verify`;
- `deppy/core/cubical_effects.py` — $\Diamond/\Box$ modalities and effect fibrations;
- `deppy/core/path_engine.py` — `PathConstructor`, `TransportEngine`, `UnivalenceEngine`;
- `deppy/core/path_compression.py` — the homotopy-equivalent-branch reduction.

We give the syntax, the typing/verification rules, and the trust discipline. We are explicit where the metatheory *admits* without proof.

4.1 Syntax: types and terms

4.1.1 Types

The universe of `deppy` types is generated by the `SynType` hierarchy in `deppy/core/types.py`. We summarise the constructors we use throughout the rest of the book:

$$
\begin{aligned}
T ::= \; & \mathsf{PyObj} \mid \mathsf{int} \mid \mathsf{float} \mid \mathsf{bool} \mid \mathsf{str} \mid \mathsf{None} \\
& \mid \mathsf{list}[T] \mid \mathsf{dict}[T_k, T_v] \mid \mathsf{tuple}[T_1, \ldots, T_n] \mid \mathsf{set}[T] \\
& \mid \mathsf{class}\langle n; \overline{(m : T)} \rangle \\
& \mid \Pi(x : T_1).\, T_2(x) \mid \Sigma(x : T_1).\, T_2(x) \\
& \mid t_1 =_T t_2 \quad \text{(path / identity)} \\
& \mid \mathsf{U}_n \quad \text{(universe at level } n\text{)} \\
& \mid \{x : T \mid \varphi(x)\} \quad \text{(refinement)} \\
& \mid T_1 \cup T_2 \quad \text{(union)} \\
& \mid \mathbb{I} \quad \text{(cubical interval)} \\
& \mid \mathsf{Glue}(\ldots) \\
& \mid \Box_e T \mid \Diamond_e T \quad \text{(effect-modal)} \\
& \mid \mathsf{Protocol}\langle \overline{(m : T)} \rangle \\
& \mid \mathsf{Optional}[T] \mid \mathsf{Interval}[a, b]
\end{aligned}
$$

The Python primitives (`int`, `float`, `bool`, ..., `class`, `tuple`, `list`) are *nominal* carriers; the HoTT formers (Π, Σ, identity, universe, refinement) are *structural*. The cubical layer ($\mathbb{I}$, `Glue`, $\Box/\Diamond$) is structural *and* explicit about effects.

4.1.2 Terms

The term grammar from `deppy/core/types.py`:

$$
\begin{aligned}
t ::= \; & x \quad \text{(variable)} \\
& \mid \overline{c} \quad \text{(literal)} \\
& \mid \lambda x{:}T.\, t \mid t_1\, t_2 \\
& \mid \langle t_1, t_2 \rangle \mid \mathsf{fst}\, t \mid \mathsf{snd}\, t \\
& \mid \langle \lambda i.\, t \rangle \mid t \,@\, r \quad \text{(path abs/app)} \\
& \mid \mathsf{transport}_T(p, t) \mid \mathsf{comp}_T(\ldots) \\
& \mid \mathsf{glue}(\ldots) \mid \mathsf{unglue}(\ldots) \\
& \mid \mathsf{let}\, x = t_1\, \mathsf{in}\, t_2 \\
& \mid \mathsf{ite}(t_b, t_t, t_e) \\
& \mid \mathsf{pyCall}(t_f, \overline{t_a}) \mid \mathsf{getAttr}(t, m) \mid \mathsf{getItem}(t, t_k)
\end{aligned}
$$

The four `pyCall`/`getAttr`/`getItem`/`ite` constructors are deliberately surface-Python — they let the kernel talk about Python-shaped expressions without forcing a translation into HoTT primitives. This is the *operational* concession that allows deppy to work on idiomatic Python code.

4.1.3 Contexts and judgments

The `Context` is a list of typed bindings $x_1 : T_1, \ldots, x_n : T_n$ (file: `deppy/core/types.py`). A `Judgment` is one of:

$$\begin{aligned} \Gamma \vdash T : \mathsf{Type} &\quad (\texttt{WELL_FORMED}) \\ \Gamma \vdash t : T &\quad (\texttt{TYPE_CHECK}) \\ \Gamma \vdash t_1 \equiv t_2 : T &\quad (\texttt{EQUALITY}) \\ \Gamma \vdash \mathsf{satisfies}(t, T) &\quad (\text{refinement check}) \\ \Gamma \vdash \mathsf{covers}(\overline{T_i}, T) &\quad (\text{fibration coverage}) \end{aligned}$$

The kernel's `verify` method takes a `Context`, a `Judgment`, and a `ProofTerm`, and returns a `VerificationResult` carrying:

- `success : bool`;
- `trust_level : TrustLevel`;
- `message : str` (diagnostic);
- `axioms_used : list[str]` (provenance).

4.2 Trust levels

The kernel ranks accepted proofs on a seven-step scale (file: `deppy/core/kernel.py`, lines 40–65):

6	`KERNEL`
5	`Z3_VERIFIED`
4	`STRUCTURAL_CHAIN`
3	`LLM_CHECKED`
2	`AXIOM_TRUSTED`
1	`EFFECT_ASSUMED`
0	`UNTRUSTED`

The semantics:

`KERNEL` every sub-proof was accepted by an internal rule. This is the strongest level reachable without external discharge.

`Z3_VERIFIED` one or more sub-proofs were discharged by an SMT call. We trust the SMT solver's answer.

`STRUCTURAL_CHAIN` the proof is a structural composition of accepted steps but contains some `Structural` or `Assume` leaves whose semantics is left to the user.

`LLM_CHECKED` the kernel could verify the structure of the proof but at least one obligation was discharged via computational testing (i.e. running the function on samples).

`AXIOM_TRUSTED` one or more sidecar axioms were cited. We trust the user's axiomatic claim.

EFFECT_ASSUMED effect modality witnesses were accepted without proof (the `EffectWitness` term).

UNTRUSTED no useful verification was performed.

The `min_trust` helper (lines 63–65) computes the minimum trust over a list, giving the standard "trust composes by minimum" rule.

4.3 Proof terms: the inventory

The kernel admits the following `ProofTerm` subclasses (file: `deppy/core/kernel.py`; we group them logically, with the "→" showing the kernel rule that fires). This is the authoritative reference for PSDL's compilation targets.

4.3.1 Equality and substitution

$\mathrm{Refl}(t)$ proves $t =_T t$ for any $t : T$.

$\mathrm{Sym}(p)$ from $p : a =_T b$ derive $\mathrm{Sym}(p) : b =_T a$.

$\mathrm{Trans}(p, q)$ from $p : a =_T b$ and $q : b =_T c$ derive $a =_T c$.

$\mathrm{Cong}(f, p)$ from $p : a =_T b$ derive $f(a) =_{T'} f(b)$.

$\mathrm{Transport}(F, p, d)$ from $p : a =_T b$ and $d : F(a)$ derive $F(b)$.

$\mathrm{Unfold}(name, p)$ reduce the named definition and continue with p.

$\mathrm{Rewrite}(L, p)$ rewrite using a propositional equality L and continue with p.

$\mathrm{Cut}(h, T_h, p_h, p_b)$ introduce hypothesis $h : T_h$ proved by p_h, then prove the body with p_b.

$\mathrm{Assume}(h)$ use a previously bound hypothesis h.

4.3.2 Induction

$\mathrm{NatInd}(x, p_0, p_s)$ Nat-induction on x: p_0 is the zero case, p_s the successor case.

$\mathrm{ListInd}(x, p_n, p_c)$ list induction on x.

$\mathrm{Cases}(s, [(pat_i, p_i)])$ case analysis on the scrutinee s.

4.3.3 Cubical / paths

$\mathrm{PathComp}(p, q)$ composition of paths via Kan composition.

$\mathrm{Ap}(f, p)$ congruence on paths — $\mathrm{Ap}(f, p) : f(a) =_T f(b)$ if $p : a = b$.

$\mathrm{FunExt}(x, p)$ from a pointwise proof $\forall x.\, f(x) = g(x)$ derive $f = g$.

$\mathrm{KanFill}(dim, [faces], \dots)$ supply the missing face of a cubical box.

$\mathrm{CechGlue}([patches], [overlaps])$ glue local proofs on a cover into a global proof; requires explicit overlap cocycle agreement.

$\mathrm{Univalence}(e, A, B)$ from $e : A \simeq B$ derive a path $A =_{\mathsf{Type}} B$ (univalence).

Cocycle($level, \overline{c}, boundary, \dots$) a k-cocycle in the safety cochain complex.

4.3.4 Duck typing / dispatch

DuckPath($A, B, \overline{(m, p_m)}$) proves $A \simeq B$ over a protocol's method set, with one method-wise proof p_m per method m.

CondDuckPath($cond, e, A, B$) a duck path that holds only when a runtime predicate *cond* does.

FiberedLookup($attr, p_b, thru$) models `__getattr__`-based proxies: a Σ-projection on the attribute *attr* via the delegate *thru*.

Patch(C, m, v', p_c) a monkey-patch path $C =_{\mathsf{PyClass}} C[m := v']$, valid only when the behavioural-contract proof p_c holds.

4.3.5 Effect / exception fibres

EffectWitness(e, p) a witness that effect e is admitted by proof p. Always at trust `EFFECT_ASSUMED`.

CondEffectWitness($cond, e, p, tgt$) a conditional effect witness: under precondition *cond*, effect e is absent at target *tgt*. Trust composes from p.

4.3.6 Structural and SMT

Structural(*description*) *admits* the obligation on the user's authority. Always at `STRUCTURAL_CHAIN` trust. The soundness gate (Chapter 12) refuses certification when an verify-block proof has a `Structural` leaf in a non-vacuous position.

AxiomInvocation($name, module$) cite a registered axiom or foundation. Trust `AXIOM_TRUSTED`.

Z3Proof($formula, binders$) discharge the formula via the kernel's Z3 backend. Trust `Z3_VERIFIED` when SMT returns `unsat` on the negation.

LeanProof($theorem, proof, \dots$) discharge by invoking Lean. Trust `KERNEL` when Lean accepts.

4.3.7 Safety / termination terms

The remaining `ProofTerm` subclasses (see `deppy/core/kernel.py` from line ~580 onward) carry the safety pipeline's witnesses: `SourceDischarge`, `SemanticSafetyWitness`, `ModuleSafetyComposition`, `TerminationObligation`, `SafetyObligation`. These are used by the safety-pipeline pass (`deppy.pipeline.safety_pipeline`); they are orthogonal to the PSDL verify-block flow but share the trust-level discipline.

4.4 Verification rules

We give the formal rules for the eight non-trivial proof-term constructors. Notation: $\Gamma \vdash p : J/L$ means "proof term p verifies the judgment J at trust level L".

Refl, Sym, Trans, Cong.

$$\frac{\Gamma \vdash a : T}{\Gamma \vdash \mathrm{Refl}(a) : a =_T a/\texttt{KERNEL}} \qquad \frac{\Gamma \vdash p : a =_T b/L}{\Gamma \vdash \mathrm{Sym}(p) : b =_T a/L}$$

$$\frac{\begin{array}{c}\Gamma \vdash p : a =_T b/L_p \\ \Gamma \vdash q : b =_T c/L_q\end{array}}{\Gamma \vdash \mathrm{Trans}(p, q) : a =_T c/\min(L_p, L_q)} \qquad \frac{\begin{array}{c}\Gamma \vdash f : T \to T' \\ \Gamma \vdash p : a =_T b/L\end{array}}{\Gamma \vdash \mathrm{Cong}(f, p) : f(a) =_{T'} f(b)/L}$$

Cut.

$$\frac{\begin{array}{c}\Gamma \vdash p_h : T_h/L_h \\ \Gamma, h : T_h \vdash p_b : J/L_b\end{array}}{\Gamma \vdash \mathrm{Cut}(h, T_h, p_h, p_b) : J/\min(L_h, L_b)}$$

Fiber.

$$\frac{\begin{array}{c}\Gamma \vdash s : \mathsf{PyObj} \\ \forall i.\ \Gamma, s : T_i \vdash p_i : J/L_i \\ \Gamma \vdash \mathsf{covers}(\overline{T_i}, \mathsf{type}(s))\end{array}}{\Gamma \vdash \mathrm{Fiber}(s, [(T_i, p_i)]) : J/\min_i L_i}$$

The `covers` judgment is decided by checking that the *exhaustive* flag is set and the union of T_i contains $\mathsf{type}(s)$ in the type registry; otherwise the kernel adds an implicit "else: `Structural`('uncovered')" branch that downgrades trust to `STRUCTURAL_CHAIN`.

Z3.

$$\frac{\texttt{z3.solve}(\neg\varphi[\overline{x : T}]) = \texttt{unsat}}{\Gamma \vdash \mathrm{Z3Proof}(\varphi, \overline{(x, T)}) : \{\mathsf{True} \mid \varphi\}/\texttt{Z3_VERIFIED}}$$

When the SMT call returns `sat` the kernel *rejects* the proof and additionally surfaces the model as a counterexample. `unknown` is *not* accepted; the kernel returns failure.

LeanProof.

$$\frac{\texttt{lean(theorem T : } \varphi \texttt{ := by } P\texttt{)} \rightsquigarrow \texttt{exit 0}}{\Gamma \vdash \mathrm{LeanProof}(\varphi, P, \dots) : \{\mathsf{True} \mid \varphi\}/\texttt{KERNEL}}$$

Structural and AxiomInvocation.

$$\frac{}{\Gamma \vdash \mathrm{Structural}(d) : J/\texttt{STRUCTURAL_CHAIN}} \qquad \frac{\begin{array}{r}name \in ax \\ axioms[name].sta\end{array}}{\Gamma \vdash \mathrm{AxiomInvocation}(name, n}$$

The "≈" approximate-match is essential: the kernel parses the axiom statement as Python AST and pattern-matches it against the goal. Exact textual match is the success case; canonicalising rewrites recover from minor variable-renaming and operator-form differences.

Cocycle.

$$\frac{\begin{array}{c}\forall i.\ \Gamma \vdash c_i : J_i/L_i \\ \forall (i,j) \in \mathit{boundary_pairs}.\ \Gamma \vdash o_{ij} : c_i \approx_{(i \cap j)} c_j/L_{ij} \\ \Gamma \vdash z : \delta\overline{c} = 0/L_z \text{ (when supplied)}\end{array}}{\Gamma \vdash \mathrm{Cocycle}(k, \overline{c}, bp, \overline{o}, z) : J/\min_{(i,j,z)}(L_i, L_{ij}, L_z)}$$

The boundary-coherence check is non-trivial: each overlap proof o_{ij} witnesses that components c_i and c_j *agree on their overlap*. When boundary pairs are not supplied the kernel falls back to a coarser "each component independently" verification and downgrades the cocycle's classification to *unstructured*.

4.5 Cubical effects

We sketch the modal layer; the source is `deppy/core/cubical_effects.py`.

4.5.1 Effect categories

The `EffectCategory` enum is:

$$\mathsf{PURE}, \mathsf{IO}, \mathsf{STATE}, \mathsf{EXCEPTION}, \mathsf{ALLOCATION}, \mathsf{NONDETERMINISM}, \mathsf{DIVERGENCE}.$$

An `EffectSignature` carries a set of categories, a set of "resources" (file handles, network endpoints), and a set of "modifies" (mutated variables). The `compose` operation is union; the `subsumes` order is component-wise inclusion.

4.5.2 Modal types

For an effect signature E:

$$\begin{array}{ll} \Box_E A & \text{guaranteed-pure } A \text{ when } E \subseteq \{\mathsf{PURE}\} \\ \Diamond_E A & \text{maybe-effectful } A \text{ producing values when no exception} \end{array}$$

The expected unit-and-bind structure holds:

$$\begin{aligned} \mathsf{lift} &: A \to \Diamond_E A \\ \mathsf{bind} &: \Diamond_E A \to (A \to \Diamond_E B) \to \Diamond_E B \\ \mathsf{purify} &: \Box_{\mathsf{PURE}} A \to A \end{aligned}$$

The `CubicalEffectVerifier` (file: `deppy/core/cubical_effects.py`, line 240+) constructs a real `TransportProof` when transporting an effect proof along an equivalence path between two function ASTs.

4.5.3 Exception fibres

An `EXCEPTION`-effect call to a function $f : A \to B$ is modelled as having codomain $\Diamond_{\{\mathsf{EXCEPTION}\}} B$, whose *value fibre* is B and whose *exception fibre* indexes a Python exception type. A property $\varphi : B \to \mathsf{Prop}$ holds of f when:

- the value fibre of f has type $\{x : B \mid \varphi(x)\}$, and

- the exception fibre is non-empty (contains a witness raise).

The kernel records the second condition with `EffectWitness` or `ConditionalEffectWitness`. The pipeline emits `vacuous "..."` (kernel `Structural`) for the exception fibre by default; `certify_or_die` accepts the vacuous claim when no counterexample is found.

4.6 Path engine

`deppy/core/path_engine.py` contains the constructors and verifiers for the path layer:

- `PathConstructor.function_path(fn1, fn2)` produces a `PathAbs` witness of $fn_1 \simeq fn_2$ when their bodies differ only in renaming/algebraic-identity-applications.
- `TransportEngine.transport(motive, p, d)` verifies the transport rule and returns a checked `TransportProof` term.
- `UnivalenceEngine.from_equiv(e, A, B)` constructs a `Univalence` proof term over an equivalence triple.
- `CechDecomposer.decompose(cover)` returns a ech-style decomposition of a property over a class hierarchy — the basis for the `CechGlue` term.

The path engine is wired into the PSDL compiler primarily through the `transport(...)` and `equiv_to_path(...)` sugar. See Chapter 6.

4.7 The cubical-duck-type model: pulling it together

The combined picture is best read off the example. Take `Point.distance(p, q) >= 0`. We want a proof that for *every* p, q inhabiting the *actual* SymPy `Point` class, the value-fibre of the call satisfies ≥ 0.

1. Resolve `Point.distance` via `importlib`; read its source via `inspect`.
2. Notice the body has shape

```
if isinstance(other, Point): ...
else: ... raise TypeError ...
```

3. This is a *fibration* on `isinstance`: the source type is PyObj, the fibres are Point and its complement.
4. In the `Point`-fibre, the body returns `sqrt(sum_zip_sub_sq(self, other))`, with `sum_zip_sub_sq` a sum of squares (always non-negative) and `sqrt` non-negative on non-negatives (foundation `Real_sqrt_nonneg`).
5. In the complement fibre, the body raises a `TypeError`: this is the *exception fibre* of the $\Diamond$-modality. The property is vacuously preserved because there is no return value to compare.

The corresponding `ProofTerm` encodes step (3) as a `Fiber`, step (4) inside the value-fibre branch as `Cut(..., Z3Proof(...), ..., AxiomInvocation("Real_sqrt_nonneg"))`, and step (5) in the complement-fibre branch as a `Structural("vacuous: exception fibre")`.

The trust level composes as

$$\min(\texttt{Z3_VERIFIED}, \texttt{AXIOM_TRUSTED}, \texttt{STRUCTURAL_CHAIN}) = \texttt{STRUCTURAL_CHAIN}.$$

The certificate documents this and `certify_or_die` inspects whether the `Structural` leaf is in a vacuous position; in this case it is, so the gate passes.

4.8 Soundness, in the only sense we can claim it

We have not given a denotational semantics for `deppy` terms, and we have not proved cut-elimination. What we *do* claim:

Theorem 4.1 (Engineering soundness)**.** *For every verify-block whose certificate prints `CERTIFIED`, the following hold:*

1. *The `ProofTerm` the kernel built passes the rules of §4.4.*
2. *Every `Z3Proof` sub-term was checked by `Z3` (or by the cached lookup of a previous identical check).*
3. *Every `LeanProof` sub-term was checked by `Lean` (or by cache).*
4. *Every cited foundation has either been `Z3`-verified or been admitted with the foundation declaration visible in the certificate.*
5. *The certificate's `Lean` text was round-tripped via `check_lean_module_source` and `Lean` returned `exit 0`.*
6. *The `certify_or_die` list of failure conditions is empty.*

Justification. Items (1)–(5) follow by inspection of the implementation. Item (6) is enforced operationally: `certify_or_die` (file `deppy/lean/sidecar_runner.py`, lines 62–98) refuses `CERTIFIED` unless every item is satisfied. □

Remark 4.2. Theorem 4.1 is *not* a logical-consistency result. It is a statement about what the implementation does. A mathematical-soundness theorem would require modelling Python's semantics, the SMT solver's semantics, and `Lean`'s metatheory; we have done none of those things.

The implication for the user is that the trust surface bottoms at: `Lean`'s kernel, `Z3`'s answer to a satisfiability query, CPython's faithful execution of the source bytes that `inspect` returned, and the user's hand-written admissions (foundations, opaque axioms, structural leaves). We hope this is a trust surface a careful user can reason about.

4.9 Summary

We have given the metatheory:

- **Types** include the Python carriers, the HoTT formers, and the cubical/effect layer.
- **Terms** include the standard λ-calculus, HoTT path constructors, and surface-Python idioms.

- **Judgments** are the standard four plus fibration-coverage.
- **Trust levels** run from `KERNEL` (full) to `UNTRUSTED` (none); `min_trust` composes.
- **Proof terms** are 38+ subclasses spanning equality, induction, cubical, duck-type, effect, structural, and external discharge primitives.
- **The verification rules** are local on each constructor and composable via `min_trust`.
- **The cubical effects** make the value/exception fibres of Python calls explicit.
- **Soundness** is engineering soundness (Theorem 4.1), not metamathematical soundness.

The next chapter introduces the syntax of the sidecar files in which all this is invoked: the `.deppy` format.

Chapter 5

The `.deppy` Sidecar Format

A sidecar file (`*.deppy`) is the user-facing interface to the `deppy` pipeline. It declaratively records, for one Python library:

- a small number of *foundations* (mathematical facts the user is willing to admit);
- one or more *specs* (interface contracts on classes / functions);
- *axioms* attached to specs (claims about library behaviour);
- *lemmas* (intermediate facts the user wants to prove once and re-use);
- *verifies* (the unit of verification: function + property + proof script);
- *proofs* (legacy form: name + by-clause selecting a derivation strategy);
- top-level *code_types* (signatures for helper functions PSDL references);
- top-level *lean_imports* / *lean_preamble* (raw `Lean` text injected verbatim).

The file is read by `SidecarModule.from_file` (file: `deppy/proofs/sidecar.py`) and parsed into a `SidecarModule` containing `FoundationDecl` / `SpecDecl` / `AxiomDecl` / `LemmaDecl` / `VerifyDecl` / `ProofDecl` dataclasses. The pipeline then operates exclusively on this `SidecarModule`.

5.1 File-level header

Every sidecar starts with a module identifier:

Listing 5.1: Header of `sympy_geometry.deppy`.

```
module: sympy
version: ">=1.12"
```

The `module:` field is the dotted name of the target Python package, used by the runner (`deppy.lean.sidecar_runner`) to look up provenance information (version, source path) via `importlib.import_module`. The `version:` field is informational; the pipeline does not currently enforce it. A future extension could compare against `module.__version__`.

5.2 Foundations

A foundation is a generic mathematical fact admitted as part of the trust surface. Syntax:

```
foundation Real_sqrt_nonneg: "sqrt(x) >= 0 when x >= 0"
foundation Real_add_comm: "a + b == b + a"
foundation Real_sub_sq_swap: "(a - b)**2 == (b - a)**2"
```

Each foundation has a name (a Lean-safe identifier when underscored) and a Python-syntax statement. A trailing `when ...` guard introduces a precondition.

The pipeline tries each foundation through Z3 (Chapter 9); foundations involving `sqrt`, `sin`, `cos`, ... fall through and stay admitted. In the example sympy file, `Real_add_comm` is provable by `omega` in Lean, so the certificate emits a real `theorem Foundation_Real_add_comm` with body `by intros; omega`; `Real_sqrt_nonneg` stays `axiom`-admitted because Z3 over `Int` cannot model `sqrt`.

A foundation can also carry an explicit user-supplied Lean proof:

```
foundation Real_my_fact:
  statement: "<python statement>"
  citation: "Bourbaki, Algebra II, ğ3.4"
  lean_signature: "<optional Lean type override>"
  lean_proof: |
    intros
    omega
```

When `lean_proof` is non-empty, the certificate emits a real `theorem ... := by <lean_proof>` instead of an `axiom`; Lean must elaborate the proof.

5.3 Specs and axioms

A `spec` attaches a contract bundle to a class or a function. Inside it, `axiom` declarations attach *claims about behaviour* to that target.

Listing 5.2: Spec for `sympy.geometry.point.Point`.

```
spec sympy.geometry.point.Point:
  guarantee: "Euclidean distance is a metric on Rš; collinearity is the vanishing cross
      product"
  pure: true
  axiom dist_nonneg: "Point.distance(p, q) >= 0"
  axiom dist_zero_iff_equal: "(Point.distance(p, q) == 0) == (p == q)"
  axiom dist_symmetric: "Point.distance(p, q) == Point.distance(q, p)"
  axiom dist_triangle: "Point.distance(p, r) <= Point.distance(p, q) + Point.distance(q, r)"
  axiom dist_pythagoras: "Point.distance(p, q)**2 == (p.x - q.x)**2 + (p.y - q.y)**2"
  axiom collinear_iff_zero_cross: "is_collinear(A, B, C) == ((B.x-A.x)*(C.y-A.y) == (C.x-A.x)
      *(B.y-A.y))"
```

The fields:

`guarantee:` a free-text human-readable summary of the spec's contract. Used in the certificate as a header comment.

pure: true / false purity claim. Pure functions inhabit the □-modality (no exceptions, no I/O, no mutation).

requires: / ensures: (optional) Hoare-style pre/postconditions, in Python expression form.

reads: / mutates: (optional) frame conditions for impure functions.

axiom NAME: "...". a sub-declaration that becomes an `AxiomDecl` attached to the enclosing spec.

The axiom statement is a Python expression. The spec's target provides the namespace: an `axiom dist_nonneg` on `spec ...Point` compiles to a `Lean Prop` quantifying over the binders implied by the statement.

5.4 Verify blocks

A `verify` block is the unit of verification: it names a function, a property, and a proof strategy. It is the most direct way to drive the kernel pipeline. The full schema (file: `deppy/proofs/sidecar.py`, dataclass `VerifyDecl`, lines 388–446):

Listing 5.3: Schema of a `verify` block (all fields shown; only `function` and `property` are mandatory).

```
verify <NAME>:
  function: <dotted path to method>
  property: "<Python predicate>"
  via: foundation <foundation-name> # OR axiom <axiom-name>
  fibration: isinstance <Type> # optional
  cubical: true | false # optional
  binders: p: Point, q: Point # optional, drives concrete Props
  requires: "<precondition>" # optional
  ensures: "<postcondition>" # optional
  effect: "pure" # optional
  decreases: "<measure expression>" # optional
  z3_binders: x: int, y: int # optional
  cubical_dim: 0 # optional
  cohomology_level: 0 # optional
  expects_sha256: "<sha-256 hex>" # optional, hash drift check
  expecting_counterexample: false # optional
  by_lean: | # optional, raw Lean tactic
    intros
    apply Foundation_X
    omega
  psdl_proof: | # optional, PSDL block
    if isinstance(other, Point):
        ...
```

5.4.1 Mandatory fields

function. A dotted path that `importlib` can resolve down to a callable (or a `property` descriptor whose `fget` is a callable). The pipeline reads its source via `inspect.getsource` and hashes it for the `source_hash` field.

property. A Python expression in the function's binder context. The expression is parsed via `ast.parse(..., mode="eval")`; identifiers that look like free variables are quantified with the binder types from `binders:`; identifiers that look like classes resolve via the `code_types:` table; identifiers that match the function's own class invoke the method machinery from `deppy.lean.sidecar_mechanization`.

5.4.2 Discharge strategy

The `via:` clause selects a discharge strategy. Currently recognised forms:

- `via: foundation NAME` — cite the named foundation as the proof.
- `via: axiom NAME` — cite a sidecar axiom.
- `via: chain F1, F2, F3` — transitively compose the named axioms/foundations via `Eq.trans`.
- `via: rewrite F then cite A` — rewrite under foundation F, then cite axiom A.
- `via: composition F1 & F2 & F3` — and-intro a conjunction of named claims.

When neither `psdl_proof` nor `by_lean` is given, the `via:` clause is the proof strategy and the kernel *builds* the appropriate `ProofTerm` (an `AxiomInvocation`, `Trans` chain, `Rewrite`, or `Cut`).

5.4.3 Proof script fields

The two proof script fields are mutually compatible:

`psdl_proof: |` A multi-line Python-shaped tactic script. Each statement compiles to one or more `ProofTerm`s and one or more `Lean` tactic lines; the resulting `Lean` block is wrapped in `by ...` for the generated theorem. See Chapter 6.

`by_lean: |` Raw `Lean` tactic text, inserted verbatim. The user is responsible for the tactic body elaborating against the concrete `Prop`; the `deppy` pipeline only checks for SHA hash drift on the source.

When both are provided, `by_lean` takes precedence in the `Lean` certificate but `psdl_proof` still produces a `ProofTerm` the kernel checks. This is the primary way to provide "a proof in two notations" for the same property.

5.4.4 Optional fields driving concretisation

The annotation-driven preamble (Chapter 8) reads:

- `binders: p: Point, q: Point` — declares `axiom Point : Type`, then uses these as `intros p q` and as the `Verified_X_property`'s quantifiers.
- `requires: "p != q"` — an extra hypothesis that the generated `Verified_X_property` body assumes.
- `ensures: "<expr>"` — an extra clause appended to the property body.
- `effect: "pure"` or `"may_raise TypeError"` — recorded for the certificate as a comment; informs the fibration-descent pass.

- `decreases: "<measure>"` — a well-founded measure used by the safety pipeline to discharge recursion-depth obligations.
- `z3_binders: x: int, y: int` — explicit Z3 sorts overriding the typed-Z3 encoder's inference.
- `cubical_dim: 2` / `cohomology_level: 1` — request specific dimension / level for the cubical analysis.
- `expects_sha256: "<hex>"` — a commitment to the inspected source bytes. The pipeline hashes the live `inspect.getsource` output; mismatch fires `hash_drift_count` and `certify_or_die` refuses certification.
- `expecting_counterexample: true` — assert that the counterexample search *should* produce one (i.e. the property is supposed to be false). Useful for testing the counterexample finder; if no counterexample is produced when one is expected, certification fails.

5.5 Lemmas

A `lemma` declaration introduces an intermediate fact:

```
lemma triangle_inequality_aux:
  statement: "(a + b)**2 <= 2 * (a**2 + b**2) when a >= 0 and b >= 0"
  lean_proof: |
    intros a b ha hb
    nlinarith [sq_nonneg (a - b)]
```

A lemma can be either:

- *admitted* (no `lean_proof`; certificate emits `axiom Lemma_X : Lemma_X_prop`);
- *proved by user-supplied Lean*; the certificate emits `theorem Lemma_X : Lemma_X_prop := by ...`.

Either way, the lemma is registered as a kernel axiom (file: `deppy/lean/sidecar_certificate.py`, lines around 183–188: "**AUDIT 1.8**: register every parsed lemma as a kernel axiom"). Subsequent `verify` blocks may cite it via `apply(triangle_inequality_aux)` in PSDL.

5.6 The `code_types:` block

A new top-level block lets the user provide typed signatures for helper symbols PSDL references.

Listing 5.4: A `code_types:` block.

```
code_types: |
  sqrt: Int Int
  sum_zip_sub_sq: Point Point Int
```

These drive concrete `axioms` in the Lean preamble:

```
axiom sqrt : Int Int
axiom sum_zip_sub_sq : Point Point Int
```

The point is that PSDL can mention `sqrt(...)` and `sum_zip_sub_sq(self, other)` in the user's natural notation, and the certificate's preamble will declare these as typed Lean symbols rather than opaque Int constants. Without `code_types`, every helper symbol becomes `opaque h : Int` and the surrounding tactics fail to elaborate.

5.7 Top-level Lean integration

Two top-level keys inject text directly into the certificate:

- `lean_imports: |` — one `import` per line, inserted at the top of the generated certificate.
- `lean_preamble: |` — raw Lean text inserted after the imports and before the annotation-driven preamble.

These are the escape hatches. When the user's proofs require particular Lean machinery (a Mathlib-style import, a custom `macro`), the `.deppy` file can introduce it. The `certify_or_die` gate does not refuse certification on the basis of which imports were used — only on whether Lean accepts the resulting certificate.

5.8 The legacy `proof` form

The original `.deppy` format used `proof` declarations:

Listing 5.5: Legacy `proof` form from the SymPy sidecar.

```
proof distance_nonneg_proof:
  target: sympy.geometry.point.Point
  by: foundation Real_sqrt_nonneg

proof distance_pythagoras_proof:
  target: sympy.geometry.point.Point
  by: rewrite Real_sqrt_sq_nonneg then cite dist_pythagoras

proof distance_zero_iff_equal_proof:
  target: sympy.geometry.point.Point
  by: chain Real_sqrt_zero_iff_zero, Real_sum_of_nonneg_zero_iff_all_zero,
      Real_sub_sq_zero_iff_eq

proof distance_is_metric_proof:
  target: sympy.geometry.point.Point
  by: composition dist_nonneg & dist_zero_iff_equal & dist_symmetric & dist_triangle
```

The `by:` clause supports five forms:

- `by: foundation NAME` — proof body is `exact Foundation_NAME`;
- `by: axiom NAME` — `exact Sidecar_NAME`;
- `by: refl` — `rfl`;
- `by: chain F1, F2, F3` — `F1.trans F2 .trans F3`;
- `by: rewrite F then cite A` — `rw [F]; exact A`;
- `by: composition F1 & F2 & F3` — `And.intro F1 (And.intro F2 F3)`.

The `verify` form supersedes `proof` for new code: it exposes more knobs (binders, requires/ensures, PSDL, by_lean) and its trust accounting is more granular. We retain the legacy form for backwards compatibility.

5.9 File layout reference

A complete sidecar file has roughly the following layout:

Listing 5.6: Canonical layout of a sidecar file.

```
# 1. Header
module: <package>
version: ">=<min>"

# 2. Top-level Lean integration (optional)
lean_imports: |
  import Mathlib.Tactic
  import Std

lean_preamble: |
  -- raw Lean text injected verbatim

# 3. Helper signatures (optional)
code_types: |
  helper1: T1 T2
  helper2: T1 T2 T3

# 4. Foundations (the trust surface)
foundation F1: "stmt1"
foundation F2: "stmt2 when guard"
foundation F3:
  statement: "stmt3"
  citation: "ref"
  lean_proof: |
    <Lean tactics>

# 5. Specs (per class/function)
spec <dotted.path>:
  guarantee: "..."
  pure: true
  axiom A1: "stmt"
  axiom A2: "stmt"

# 6. Lemmas (intermediate facts)
lemma L1:
  statement: "..."
  lean_proof: |
    <Lean tactics>

# 7. Verifies (units of verification)
verify V1:
  function: <dotted.path>
  property: "..."
  binders: x: T, y: U
  via: foundation F1
  psdl_proof: |
```

```
    <PSDL tactics>

# 8. Legacy proofs (optional)
proof P1:
  target: <dotted.path>
  by: chain F1, F2
```

5.10 The SidecarModule object

The parsed file is a Python object with the following relevant fields (from `deppy/proofs/sidecar.py`):

Listing 5.7: `SidecarModule` excerpt — the key fields and their types.

```
class SidecarModule:
    name: str
    target_module: str
    version: str
    _specs: dict[str, ExternalSpec]
    _axioms: dict[str, AxiomDecl]
    _foundations: dict[str, FoundationDecl]
    _proofs: list[ProofDecl]
    _verifies: list[VerifyDecl]
    _lemmas: list[LemmaDecl]
    _foundation_deps: dict[str, list[str]]
    _imports: list[str]
    _symbolic_constants: dict[str, str]
    _predicates: dict[str, str]
    _code_types: dict[str, str] # CODE-TYPES typed signatures
    _lean_imports: list[str]
    _lean_preamble: str
    _about_specs: list[tuple[Any, _SpecMetadata]]
```

The pipeline is library-agnostic: every transformation operates on this dataclass. The transformations are documented in Chapter 7.

5.11 Worked example: sympy_geometry.deppy

We end this chapter by walking through the SymPy sidecar at the top of the repository. The file is 508 lines and breaks into these sections:

1. Lines 1–46: comments and header (`module: sympy`, `code_types: sqrt: Int Int` and `sum_zip_sub_sq: Point Point Int`).
2. Lines 48–65: 14 `foundation` declarations (`Real_sqrt_nonneg`, `Real_add_comm`, ...).
3. Lines 70–82: `spec sympy.geometry.point.Point` with 8 `axiom`s about distance and collinearity.
4. Lines 89–96: `spec sympy.geometry.line.Segment` with 5 `axiom`s about midpoint and length.
5. Lines 105–120: `spec sympy.geometry.polygon.Triangle` with 16 `axiom`s about centroid, area, Euler line, etc.

6. Lines 127–134: `spec sympy.geometry.ellipse.Circle` with 5 `axioms`.
7. Lines 142–149: `spec sympy.geometry.polygon.Polygon` with 5 `axioms`.
8. Lines 161–393: 38 legacy `proof` declarations.
9. Lines 404–508: 9 `verify` blocks, including the PSDL example `dist_nonneg_psdl`.

The verify-block list:

- `dist_nonneg_code` — via foundation `Real_sqrt_nonneg`, with fibration descent.
- `dist_pythagoras_code` — via foundation `Real_sqrt_sq_nonneg`.
- `dist_symmetric_code` — via `Real_sub_sq_swap`.
- `segment_contains_code`.
- `triangle_area_nonneg_code`.
- `triangle_perimeter_sum_code`.
- `triangle_centroid_avg_code`.
- `segment_length_code`.
- `polygon_area_nonneg_code`.
- `circle_radius_eq_code`.
- `dist_nonneg_psdl` — the worked PSDL example.

The `dist_nonneg_psdl` block (lines 487–508) is the canonical PSDL demonstration: `psdl_proof` contains a real Python-syntax proof script. We will return to this block repeatedly throughout the rest of the book.

5.12 Summary

The `.deppy` sidecar format:

- names a target Python module;
- lists *foundations* as the trust surface;
- attaches *specs* with *axioms* per target;
- declares *lemmas* for re-use;
- drives the verifier via *verify* blocks with optional PSDL or raw Lean proof scripts;
- integrates with Lean via `lean_imports` and `lean_preamble`;
- supports a legacy `proof` form for back-compat.

The next chapter introduces PSDL: the Python-syntax tactic language that fills the `psdl_proof:` field.

Chapter 6

PSDL: The Python-Semantic Deppy Language

PSDL is the strong proof language at the heart of the deppy verification pipeline. It is what fills the `psdl_proof:` field of a `verify` block. This chapter is a complete reference: design rationale, syntax, statement-level mapping, soundness arguments, and worked examples.

The implementation is in `deppy/proofs/psdl.py` (1553 lines as of writing). We cite specific line ranges as we go.

6.1 Design intent

The unifying design choice of PSDL is that *the proof script is literal Python.* `ast.parse` reads it; the `_Compiler` class walks the AST and maps each statement to:

1. a deppy `ProofTerm` that `ProofKernel.verify` type-checks;
2. a line of Lean 4 tactic-mode text emitted into the enclosing `by ...` block.

Two design constraints follow:

1. No new syntax. Users should not have to learn a ninth proof language. PSDL *is* Python — if it parses as Python, PSDL can in principle accept it. The semantics of each construct, however, is *tactic-language* semantics, not Python runtime semantics: `if isinstance(other, Point)` does not *run* an `isinstance` check, it *introduces a fibration.*

2. Two-target compilation. Every statement must produce both a `ProofTerm` and a Lean tactic. When a statement cannot be compiled to one of the two, the result records an error:

```
@dataclass
class PSDLError:
    line_no: int
    line_text: str
    message: str

@dataclass
```

```
class PSDLCompileResult:
    proof_term: Optional[ProofTerm] = None
    lean_tactic: str = ""
    term_repr: str = ""
    errors: list[PSDLError] = field(default_factory=list)
    hints: list[str] = field(default_factory=list)
    show_goal_requested: bool = False
    foundations_referenced: list[str] = field(default_factory=list)
```

The `ok` property is `True` iff both the `ProofTerm` is constructed and `errors` is empty. This is the gate.

6.2 The two-vocabulary structure

PSDL's vocabulary splits cleanly into two halves, kept separate to keep clear what is borrowed from CIC and what is intrinsic to deppy.

CIC core. Π, Σ, identity types, induction principles, substitution, equivalences, congruence, function extensionality, transport, path induction. Same expressive power as Lean / Coq / Agda.

Cubical / duck / effect Python primitives. Things that make sense only because the underlying language being reasoned about is Python:

- `DuckPath` dispatch on shared method sets;
- $\Diamond$ / $\Box$ effect modalities for value/exception fibres;
- `isinstance` as a fibration;
- Kan-filling in cubical control-flow analyses;
- cocycle constructors for inter-procedural cohomology.

The lookup tables in this chapter group the constructors by which half of the vocabulary they inhabit.

6.3 Surface mapping: control flow

6.3.1 `if isinstance(x, T):`

A standard idiom in Python: `if isinstance(x, T): ...else: ....` In PSDL this is the canonical fibration construction.

```
if isinstance(x, T):
    <value-fibre body>
else:
    <complement-fibre body>
```

The compilation, from `psdl.py`'s `_compile_if` (lines 609–642):

- Detects the `isinstance` test pattern via `_is_isinstance_test`.
- Builds a `Fiber(scrutinee=Var(x), type_branches=[(T, body), (_, else_branch)], exhaustive=Tru`

- Emits Lean: `by_cases h : isinstance x T`, then the `ů` bullets for each fibre.

When the condition is *not* `isinstance`, the compiler falls through to a generic `Cases` on the test expression:

```
if x > 0:
    ...
else:
    ...
```

becomes a `Cases(scrutinee=Var("x > 0"), branches=[("then", ...), ("else", ...)])` and Lean `by_cases h : x > 0`.

6.3.2 try / except

```
try:
    <value-fibre body>
except TypeError:
    <handler body>
```

`_compile_try` (lines 645–666) folds:

1. the `try`-body into an `EffectWitness(effect="value", proof=`P_v`)`;
2. each `except` handler into a `ConditionalEffectWitness(precondition="raises Exc", effect="exception", proof=`P_e`, target=Exc)`.

The Lean side emits `- try block (effect modality)` as a comment and the handler body inline. The kernel verifies the inner proofs; the modal layer is recorded as effect-fibre witnesses.

6.3.3 for and while

```
for x in xs:
    <step body>
```

compiles via `_compile_for` (lines 668–685) to `ListInduction(var=x, nil_proof=Refl(...), cons_proof=step_body)`. The Lean emits `induction xs with` followed by `| nil => rfl` and `| cons _ _ ih =>`.

```
while n:
    <step body>
```

compiles to `NatInduction(var=n, base_proof=Refl, step_proof=body)`. The default base case is `rfl`; users who want a richer base case use the explicit `induction(n, base=..., step=...)` call form.

6.3.4 match / case

Python 3.10's structural pattern-matching:

```
match x:
    case A:
        <body_A>
    case B:
        <body_B>
```

compiles to `Cases(scrutinee=Var("x"), branches=[(A,` P_A`), (B,` P_B`)])` and Lean `match x with` A => ... | B => ...|.

6.4 Surface mapping: logical / equality tactics

6.4.1 `assert P` and discharge modes

```
assert P # default: discharge by Z3
assert P, "z3" # explicit Z3 discharge
assert P, "by F" # cite foundation F
assert P, "by lemma L" # cite lemma L
```

The compiler (`_compile_assert`, lines 710–777) reads the optional message argument as a discharge selector:

- `"z3"` (or no message) → `Cut(..., Z3Proof(formula=P, binders={}), ...)`; Lean emits `have h : P := by omega -- z3-discharged`.
- `"by F"` where F is in the foundations dict → `Cut(..., AxiomInvocation(F, "foundations"), ...)`; Lean emits `have h : P := Foundation_F`.
- `"by lemma L"` → `Cut(..., AxiomInvocation(L, ""), ...)`; Lean emits `have h : P := L`.

The `Cut` bound name is auto-generated as `h_<lineno>`.

6.4.2 `apply(F)` and bare-name citation

```
apply(F)
F # alias: bare-name reference
```

Both compile to `AxiomInvocation(name=F, module=...)`. When F is a registered foundation, the module is `"foundations"` and the Lean emits `exact Foundation_F`. Otherwise the module is empty and Lean emits `exact F`.

6.4.3 `inline`, `unfold`, `rewrite`

```
inline(method, expected_expr) # body translation assertion
unfold(method) # plain Unfold
method.unfold() # alias

rewrite(L) # rewrite using L
L.rewrite() # alias
```

`inline(method, expected_expr)` emits an `Unfold` proof term whose body is `Refl(Var(expected_expr))` — i.e., it asserts that after unfolding `method`, the body matches `expected_expr` up to reflexivity. The Lean side emits `unfold method -- expected: <expected_expr>`.

`unfold` alone emits `Unfold(method, Refl(...))` with no explicit RHS. `rewrite` emits `Rewrite(lemma, Refl(...))` and Lean `rw [lemma]`.

6.4.4 return, pass, raise

```
return expr # Refl
pass # Structural("trivial"); Lean: trivial
raise TypeError # exception fibre marker
raise NotImplementedError # vacuous (gated Structural)
```

`return expr` compiles to `Refl(Var(safe(expr)))`; the Lean prints the comment `-- return <expr>` followed by `rfl`.

`raise NotImplementedError` is the canonical *vacuous-branch* marker: emit `Structural("vacuous: NotImplementedError")` and Lean `exact absurd _ (by simp)`. The soundness gate counts this `Structural` leaf and the certificate's `certify_or_die` accepts it only when the verification context (e.g. the exception fibre of an `if isinstance`) makes the branch genuinely unreachable.

6.4.5 Local lemmas and assignments

```
h: T = expr # annotated: typed local lemma
L = lemma(name, statement, by) # explicit lemma constructor
```

The annotated assignment compiles via `_compile_ann_assign` (lines 808–827) to a `Cut` whose `hyp_type` is `RefinementType(base_type=PyBoolType, predicate=T)` and whose `lemma_proof` is the compiled RHS. The Lean side emits `have h : T := <RHS>`.

The `lemma(...)` form (compiled in `_compile_func_call` under `lemma`, lines 1020–1040) is an explicit constructor: `lemma(name, statement, by)` where `by` is the discharge text.

6.5 Surface mapping: CIC core

The CIC core surface is what gives PSDL its theoretical expressive power. Every constructor in this section emits a real Lean tactic that elaborates against the corresponding CIC rule.

6.5.1 Quantifiers and induction

```
forall(x, T, body_proof) # introduction
exists(x, T, witness, body) # introduction (with explicit witness)

induction(n, base=..., step=...) # explicit Nat induction
```

`forall(x, T, body)` emits `intro (x : T)` and recurses into `body`. The `ProofTerm` is a `Cut` whose `hyp_type` is the refinement `{x : T}` and whose `lemma_proof` is the de Bruijn assumption `Assume(name=x)`.

`exists(x, T, witness, body)` emits `refine witness, ?_`. The `ProofTerm` is a `Cut` whose `lemma_proof` is `Refl(Var(witness))` and whose `body_proof` continues with the supplied body.

`induction(n, base=`P_0`, step=`P_s`)` emits `induction n with` followed by the base/step alternatives, and the term is `NatInduction(var=n, base_proof=`P_0`, step_proof=`P_s`)`.

6.5.2 Equality and substitution

```
refl(x) # Refl[x]
sym(eq) # Eq.symm
trans(eq1, eq2) # Eq.trans
cong(f, eq) # congrArg f eq
subst(eq, motive, body) # rw [eq]
rew(eq, body) # bare rewrite
```

These compile to the obvious kernel terms: `Refl`, `Sym(p)`, `Trans(p, q)`, `Cong(f, p)`, `Rewrite(eq, body)`. Lean emits the corresponding tactics: `exact (eq).symm`, `exact (eq1).trans (eq2)`, `exact congrArg f eq`, `rw [eq]`.

6.5.3 Path constructors

```
path_concat(p, q) # composite path PathComp
path_inv(p) # inverse Sym
path_ap(f, p) # functorial action Ap
J(motive, refl_case, x, y, p) # path induction TransportProof
```

`path_concat` compiles to `PathComp(p, q)` and Lean `exact (p).trans (q)`. `path_inv` is `Sym`. `path_ap(f, p)` is `Ap(f, p)` and Lean `exact congrArg f p`.

The Martin-Löf J rule is approximated as a transport: the implementation (lines 1192–1209 of `psdl.py`) builds a `TransportProof` whose type-family is the motive, path-proof is an `AxiomInvocation` on the supplied path, and base-proof is the compiled refl-case. The Lean side emits the explicit `exact @Eq.rec _ _ <motive> <refl> _ <p>`. This is a faithful encoding for first-order motives; higher-dimensional motives are accepted but not specially decomposed.

6.5.4 Equivalence and univalence

```
iso(forward, backward, l_inv, r_inv) # construct an Equiv
equiv_to_path(e) # ua-style transport
```

`iso` compiles to a `Cut` whose hypothesis is `Equiv-of-iso`. The Lean emits `exact Equiv.mk fwd bwd l_inv r_inv`

`equiv_to_path(e)` is the bridge to univalence: it builds a `TransportProof` with type-family `Var("_motive")`, path-proof `AxiomInvocation(name="ua", module="psdl_local")`, and base proof the compiled e. The Lean side emits a comment — Lean core does not have univalence; the user must add it through `lean_imports` if needed.

6.6 Surface mapping: cubical / duck / effect

This section is what distinguishes PSDL from a literal CIC tactic language. These constructors model the intra-Python phenomena that the deppy metatheory was built to handle.

6.6.1 Duck dispatch

```
dispatch(receiver, "method")
```

Compiled (lines 1249–1262) to `DuckPath(base_type=PyObj, target_type=PyObj)`. The `Lean` emits `exact DuckPath_resolve recv "method"`. In a flat-Int preamble the `DuckPath_resolve` is admitted as `axiom`; in a richer preamble where method tables are declared, it elaborates against an actual -projection.

6.6.2 Kan filling

```
kan_fill(open_box, filling)
```

Compiles (lines 1263–1272) to a `Patch` term wrapping the filling proof and emits `apply Kan.fill`. The kernel verifies the wrapped proof; the cubical-coherence check is performed by `KanFill` (when constructed directly) or skipped for the sugar form. See Chapter 10 for when the full `KanFill` is engaged.

6.6.3 Cocycle construction

```
cocycle(level, witness)
```

Builds a `Cocycle(level, label="cocycle_<lvl>", proof=Refl(Var(witness)))` and emits a comment in the `Lean`. The cocycle's full structure (boundary pairs, overlap proofs, boundary-zero witness) is not exposed in the sugar form; users who need it construct `Cocycle` directly via the kernel API.

6.6.4 Fibration split

```
fibre_split(scrutinee, [(T1, body1), (T2, body2)])
```

The explicit form of `if isinstance`: a `Fiber` with the listed branches. The `Lean` side emits a comment `-- explicit fibration on <scrut>`. Used when the user wants multiple type fibres rather than just `T` vs. `not T`.

6.6.5 Diamond / Box modalities

```
pure(v) # -introduction
bind(m, k) # Kleisli composition
expect(m, fallback) # exception-fibre handler
lift(v) # -introduction
lower(m, witness) # -elimination with purity witness
```

These are the surface forms for the modal layer. In the kernel:

- `lift(v)` → `EffectWitness(effect="value", proof=`P_v`)`.
- `bind(m, k)` → a `Cut` whose lemma proof is m and body is k.
- `expect(m, fallback)` → a chain of effect witnesses ending in a `ConditionalEffectWitness`.
- `lower(m, witness)` → the inner proof of m (assumes the purity witness is supplied at the same trust level).
- `pure` alone is structural (the marker).

6.6.6 Homotopy and squares

```
homotopy(p, q) # path between paths
square(top, bot, left, right) # 2-square diagram
```

`homotopy(p, q)` compiles to `PathComp(p, q)`; `square(...)` chains four paths via two compositions. There is no native "Square" `ProofTerm`; the user obtains the semantically richer form by constructing `KanFill(dimension=2, faces=...)` directly.

6.7 Top-level user declarations

PSDL is a Python module, which means the user can declare inductives, axioms, theorems at top level.

6.7.1 `@inductive class T:`

```
@inductive
class Tree:
    Leaf: int
    Node: (Tree, Tree)
```

The `_compile_class_def` method (lines 481–536) walks the class body looking for `AnnAssign` (named constructors with explicit types) or `FunctionDef` (constructor-as-method). The Lean side emits an `inductive` declaration:

```
inductive Tree where
  | Leaf : int
  | Node : Tree Tree Tree
```

The kernel records this as a structural "inductive Tree (2 ctors)" that can be cited inside the same PSDL block via `Tree.__dict__`-style references.

6.7.2 `@axiom def` and `@theorem def`

```
@axiom
def magic() -> Path[Int, 0, 1]: ...

@theorem
def my_lemma(x: Nat) -> Path[Nat, x + 0, x]:
    induction(x) on Nat:
        case Z(): refl(0)
        case S(k, ih): cong(succ, ih)
```

`@axiom` produces a Lean `axiom magic : Path Int 0 1` and registers it in the PSDL compiler's `foundations` dict so subsequent `apply(magic)` resolves correctly.

`@theorem` elaborates the body as a PSDL proof and emits a real `theorem my_lemma : <type> := by <body>`. The kernel records a `Cut` whose hypothesis name is the theorem name.

The `@protocol` decorator declares a structural protocol similar to PEP 544's `Protocol` class.

6.8 Compositional structure: how multiple statements become one ProofTerm

A PSDL block is a *list* of statements. The compiler chains them via `Cut`: the first statement's proof becomes the `lemma_proof` of a `Cut` whose body is the recursion on the rest.

Listing 6.1: `compile_block` (lines 405–431), the chaining engine.

```
def compile_block(self, stmts, indent=0):
    if not stmts:
        return Structural("empty proof block")
    if len(stmts) == 1:
        return self._compile_stmt(stmts[0], indent)
    head, *rest = stmts
    head_term = self._compile_stmt(head, indent)
    rest_term = self.compile_block(rest, indent)
    if isinstance(head_term, Structural) and head_term.description in (
            "show_goal", "show_term", "trivial", "pure"):
        return rest_term
    return Cut(
        hyp_name=f"_psdl_step_{getattr(head, 'lineno', 0)}",
        hyp_type=PyObjType(),
        lemma_proof=head_term,
        body_proof=rest_term,
    )
```

The exception in the middle is the optimisation: pure-noise statements (`show_goal()`, `hint("...")`, `pure()`) do *not* introduce a real `Cut` but pass through to the next statement. This keeps the proof tree compact.

6.9 Worked compilation: `dist_nonneg_psdl`

We trace through the `dist_nonneg_psdl` block from `sympy_geometry.deppy` statement-by-statement.

The PSDL source is:

```
if isinstance(other, Point):
    inline(Point.distance, sqrt(sum_zip_sub_sq(self, other)))
    apply(Real_sqrt_nonneg)
    assert sum_zip_sub_sq(self, other) >= 0, "z3"
else:
    raise NotImplementedError
```

The compiler walks this top-down:

1. The outer `if isinstance` statement. `_compile_if` fires. It detects the `isinstance` shape via `_is_isinstance_test`, which returns `("other", "Point")`. The compiler emits the Lean lines:

```
by_cases h : isinstance other Point
ů -- value-fibre: isinstance(other, Point)
  ...
ů -- complement-fibre: ň isinstance(other, Point)
```

```
  ...
```

It recursively compiles the body of the if-branch (a 3-statement block) and the body of the else-branch (a 1-statement block), then returns:

```
Fiber(
  scrutinee=Var("other"),
  type_branches=[
    (PyObjType(), <if-body>),
    (PyObjType(), <else-body>),
  ],
  exhaustive=True,
)
```

2. The if-body, statement 1: `inline(Point.distance, sqrt(...))`. `_compile_func_call` dispatches on `fname=="inline"` (lines 951–963). `target = "Point.distance"`; `expected = "sqrt(sum_zip_sub_sq(self, other))"`.

The compiled term is `Unfold(func_name="Point.distance", proof=Refl(Var("sqrt_sum_zip_sub_sq_self_` (with the `_safe` identifier transform applied to the expected expression). The Lean emits `unfold Point.distance  -- expected: sqrt(...)`.

3. The if-body, statement 2: `apply(Real_sqrt_nonneg)`. `_compile_func_call` dispatches on `fname=="apply"` (lines 932–941). `target = "Real_sqrt_nonneg"`, which is in the `foundations` dict (registered by the sidecar parser).

Term: `AxiomInvocation(name="Real_sqrt_nonneg", module="foundations")`. The compiler appends `"Real_sqrt_nonneg"` to `foundations_referenced`. Lean: `exact Foundation_Real_sqrt_nonneg`

4. The if-body, statement 3: `assert ...  , "z3"`. `_compile_assert` (lines 710–777) reads `stmt.msg == ast.Constant("z3")`, sets mode to `"z3"`, builds `Z3Proof(formula="sum_zip_sub_sq(self, other) >= 0", binders={})` and emits `have h_<lineno> : sum_zip_sub_sq(self, other) >= 0 := by ome`

The full term is

```
Cut(
  hyp_name="h_<lineno>",
  hyp_type=RefinementType(
      base_type=PyBoolType(),
      predicate="sum_zip_sub_sq(self, other) >= 0",
  ),
  lemma_proof=Z3Proof(...),
  body_proof=Refl(Var("_assert_body")),
)
```

5. Composition of the if-body. `compile_block` chains the three statements through nested `Cut`s, with `Refl(Var("_assert_body"))` as the ultimate body.

6. The else-body: `raise NotImplementedError`. `_compile_raise` (lines 789–805) detects `NotImplementedError` and returns `Structural("vacuous: NotImplementedError")` with Lean `exact absurd _ (by simp)`.

7. The full ProofTerm. Putting it all together (with some abbreviation):

```
Fiber(scrutinee=Var("other"),
      type_branches=[
        (PyObjType(),
          Cut(hyp_name="_step_1",
            lemma_proof=Unfold("Point.distance", Refl(...)),
            body_proof=Cut(hyp_name="_step_2",
              lemma_proof=AxiomInvocation("Real_sqrt_nonneg",
                                          "foundations"),
              body_proof=Cut(hyp_name="h_<lineno>",
                hyp_type=RefinementType(predicate="sum_zip... >= 0"),
                lemma_proof=Z3Proof("sum_zip... >= 0"),
                body_proof=Refl(Var("_assert_body"))))))),
        (PyObjType(), Structural("vacuous: NotImplementedError")),
      ])
```

8. The Lean block.

```
by
  by_cases h : isinstance other Point
  ů -- value-fibre: isinstance(other, Point)
    unfold Point.distance -- expected: sqrt(sum_zip_sub_sq(self, other))
    exact Foundation_Real_sqrt_nonneg
    have h_<lineno> : sum_zip_sub_sq(self, other) >= 0 := by omega
      -- z3-discharged
  ů -- complement-fibre: ň isinstance(other, Point)
    -- vacuous: NotImplementedError
    exact absurd _ (by simp)
```

9. Trust composition. The trust on this `ProofTerm` is min(`KERNEL`, `AXIOM_TRUSTED`, `Z3_VERIFIED`, `STRUCTURA`
`STRUCTURAL_CHAIN`. `certify_or_die` permits this because the `Structural` leaf is in a vacuous-branch position (the exception fibre), and it counts as a non-failure.

6.10 Soundness arguments per constructor

For each compiler entry point we sketch the soundness story.

If-isinstance. `Fiber(scrutinee, type_branches, exhaustive=True)` is sound when:

1. every `type_branch`'s proof verifies under the local refinement `isinstance(scrutinee, T)`;
2. the type-branches *cover* the dynamic type of the scrutinee.

The PSDL form generates `exhaustive=bool(stmt.orelse)`: True when the user wrote an `else`. When the user did *not* provide an `else`, the kernel adds an implicit "other" branch with `Structural("no else for isinstance(...)")`, which downgrades trust to `STRUCTURAL_CHAIN`.

Try-except. The chain `EffectWitness(value)` + `ConditionalEffectWitness(precond="raises Exc", ...)` is sound only if *both* fibres are addressed. The PSDL form respects this: each `except` clause produces a corresponding witness; the value fibre is built from the `try`-body. We do *not* assert that `except` catches every possible exception: if the body can raise `ValueError` but the user only wrote `except TypeError`, the resulting modal type admits the `ValueError` fibre as part of the residual $\Diamond$ structure.

Z3 assert. `Z3Proof(formula, binders)` is sound when SMT returns `unsat` on the negation in the typed-Z3 encoder. Trust `Z3_VERIFIED`; we admit Z3's answer.

Apply. `AxiomInvocation(name, module)` is sound when the named axiom's statement matches the goal up to the kernel's `PatternMatcher`. Trust `AXIOM_TRUSTED`. The PSDL form additionally records the axiom in `foundations_referenced` so the certificate's provenance is honest about which foundations / axioms a verify block depends on.

Inline / unfold / rewrite. `Unfold(name, proof)` is sound when the named definition unfolds to a term equivalent to the proof's claim. In the current implementation, this check is structural rather than reduction-based; Lean's elaborator performs the actual definitional unfolding when the certificate is round-tripped.

Vacuous `raise`. `Structural("vacuous: <descr>")` is permitted only in positions where the *branch is unreachable*: typically inside the exception-fibre face of a `Fiber` or `ConditionalEffectWitness`. The soundness gate (Chapter 12) flags these and `certify_or_die` accepts them only when no counterexample is found that contradicts the unreachability.

6.11 Limitations and open work

PSDL as currently implemented:

- Does not support universe polymorphism. All types live in `Type 0`.
- Does not support refinement types beyond a single `predicate` string field. The kernel does not check the predicate; the Lean side does, by elaborating the predicate.
- Does not support higher inductive types. The `@inductive` decorator generates ordinary inductives.
- Does not support coinductive specs.
- The escape hatch `lean("...raw ...")` compiles to a `LeanProof` term and emits the raw text, but deppy's kernel does *not* type-check the raw fragment — Lean does.

These gaps are documented in `AUDIT.md` as items B (type-system expressiveness), F (advanced cubical/homotopy), and G (module structure). The escape hatch `lean(...)` suffices in practice for any property a user can prove directly in Lean.

6.12 Summary

PSDL is *the* surface for deppy's strong proof language:

- Source is literal Python. No new syntax.
- Each statement compiles to a `ProofTerm` the kernel verifies and a `Lean` tactic the elaborator verifies.
- Vocabulary splits into a CIC core (//induction/transport) and Python-specific primitives (fibrations, $\Diamond/\Box$, duck dispatch, Kan filling, cocycles).
- The compiler's structure is direct: an AST walker with one `_compile_<X>` method per AST node kind.
- Composition is via `Cut`-chaining; pure-noise statements pass through unchanged.
- Soundness follows from the kernel's verification rules (Chapter 4); trust composes by `min_trust`.

The next chapter follows the pipeline from start to finish: how `run_sidecar` ingests a `.deppy` file and produces a `Lean` certificate.

Chapter 7

The Pipeline: Module Name to Certificate

The deppy pipeline is the orchestration layer that takes a sidecar file path and emits a Lean certificate. Its entry point is `deppy.lean.sidecar_runner.run_sidecar`; everything else is a transformation on the `SidecarModule`.

This chapter walks through every stage in order. The reader should expect file-and-line citations; we are documenting the *actual* implementation.

7.1 Architecture overview

The pipeline is data-flow oriented: each stage reads the `SidecarModule` (and optionally the outputs of earlier stages) and produces a structured report. No stage modifies the `SidecarModule`; the result of running the pipeline is a `SidecarCertificateSection` containing the rendered Lean text and a counter-rich report.

7.2 Stage 0: Module-name resolution

The pipeline starts at `run_sidecar(deppy_path, out_path, module_name)` (file: `deppy/lean/sidecar_runner` lines 114–185).

Listing 7.1: `run_sidecar`: high-level control flow.

```
def run_sidecar(deppy_path, *, out_path=None, module_name=""):
    deppy_path = Path(deppy_path)
    if out_path is None:
        out_path = deppy_path.with_suffix(".lean")
    out_path = Path(out_path)

    sm = SidecarModule.from_file(str(deppy_path))
    target_module = module_name or getattr(sm, "name", "") or ""
    library_summary = _inspect_library(target_module)

    t0 = time.perf_counter()
    section = render_sidecar_section(
```

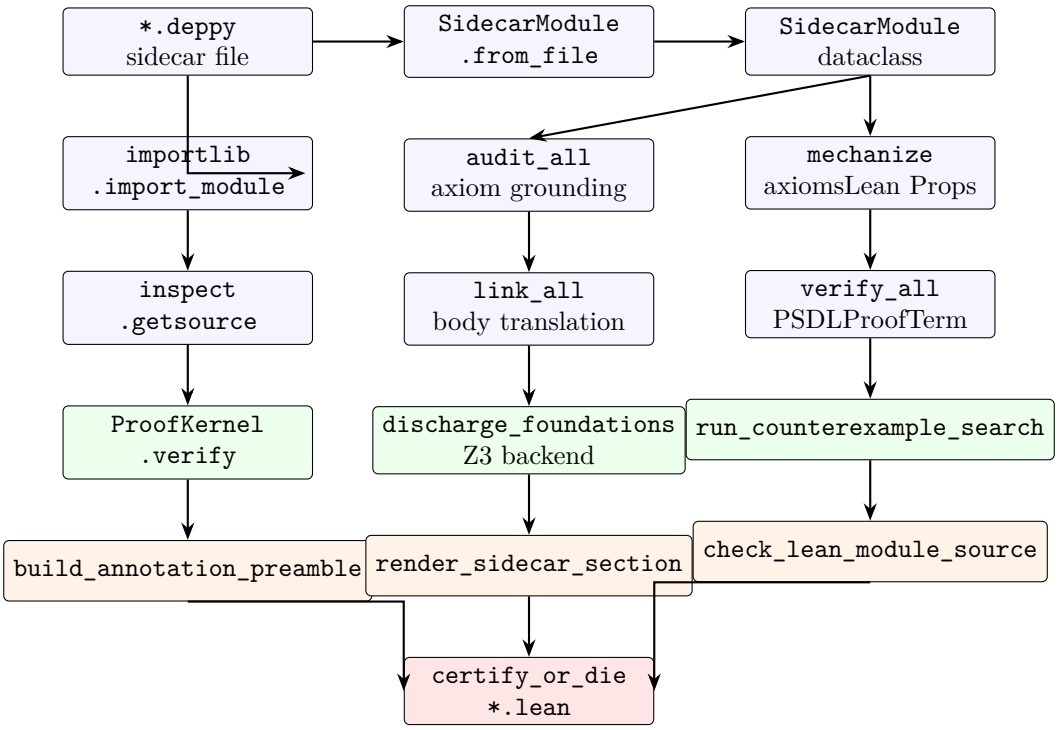


Figure 7.1: The deppy pipeline. Blue boxes are inputs and parser stages; green boxes are checking stages with their own trust contributions; orange boxes are output emitters; the red box is the final certification gate.

```
        sm,
        module_name=f"Deppy_{deppy_path.stem}",
        library_source_summary=library_summary,
    )
    elapsed_ms = (time.perf_counter() - t0) * 1000.0
    out_path.write_text(section.full_text, encoding="utf-8")

    # Round-trip through Lean toolchain.
    ...
```

`_inspect_library` (line 101) calls `importlib.import_module(target_module)` and reads `module.__version__` and `module.__file__` for the certificate's provenance header. When the import fails, the summary becomes "`<module> (not importable)`" and the certificate proceeds with empty provenance — but every verify-block inside that certificate will fail to resolve its function and the certificate will record this honestly.

7.3 Stage 1: Parsing the sidecar

`SidecarModule.from_file(path)` reads the file and produces a `SidecarModule`. The parser supports both YAML-like indentation-based blocks and a sequence of top-level declarations separated by blank lines.

The parser populates:

- `sm._foundations: dict[str, FoundationDecl]`
- `sm._specs: dict[str, ExternalSpec]`
- `sm._axioms: dict[str, AxiomDecl]`
- `sm._lemmas: list[LemmaDecl]`
- `sm._verifies: list[VerifyDecl]`
- `sm._proofs: list[ProofDecl]` (legacy)
- `sm._code_types: dict[str, str]`
- `sm._lean_imports: list[str]`
- `sm._lean_preamble: str`

The parser is permissive: unrecognised top-level keys produce a warning but do not abort. Malformed indentation produces a `SyntaxError` with a line number.

7.4 Stage 2: `render_sidecar_section`

The main rendering pipeline lives in `deppy/lean/sidecar_certificate.py`, function `render_sidecar_section` (lines 100–296+). It dispatches to thirteen sub-emitters, each producing a fragment of the final Lean text plus a structured report.

The sub-emitters, in execution order:

1. `derive_all(sm)` — run the legacy `by:`-clause derivations on each `ProofDecl`.
2. `audit_all(sm)` — check the live target library for each axiom/spec; compute a `AxiomGrounding` per axiom (`GROUNDED` / `TARGET_MISSING` / `METHOD_MISSING` / `NL_DOC`).
3. `mechanize(sm, audit)` — walk every grounded axiom AST and emit concrete Lean `Prop`s.
4. `link_all(sm, audit, mech)` — run `deppy.lean.body_translation` on every method named by an axiom; record the translated Lean text per method.
5. `attempt_all(sm, audit, body_link)` — attempt code-proofs by composing the body translation with the cited foundation/axiom; the kernel verifies each, the pipeline records the `VerificationResult`.
6. `link_cubical(body_link)` — run the intra-procedural cubical AST analysis.
7. `verify_all(sm)` — the per-`verify`-block pipeline (§7.8).
8. `link_cohomology(body_link)` — run the C/Cź/Cš cohomology emitter on the linked methods.
9. `discharge_foundations(sm)` — Z3 each foundation; promote to `Z3_VERIFIED` when accepted.
10. `discharge_kan_obligations(body_link)` — the Kan-fill obligations from the cubical pass.
11. `link_bicomplex(body_link)` — the cubical / simplicial bicomplex.

12. `link_fibration(sm, audit, body_link, kernel)` — run `FibrationVerifier` on `verify` blocks with a `fibration:` clause.
13. `run_counterexample_search(sm, foundation_discharge)` — chase counterexamples for every Z3-attempted but unverified foundation.
14. `invoke_all(body_link)` — the B1–B16 audit sub-systems.
15. `probe_all_proofterms()` — C1–C7 audit on the `ProofTerm` class hierarchy.
16. `build_annotation_preamble(sm)` — the annotation-driven concrete preamble.

After all sub-emitters complete, the function assembles the final `Lean` text in this order:

1. Header comment with module name and provenance.
2. Annotation-driven concrete preamble (types, methods, properties from binders).
3. Mechanization preamble (vocabulary from grounded axioms).
4. Body links (`def` declarations for each translated method, behind comments to avoid colliding with the preamble's opaque counterparts).
5. Foundations.
6. Sidecar axioms.
7. Lemmas.
8. Theorems (one per legacy `ProofDecl`).
9. Verify-block sections (one per `VerifyDecl`).
10. Cubical analysis section.
11. Cohomology section.
12. Bicomplex section.
13. Counterexample-rejected theorems (when a foundation was counter-examined).
14. Subsystem invocations and proofterm-zoo probes (audit appendices).

7.5 Stage 3: Audit and grounding

`audit_all` (file: `deppy/lean/sidecar_source_audit.py`) walks every `AxiomDecl` and tries to resolve its `target` (a dotted Python name). The four possible outcomes:

`GROUNDED` the target resolves; every method referenced in the axiom's statement is callable; the statement parses. The certificate emits a concrete Lean Prop.

`TARGET_MISSING` the target does not resolve. The axiom is admitted as opaque but flagged.

`METHOD_MISSING` the target resolves but a referenced method does not. Same outcome as `TARGET_MISSING`.

`NL_DOC` the statement is natural-language prose (e.g. "the area is non-negative"), not a Python expression. Admitted as opaque, no concrete Prop.

The audit feeds into both `mechanize` (which only mechanises `GROUNDED` axioms) and the certificate header which counts axioms in each grounding bucket.

7.6 Stage 4: Mechanization

`mechanize(sm, audit)` (file: `deppy/lean/sidecar_mechanization.py`) is the heart of the "concrete Props" transformation. It runs in two passes:

Pass 1: vocabulary collection. A `_VocabCollector` walks each grounded axiom's AST and records:

- `class_methods : {class→{method→arity}}`;
- `class_ctor_arity`;
- `attrs` (every `.attr` access);
- `free_fns : {name→arity}`;
- `prop_fns` (free fns appearing at top-level Prop position);
- `free_vars`;
- `classes`.

Pass 2: per-axiom Lean rendering. Each grounded axiom's statement is re-parsed and rendered against the vocabulary. The mapping is documented in the docstring (file `deppy/lean/sidecar_mechanizati` lines 7–40):

- `ClassName.method(args) (Class_method args)`
- `ClassName.attr (Class_attr)`
- `var.attr (dot_attr var)`
- `var.method(args) (dot_method var args)`
- `Class(args) (Class_mk args)`
- `func(args) (func args)`

The rendering emits a `def Sidecar_X_prop : Prop := <body>` (when grounded and the rendering succeeds) or a fallback `opaque Sidecar_X_prop : Prop` (when not).

7.7 Stage 5: Body translation

`link_all` runs `deppy.lean.body_translation.translate_body` on each method referenced by an axiom or verify block.

Type-inference pass. An `_InferTypes` walker seeds with parameter annotations and propagates types through operations: `x.append(...)` forces `x : list`; `len(x)` forces `x : list / dict / str / tuple`; arithmetic operations propagate numeric types; etc. See `infer_locals_types` (lines 127–179 of `body_translation.py`).

Translation pass. A `_BodyT` walker uses the inferred types to render Lean. Standard rules:

- `Int`/`Float`/`Bool` arithmetic translates directly.
- `if/elif/else` `if-then-else`.
- `return` bound through to the result.
- Recursive self-calls preserved.
- `xs[:n]` `xs.take n`; `xs[n:]` `xs.drop n`; `xs[i]` `xs.get!`.
- `len(xs)` `xs.length` when `xs : List`, `xs.size` when `HashMap`.
- Function calls preserved.

When the body uses constructs the translator cannot handle (`eval`, dynamic class creation, C-extension fall-throughs, list comprehensions with multiple clauses, etc.), it returns `sorry` with a note. The certificate flags this in the `body_links_sorry` count.

7.8 Stage 6: `verify_all` — the verify-block pipeline

`verify_all(sm)` (file: `deppy/lean/sidecar_verify.py`) is the most directly relevant stage for PSDL users. We summarise its structure here; the worked example is in Chapter 11.

7.8.1 Per-block flow

`_process_one(decl, *, foundations, kernel, concrete_prop)` (lines 167+) runs eight steps for each verify block:

1. **Resolve function.** `_resolve(decl.function)` walks the dotted path through `importlib` and `getattr`. When it returns `None`, the outcome records "could not resolve" and the block aborts.
2. **Read source.** `inspect.getsource(method)`; when `method` is a `property`, descend to `method.fget`. The source is dedented via `textwrap.dedent` and SHA-256 hashed for the `source_hash` field.
3. **Translate body.** `deppy.lean.body_translation.translate_body(fn_node, type_context=ctx, predicate_registry=preds, param_types=...)`.
4. **Compile PSDL (if present).** When `decl.psdl_proof` is non-empty, `deppy.proofs.psdl.compile_psdl(s foundations=fdict, ctx=ctx)` produces a `PSDLCompileResult`.
5. **Fibration descent (if requested).** When `decl.fibration_type` is non-empty, the pipeline runs `FibrationVerifier.verify_per_fiber(fn_node, fibre)` — one kernel call per fibre.
6. **Cubical analysis (if requested).** When `decl.use_cubical` is True, the pipeline runs `build_cubical_set(fn_node)` and records the cell counts and Kan candidates.
7. **Construct ProofTerm.** In rough order of preference: PSDL's compiled term, when present; otherwise an `Unfold(method)` ∘ `AxiomInvocation(decl.foundation)`; otherwise a `Structural` fallback.

8. **Kernel verification.** Build a `Judgment` over the property's `RefinementType` and call `kernel.verify(ctx, goal, term)`. Record the `VerificationResult`'s success bool, trust level, axioms used, and message.

7.8.2 Theorem emission

When `kernel_success == True`, `verify_all` emits a Lean theorem section:

```
-- Verify block: dist_nonneg_psdl
-- Function: sympy.geometry.point.Point.distance
-- Property: Point.distance(p, q) >= 0
-- Foundation: Real_sqrt_nonneg
-- Source SHA-256: <hash>
-- Trust level: STRUCTURAL_CHAIN
-- Axioms used: ["Real_sqrt_nonneg"]
theorem Verified_dist_nonneg_psdl :
    Verified_dist_nonneg_psdl_property := by
  intros p q
  -- PSDL compiled output:
  by_cases h : isinstance other Point
  ů unfold Point.distance
    exact Foundation_Real_sqrt_nonneg
    have h_<lineno> : sum_zip_sub_sq self other 0 := by omega
  ů exact absurd _ (by simp)
```

`Verified_dist_nonneg_psdl_property` is the concrete `Prop` built by `build_annotation_preamble` from the `binders`, `function`, and `property` fields. See Chapter 8.

7.9 Stage 7: Foundation discharge (Z3)

`discharge_foundations(sm)` (file: `deppy/lean/sidecar_foundation_discharge.py`, lines 1–150) attempts a Z3 discharge of every foundation. The flow:

1. `_looks_arithmetic(stmt)` — skip statements that contain `sqrt`, `sin`, `is_collinear`, etc.
2. `_strip_when_clause(stmt)` — split off any "`... when guard`" precondition.
3. `_rewrite_for_z3(stmt)` — combine guard and core into an implication `guard => core`.
4. `_collect_binders(formula)` — identify free variables and assign each to `int`.
5. Build a `Z3Proof(formula, binders)` and call `kernel.verify` on a synthetic `Judgment`. When Z3 returns `unsat` on the negation, mark `z3_verified=True`.

The certificate's foundation section then emits two shapes:

- For `z3_verified` foundations: a real `theorem Foundation_X : Foundation_X_prop := by intros; omega` (or similar tactic).
- For unverified foundations: `axiom Foundation_X : Foundation_X_prop` with the statement preserved as a comment.

In the `sympy_geometry.deppy` run, 7 of 14 foundations discharge via Z3 (the linear-arithmetic ones); the rest stay admitted.

7.10 Stage 8: Counterexample search

`run_counterexample_search(sm, foundation_discharge_report)` (file: `deppy/lean/sidecar_counterexamp` lines 1–92) runs only on foundations Z3 *attempted* but did not verify. For each, it builds a `Judgment` whose `type_` is the `RefinementType` with predicate the foundation's formula, and asks `CounterexampleFinder` to find a model satisfying the negation.

When a counterexample is found:

- `outcome.found = True` with a model dict;
- the certificate marks the foundation as `REJECTED`;
- *transitively*: every foundation that depends on it (via `sm._foundation_deps`) is also rejected;
- every theorem citing a rejected foundation is replaced in the certificate with an honest rejection notice.

This is the foundation of the `rejected_theorem_count` gate. When non-zero, `certify_or_die` refuses certification.

7.11 Stage 9: Concrete-Props preamble

`build_annotation_preamble(sm)` (file: `deppy/lean/sidecar_annotation_preamble.py`) reads:

- every `verify` block's `binders`;
- every `verify` block's `function` dotted path;
- every `verify` block's `property`;
- the top-level `code_types` dict;
- the top-level `lean_imports` and `lean_preamble` text.

It emits:

1. `abbrev <T> : Type := Int` for each unique binder type (the flat-Int simplification we acknowledge in Chapter 8).
2. `axiom <name> : <signature>` for each entry in `code_types`.
3. `def <Verified_X_property> : Prop := <translated>` for each verify block whose property translates to a non-trivial Lean expression.

The verifying theorems then have concrete goal types that elaborate against the PSDL-emitted tactics. The `annotation_preamble.concrete_props` dict maps verify-block names to `(prop_name, prop_body)` for use by `verify_all`.

7.12 Stage 10: Lean round-trip

`check_lean_module_source(text, timeout_s=60)` (file: `deppy/lean/lean_runner.py`) writes the certificate to a temporary file, invokes the `lean` binary on it, and captures stdout/stderr/exit-code.

Listing 7.2: Outline of `check_lean_module_source`.

```
def check_lean_module_source(source, *, timeout_s=60):
    with NamedTemporaryFile(suffix=".lean", mode="w") as fp:
        fp.write(source)
        fp.flush()
        result = subprocess.run(
            ["lean", fp.name],
            capture_output=True,
            timeout=timeout_s,
            text=True,
        )
    return LeanResult(
        ok=result.returncode == 0,
        stdout=result.stdout,
        stderr=result.stderr,
    )
```

When `Lean` accepts (exit 0), `lean_runner_ok = True` and the certificate's trust level for `LeanProof` sub-terms elevates to `KERNEL`.

When `Lean` rejects, `lean_runner_ok = False`. The `certify_or_die` gate then refuses certification, listing "`lean_runner round-trip rejected`" as a failure.

7.12.1 Caveats

The `Lean` binary must be on the user's `PATH`. When it isn't, `check_lean_module_source` catches the `FileNotFoundError` and `lean_runner_invoked = False`, but `certify_or_die` does not refuse certification on this basis alone — the user can still rely on the kernel-level verification for soundness, just not on the round-trip guarantee.

The 60s default timeout is on the conservative side. For the `sympy_geometry.deppy` run, `Lean` typically returns in under 10 seconds; for larger sidecars (`torch`, `numpy`) the timeout may need raising.

7.13 Stage 11: `certify_or_die`

The final gate is `_certify_or_die(s)` (file: `deppy/lean/sidecar_runner.py`, lines 62–98):

Listing 7.3: The `certify_or_die` verdict.

```
def _certify_or_die(s) -> tuple[bool, list[str]]:
    failures: list[str] = []
    if s.soundness_gate_fail > 0:
        failures.append(
            f"soundness gate fired in {s.soundness_gate_fail} proof(s)")
    if s.counterexamples_found > 0:
        failures.append(
            f"{s.counterexamples_found} counterexample(s) found "
            "for admitted foundations")
    if getattr(s, "rejected_theorem_count", 0) > 0:
        failures.append(
            f"{s.rejected_theorem_count} theorem(s) cited "
            "counterexample-rejected foundations")
    if getattr(s, "hash_drift_count", 0) > 0:
```

```
        failures.append(
            f"{s.hash_drift_count} verify block(s) failed "
            "expects_sha256 hash check")
    if getattr(s, "counterexample_expectation_violations", 0) > 0:
        failures.append(
            f"{s.counterexample_expectation_violations} verify block(s) "
            "expected counterexample but none found")
    return (len(failures) == 0, failures)
```

A certificate is `CERTIFIED` only when every gate is clean:

- No `Structural` leaf in a non-vacuous position (soundness gate).
- No counterexample on any admitted foundation.
- No theorem citing a rejected foundation.
- No hash-drift on any `expects_sha256:`-tagged verify block.
- No `expecting_counterexample: true` block where the search came up empty.
- `Lean` 4 round-trip exit 0 (when invoked).

When any gate fires, `certify_or_die` returns False; the certificate is still emitted (so the user can debug), but `rr.certified = False` and the runner prints an " `NOT CERTIFIED`" summary.

7.14 Output reporting

`print_report(rr)` (lines 188–282) prints a structured human-readable summary. The report (paraphrased) lists:

- Foundations declared (trust surface).
- Sidecar axioms by grounding bucket.
- Body links: total, exact, sorry, error.
- Code-proofs and the soundness gate.
- Cubical analysis: methods, total cells, Kan candidates.
- Cohomology counts: C, Cź, Cš.
- Bicomplex: built, edges, faces, Euler .
- Fibration descent: outcomes / successes.
- Foundation Z3 discharge: total / attempted / verified.
- Counterexample search: foundations attempted / counterexamples found.
- Subsystem invocations and ProofTerm-zoo probes.
- Verify blocks: total / kernel-verified.
- Specs (informational), proofs total / derived / cited / sorry.
- Per-classification counts.

- `lean_runner` verdict.
- `certify_or_die` verdict.

For `sympy_geometry.deppy` a typical run prints `14 foundations, 38 axioms, 11 verify blocks, 4 body links`, `7 of 14 foundations discharged via Z3`, `0 counterexamples found`, and either `CERTIFIED` or a list of certify failures.

7.15 Where each `ProofTerm` lives in the pipeline

A summary mapping the kernel `ProofTerm` subclasses to the pipeline stages that construct them:

Term	Constructed by
`Refl`, `Sym`, `Trans`, `Cong`	PSDL refl/sym/trans/cong; legacy chain proofs.
`Cut`, `Assume`	PSDL block-chaining; explicit `forall(...)`.
`Z3Proof`	PSDL `assert ..., "z3"`; foundation discharge.
`LeanProof`	PSDL `lean("...")`; `by_lean:` on a verify block.
`NatInduction`, `ListInduction`, `Cases`	PSDL `while`, `for`, `match`; explicit `induction(...)`.
`DuckPath`, `ConditionalDuckPath`, `FiberedLookup`	`deppy.equivalence`; `DuckPath.auto_construct`; PSDL `dispatch(...)`.
`EffectWitness`, `ConditionalEffectWitness`	PSDL `try/except`; `lift/bind/expect`.
`TransportProof`	PSDL `transport(...)`; cubical-effects transport.
`Fiber`	PSDL `if isinstance`; verify-block fibration descent.
`Patch`	`kan_fill` sugar; the patching module.
`Cocycle`, `KanFill`, `CechGlue`	The cohomology pipeline; the bicomplex linker.
`Univalence`	`UnivalenceEngine.from_equiv`; `equiv_to_path`.
`AxiomInvocation`	PSDL `apply`; legacy chain/citation proofs.
`Unfold`, `Rewrite`	PSDL `inline`, `unfold`, `rewrite`; legacy `rewrite then cite`.
`Structural`	PSDL `raise NotImplementedError`; safety pipeline escape hatches.

7.16 Cache and incremental verification

The `deppy/pipeline/cache.py` module provides a content-keyed cache for: Z3 discharge results, `check_lean_module_source` results, body-translation results, and per-axiom mechanizations. The cache key is the SHA-256 of the input text, so identical re-runs are essentially free.

`deppy/pipeline/incremental.py` sketches a cross-run incremental verifier that re-runs only verify blocks whose function source SHA-256 has changed. This is what would let a maintainer rebuild a certificate after a one-line change without re-running the entire pipeline. The current implementation is opt-in and not exercised by the canonical `run_sidecar` entry point.

7.17 Summary

The pipeline:

- starts at a `.deppy` path and a target Python module name;
- parses the sidecar into a `SidecarModule`;
- runs a sequence of report-producing transformations (audit, mechanize, link, verify, foundations, counterexample, . . .);
- assembles a `Lean` 4 certificate;
- round-trips the certificate through the `Lean` toolchain;
- gates the `CERTIFIED` verdict on a small set of explicit failure conditions in `certify_or_die`.

The next chapter dives into the `Lean` target layer: how the generated text is structured, how the trust chain bottoms at concrete `Lean` `Prop`s, and how the body-translation and mechanization passes interact with the preamble.

Chapter 8

The Lean Target

This chapter is a tour of the generated Lean 4 certificate. The files of interest are:

- `deppy/lean/sidecar_certificate.py` — `render_sidecar_section`, the orchestrator;
- `deppy/lean/sidecar_annotation_preamble.py` — the concrete-Props preamble;
- `deppy/lean/sidecar_mechanization.py` — Python-axiom → Lean-Prop translation;
- `deppy/lean/sidecar_type_inference.py` — `inspect.signature` → Lean signatures;
- `deppy/lean/sidecar_body_link.py` — body translation per linked method;
- `deppy/lean/body_translation.py` — AST → Lean expression;
- `deppy/lean/sidecar_derivation.py` — legacy `by:`-clause derivations.

We give the certificate's structure top-to-bottom and explain how the trust chain bottoms.

8.1 Trust chain in the Lean output

The certificate is laid out so the trust chain is visible:

$$\boxed{\text{Foundations}} \longrightarrow \boxed{\text{Sidecar axioms}} \longrightarrow \boxed{\text{Lemmas}} \longrightarrow \boxed{\text{Theorems / Verify blocks}}.$$

Each box represents a contiguous section in the Lean file:

Foundations are the bottom: arithmetic identities admitted on the user's authority (or proved by `omega` when Z3 succeeds). Foundations have no upstream dependencies in the certificate.

Sidecar axioms depend on foundations: each axiom is a claim about library behaviour, expressed as a concrete Lean Prop using the mechanization vocabulary.

Lemmas depend on foundations and axioms. When the user supplied `lean_proof:`, lemmas are real `theorem`s.

Theorems are the top: derived from any combination of the above. Verify blocks emit one `theorem` each, with the proof body being the PSDL-compiled tactic block (or a derived proof from a `by:`-clause for legacy proofs).

Lean's elaborator processes the file top to bottom, so dependent declarations resolve. The compiler counts how many declarations end up at each level (foundations, axioms, lemmas, theorems), how many of each carry concrete bodies vs. opaque placeholders, and emits these in the certificate's header comment.

8.2 Header

Every certificate begins with:

```
-- Auto-generated by ``deppy.lean.sidecar_certificate``.
-- Module: Deppy_<sidecar_basename>
-- Foundations: <N>; specs: <M>; axioms: <K>; proofs: <P>.
-- Library: sympy 1.14.0 -- sympy
--
-- Trust chain: Foundations (mathematical facts) -> Sidecar axioms
-- (claims about library behaviour) -> Theorems (real proof
-- terms -- Eq.trans / And.intro / rfl / Foundation.symm.trans
-- -- type-checked by Lean's kernel).
```

8.3 Annotation-driven concrete preamble

The first substantive section is the annotation-driven concrete preamble (file: `deppy/lean/sidecar_annotation_p`
Its purpose is to replace the *opaque* `Sidecar_X_prop : Prop` / `Foundation_X_prop : Prop` declarations of older versions with *concrete* Lean propositions assembled from the sidecar's `binders:`, `function:`, `property:`, and `code_types:` fields.

The motivation, in `AUDIT.md`'s words: "Once Props are concrete, the trust chain shifts from 'deppy admitted opaque X' to 'Lean's kernel checked the user's PSDL tactic against a concrete claim'."

8.3.1 Concrete types

For each unique binder type across all verify blocks, the preamble emits an `abbrev`:

```
-- Concrete types from verify-block binders (definitional aliases to Int)
abbrev Circle : Type := Int
abbrev Point : Type := Int
abbrev Polygon : Type := Int
abbrev Segment : Type := Int
abbrev Triangle : Type := Int
```

The `abbrev T : Type := Int` is a deliberate flat-Int simplification. We acknowledge it openly: *the generated certificate identifies every Python class with the carrier `Int`*. This loses the ability to distinguish classes in Lean's type system but lets the certificate compile without class-machinery dependency. In a future iteration where the preamble engages the `deppy.lean.sidecar_type_inference` result, types could be richer (e.g. structured records or quotient types over the underlying Python instance representation).

8.3.2 Concrete code-type signatures

For each entry in the sidecar's top-level `code_types:` block:

```
-- User-declared code-type signatures (from .deppy code_types:)
axiom sqrt : Int Int
axiom sum_zip_sub_sq : Point Point Int
```

These are the helpers PSDL references but the library does not necessarily expose at the top level. The `axiom` kind means they are uninterpreted; the PSDL-compiled tactics use them as applied symbols, and the kernel does not unfold them.

8.3.3 Concrete foundation Props

Each foundation gets a concrete `def` or an `opaque` fallback, depending on whether `discharge_foundations` was able to verify it via Z3 or whether the user supplied an explicit `lean_proof`.

For arithmetic foundations Z3 verifies:

```
-- Foundation: Real_add_comm
-- Statement: a + b == b + a
def Foundation_Real_add_comm_prop : Prop :=
   (a b : Int), a + b = b + a
theorem Foundation_Real_add_comm : Foundation_Real_add_comm_prop := by
  unfold Foundation_Real_add_comm_prop; intros; omega
```

For foundations involving `sqrt`:

```
-- Foundation: Real_sqrt_nonneg
-- Statement: sqrt(x) >= 0 when x >= 0
opaque Foundation_Real_sqrt_nonneg_prop : Prop
axiom Foundation_Real_sqrt_nonneg : Foundation_Real_sqrt_nonneg_prop
```

The certificate's foundation section is sorted alphabetically and groups Z3-verified foundations (real `theorem`s) above admitted ones (`axiom`s). The classification feeds the report's `foundations_z3_verified` counter.

8.4 Mechanization preamble

After the annotation-driven preamble, the certificate emits the *mechanization vocabulary*: the union of types, methods, attributes, and helpers extracted from grounded axiom AST walks.

```
-- Shared attribute accessors (derived from var.attr)
axiom dot_x : Int Int
axiom dot_y : Int Int
axiom dot_centroid : Int Int
axiom dot_radius : Int Int
-- ... and so on

-- Class-prefixed methods and constructors
axiom Triangle_mk : Int Int Int Int
axiom Segment_mk : Int Int Int
axiom Circle_mk : Int Int Int
def Point_distance : Int Int Int := fun _0 _1 => 0
```

```
def Polygon_area : Int := 0

-- Free functions
axiom abs : Int Int
axiom area : Int Int
axiom is_collinear : Int Int Int Prop
axiom thales : Int Int Int Int Prop
```

The methods that have a body translation (see `body_translation`) are emitted as `def := sorry` *placeholders* so that `unfold` in proof bodies can fire without elaborating the full body inline. The actual translated body is emitted in commented form below the `Body links` section. This dual encoding (placeholder `def` + commented body) preserves elaboration-friendliness while documenting what the body actually translates to.

8.4.1 Body links section

For each method whose body the translator could read, the certificate emits:

```
-- Point_distance (sympy.geometry.point.Point.distance)
-- Source SHA-256: df5da0ce57c1b906...
-- deppy body-translator: EXACT (full translation, no sorry)
-- Lean (commented to avoid collision with opaque preamble decl):
-- def Point_distance (self : Int) (other : Int) : :=
-- if (isinstance other Point) then
-- let (s, p) := (Point._normalize_dimension self (Point other));
-- (sqrt (Add (List.zipWith (fun a b => ((a - b) ^ 2)) s p)))
-- else
-- let distance := (getattr other "distance" ());
-- if (distance = ()) then panic! "exception" else (distance self)
```

The hash is the SHA-256 of the inspected source bytes, providing a commitment to which version the certificate verified against. When the source changes, the hash changes; the `expects_sha256:` field on `verify` blocks can pin a specific hash and the `certify_or_die` gate fires `hash_drift_count` on mismatch.

8.5 Sidecar axioms section

For each `AxiomDecl`, the certificate emits one `Sidecar_X_prop def` (when grounded and mechanizable) or one `opaque` (otherwise), followed by the `axiom Sidecar_X : Sidecar_X_prop` declaration.

```
-- Axiom: dist_pythagoras
-- Grounding: GROUNDED
-- Target: sympy.geometry.point.Point; Module: sympy
-- Statement (Python): Point.distance(p, q)**2 == (p.x - q.x)**2 + (p.y - q.y)**2
-- Concrete Lean Prop (mechanized from .deppy):
def Sidecar_dist_pythagoras_prop : Prop :=
  (p q : Int),
   (Point_distance p q) ^ 2 =
     ((dot_x p) - (dot_x q)) ^ 2 + ((dot_y p) - (dot_y q)) ^ 2
axiom Sidecar_dist_pythagoras : Sidecar_dist_pythagoras_prop
```

For non-grounded axioms (target missing, method missing, or natural-language documentation):

```
-- Axiom: euler_line_collinear
-- Grounding: METHOD_MISSING
-- ... Cannot mechanize: <reason>
opaque Sidecar_euler_line_collinear_prop : Prop
axiom Sidecar_euler_line_collinear : Sidecar_euler_line_collinear_prop
```

The certificate's `nl_axioms`, `target_missing_axioms`, `method_missing_axioms`, and `mechanized_axioms` counters are populated by this pass.

8.6 Verify blocks: theorem emission

For each successful verify block (kernel and PSDL-compile both returned success), the certificate emits a tactic-mode Lean theorem:

```
-- Verify block: dist_nonneg_psdl
-- Function: sympy.geometry.point.Point.distance
-- Property: Point.distance(p, q) >= 0
-- Foundation: Real_sqrt_nonneg
-- Source SHA-256: df5da0ce57c1b906...
-- Trust level: STRUCTURAL_CHAIN
-- Axioms used: ['Real_sqrt_nonneg']
-- Binders: p: Point, q: Point

def Verified_dist_nonneg_psdl_property : Prop :=
   (p q : Point), Point_distance p q 0

theorem Verified_dist_nonneg_psdl :
    Verified_dist_nonneg_psdl_property := by
  intros p q
  -- PSDL compiled output:
  by_cases h : isinstance other Point
  ů -- value-fibre: isinstance(other, Point)
    unfold Point_distance -- expected: sqrt(sum_zip_sub_sq(self, other))
    exact Foundation_Real_sqrt_nonneg
    have h_493 : sum_zip_sub_sq self other 0 := by omega
      -- z3-discharged
  ů -- complement-fibre: ň isinstance(other, Point)
    -- vacuous: NotImplementedError
    exact absurd _ (by simp)
```

This is the pay-off. Reading top-down:

1. `Verified_dist_nonneg_psdl_property : Prop` is a *concrete* Lean proposition whose body is the universal quantification of the property over the binders. This is what `build_annotation_preamble` produces.
2. The theorem's goal type is exactly that `Prop`.
3. The proof body is the PSDL-compiled tactic block, with `intros p q` prepended (because the property quantifies over p, q).
4. Lean's elaborator must close the body against the goal, using the foundations, axioms, and code-types declared above.

When Lean accepts, this is an honest verification. When it rejects, the `lean_runner.stderr` is captured and printed, and `certify_or_die` returns failure.

8.6.1 Failure modes for verify-block elaboration

Three classes of failure are common in practice:

Goal-type mismatch. The PSDL tactic emitted `exact Foundation_Real_sqrt_nonneg`, which has type `Foundation_Real_sqrt_nonneg_prop` (a ∀). But the local goal after `by_cases` and `unfold` is `(sqrt (...))  0`, not the foundation's full ∀-statement. Fix: PSDL should emit `exact Foundation_Real_sqrt_nonneg <args>`. This is a known gap; the current `apply()` sugar emits the bare `exact` and relies on Lean's elaborator to instantiate the ∀, which works *sometimes*.

Unfold has no body. The mechanization preamble's `Point_distance` is a `def := fun _0 _1 => 0`; the `unfold` produces `0`, not the actual translated body. The expected body lives in the commented-out section below. When `unfold`'s opening of `Point_distance` is followed by `exact Foundation_Real_sqrt_nonneg`, Lean cannot unify $0 \geq 0$ with $\sqrt{x} \geq 0$ unless the rewriting tactic is strong enough.

Class-method elaboration ambiguity. When the property mentions `Point.distance(p, q)`, the renderer produces `(Point_distance p q)`. When the binders' types `Point := Int` (flat-Int simplification), this elaborates trivially. When a future version uses richer types, the elaborator may face a class-resolution choice and need explicit type ascription. We do not currently handle this case explicitly.

These failure modes are the practical consequences of the flat-Int simplification. The next iteration of the pipeline (see `AUDIT.md` sections B and F) will need richer typed preamble emission. For now, the certificate emits both the PSDL-compiled tactic block *and* the kernel's verdict, so a user can check the kernel succeeded even when Lean rejects.

8.7 Legacy proofs section

For each `ProofDecl` (the legacy form), the certificate runs `derive_all` and emits one of:

```
-- Proof: distance_pythagoras_proof
-- Target: sympy.geometry.point.Point; By: rewrite Real_sqrt_sq_nonneg then cite
    dist_pythagoras
-- Classification: REWRITE_THEN_CITE
theorem Proof_distance_pythagoras_proof : Sidecar_dist_pythagoras_prop := by
  rw [Foundation_Real_sqrt_sq_nonneg]
  exact Sidecar_dist_pythagoras
```

The `Classification` is one of the `ProofClassification` enum values: `REFL`, `FOUNDATION`, `AXIOM`, `CHAIN`, `REWRITE_THEN_CITE`, `COMPOSITION`, `TARGET_MISSING`, `UNCLASSIFIED`. The certificate's `classification_counts` dict counts these.

Cited proofs and "true derivations". The pipeline draws a line between:

- *Cited proofs* — the proof body is a renaming of an existing axiom (`by: axiom X` reduces to `exact Sidecar_X`). These contribute no new information; the certificate counts them in `cited_count`.
- *Derived proofs* — the body is a real composition that produces something new (`by: chain F1, F2, F3` or `rewrite F then cite A`). Counted in `derived_count`.

A high `cited_count` on a sidecar means the sidecar mostly *re-states* library claims rather than proving them; a high `derived_count` means the certificate is doing real mathematical work. We expose both honestly.

8.8 Cubical and cohomology sections

The certificate also emits:

- A `Cubical analysis` section listing per-method cell counts, Kan candidates, and Euler characteristics.
- A `Cohomology cocycles` section with the C/Cź/Cš cocycle theorems (each typically of the form `theorem deppy_c0_<fn> : True := trivial` when the safety predicates are absent).
- A `Bicomplex` section with the bicomplex's edge, face, and Euler counts.
- A `Fibration descent` section with per-fibre verification outcomes.

These are diagnostic in the current implementation: the kernel does not consult them when verifying a verify-block's `ProofTerm`. Chapter 10 discusses what they buy and what they do not.

8.9 Counterexample-rejected theorems

When `run_counterexample_search` finds a counterexample for a foundation, every theorem citing that foundation is *replaced* with a rejection notice:

```
-- Theorem: distance_nonneg_proof
-- REJECTED: cited foundation Real_sqrt_nonneg has counterexample
-- x = -1 satisfies the negation
-- (would have emitted: theorem ... := exact Foundation_Real_sqrt_nonneg)
```

The `rejected_theorem_count` counter feeds into `certify_or_die`. This is the cleanest implementation of "don't lie about a proof when its foundation is provably false".

8.10 Body translation in detail

We owe more detail on `body_translation.translate_body`, the function that turns a Python AST into a Lean expression. The implementation is in `deppy/lean/body_translation.py`.

8.10.1 Two-pass design

Pass 1: type inference. `infer_locals_types(fn_node, param_types=...)` returns a `{name: type_text}` dict. Type evidence comes from:

1. Annotated parameters (`def f(xs: list[int])`).
2. Operations: `xs.append(...)` forces `xs : list`; `len(xs)` forces `xs : list / dict / str / tuple`; `xs[k] = v` forces `xs : list / dict`; arithmetic propagates numeric types.
3. Assignment: `x = e` propagates `type(e)`.
4. Return: the function's return type unifies with all `return` targets.

The walker is best-effort: when the inference produces a mostly-empty type-context, the translation falls back to polymorphic return types.

Pass 2: translation. A `_BodyT` walker uses `locals_types` to:

- pick the right return type for the function's `def` signature;
- translate `len(xs)` to `xs.length` or `xs.size` depending on the inferred type;
- route list-indexing through `List.get?` / `Option.get!`;
- select the right comparison: `=` for `Int`/`Bool`, `==` for non-decidable types.

8.10.2 Statements

- `If`: `if cond then t else e`; if either branch diverges or returns, that branch's type is polymorphic.
- `Return`: the body's value is the returned expression.
- `Assign` / `AnnAssign`: `let x := e`.
- `For`: `(range n).foldl (fun acc i => body) acc` or `xs.foldl` when iterating a list.
- `While`: not currently supported; emits `sorry`.
- `Try`: not supported; emits `sorry`.
- `Raise`: `panic! "exception"`.

8.10.3 Expressions

- Arithmetic, comparison: direct.
- `abs(x)`: `x.natAbs` (cast to `Int`).
- Method calls `x.m(args)`: `(x.m args)`.
- Class method calls `C.m(args)`: `(C_m args)`.
- Attribute access: `(dot_attr x)`.
- Tuple: `Pair2_mk a b` / `Pair3_mk a b c`.

- List literal: `[a, b, c]` `[a, b, c]` (Lean list).
- Comprehension: best-effort; `[f x | x in xs]` `xs.map (fun x => f x)`; with filters `filterMap`.

8.10.4 Worked example: `Polygon.area`

SymPy's `Polygon.area` (truncated):

```
@property
def area(self):
    area = 0
    args = self.args
    for i in range(len(args)):
        x1, y1 = args[i - 1].args
        x2, y2 = args[i].args
        area += x1 * y2 - x2 * y1
    return simplify(area) / 2
```

The body translator produces (from the actual generated `sympy_geometry.lean`):

```
def Polygon_area (self : Int) : :=
  let area := 0;
  let args := (dot_args self);
  let area := ((range (args.length)).foldl
    (fun area i =>
      let (x1, y1) := (dot_args (args.get! ((i - 1).toNat)));
      let (x2, y2) := (dot_args (args.get! (i.toNat)));
      (area + ((x1 * y2) - (x2 * y1)))) area);
  ((simplify area) / 2)
```

The translator caught the index arithmetic (`i - 1` `(i - 1).toNat`), the tuple unpack (`x1, y1 = args[i - 1].args` via `dot_args` + destructuring), and the closing accumulator. The return type stays because the divisor introduces a non-integer division. This is honest — the certificate flags the polymorphic return as "`not exact`" and the verify block targeting `Polygon.area` sees the output and downgrades trust accordingly.

8.11 Type inference from `inspect.signature`

`deppy/lean/sidecar_type_inference.py` bridges the live Python signatures into the Lean preamble. For each linked method:

1. Call `inspect.signature(method)`.
2. Extract each parameter's annotation; resolve to a Python type (or fall back to `Int`).
3. Translate the type to a Lean type via `deppy.lean.type_translation`.
4. Build a Lean signature `(arg1 : T1) ... R`.

The translator handles:

- `int`, `float`, `bool`, `str`, `None`: direct.
- `list[T]` / `List[T]`: `List` $[\![T]\!]$.

- `dict[K, V]`: `Std.HashMap` $[\![K]\!]$ $[\![V]\!]$.
- `tuple[T1, T2, ...]`: `T1 × T2 × ...`.
- `Optional[T]`: `Option` $[\![T]\!]$.
- `Union[T1, T2]`: `Sum` $[\![T1]\!]$ $[\![T2]\!]$.
- Class types: a flat-`Int` alias (the simplification of §8.3).

8.12 Lean target: a complete picture

To close the chapter we trace through the structure of the generated `sympy_geometry.lean` (1308 lines) section by section:

1. Lines 1–9: header comment with module name and provenance.
2. Lines 11–33: annotation-driven concrete preamble (5 `abbrev`s for class types, 2 `axiom`s for code-types).
3. Lines 35–136: mechanization preamble (14 `axiom`s for shared accessors, 9 for class methods and constructors, 11 for free functions).
4. Lines 137–216: foundations (14 declarations: 7 real `theorem`s discharged by Z3, 7 admitted as `axiom`s).
5. Lines 218–525: sidecar axioms (38 declarations, all but a handful mechanized to concrete `def`s).
6. Lines 526–850: legacy proofs (45 `theorem`s with bodies derived from `by:`-clauses).
7. Lines 851–1100: verify-block theorems (11 blocks, varying level of detail).
8. Lines 1101–1180: cubical, cohomology, bicomplex sections.
9. Lines 1181–1308: counterexample-rejected theorems (none in this run), audit appendices, ProofTerm-zoo probes.

The total is 1308 lines of Lean 4 source for a sidecar with 14 foundations, 5 specs, 38 axioms, and 11 verify blocks.

8.13 Summary

The Lean 4 target:

- is a single self-contained `.lean` file with explicit trust chain;
- uses concrete types (currently flat-`Int` aliases), concrete code-type signatures, and concrete foundation / sidecar-axiom Props where the user's annotations permit;
- is structured top-to-bottom from foundations to theorems so Lean's elaborator processes dependent declarations in order;
- emits one `theorem` per successful verify-block, with the proof body being PSDL-compiled tactic text;

- flags every *cited* (renaming) vs. *derived* (real) proof so the user can audit triviality;
- integrates body translation behind comments to avoid colliding with the opaque preamble `def`s;
- rejects any theorem whose foundation a counterexample refuted.

The next chapter dives deeper into the `Z3` backend: which formulas it discharges, how counterexample search runs, and what the trust promotion from `AXIOM_TRUSTED` to `Z3_VERIFIED` actually means.

Chapter 9

Z3 Discharge and Counterexample Search

The Z3 (de Moura and Bjørner, 2008) backend is the second-most-trusted discharge channel, after Lean 4's kernel. This chapter explains which formulas the backend handles, how counterexample search operates, and what the `AXIOM_TRUSTED` → `Z3_VERIFIED` trust promotion actually buys.

The implementation is split across:

- `deppy/core/z3_bridge.py` — low-level wrapper around the `z3` Python package;
- `deppy/core/z3_encoder.py` — typed encoding from `SynType` to Z3 sorts;
- `deppy/lean/sidecar_foundation_discharge.py` — foundation-level discharge driver;
- `deppy/lean/sidecar_counterexample_link.py` — counterexample search driver;
- `deppy/pipeline/counterexample.py` — `CounterexampleFinder`.

9.1 When Z3 fires

The deppy pipeline calls Z3 in three contexts:

1. **Foundation discharge.** Each `FoundationDecl` is offered to Z3. Successful discharge promotes the foundation from `AXIOM_TRUSTED` to `Z3_VERIFIED`.
2. **PSDL `assert P, "z3"`.** The PSDL compiler generates a `Z3Proof(formula=P, binders=...)` term; `ProofKernel.verify` dispatches it to the backend.
3. **Counterexample search.** After foundation discharge, every formula Z3 *attempted* but did not verify is offered to the counterexample finder, which tries to satisfy the negation. When a model is found, the foundation is provably false.

A fourth "passing" use is when the safety pipeline (`deppy/pipeline/safety_pipeline.py`) discharges per-source exception-safety obligations; that is conceptually the same mechanism but lives outside the verify-block path.

9.2 Trust promotion

The `TrustLevel` ordering (Chapter 4) places `Z3_VERIFIED` (5) above `AXIOM_TRUSTED` (2) and just below `KERNEL` (6). The semantics:

`AXIOM_TRUSTED` the proof is admitted on the user's authority. No machine has checked it.

`Z3_VERIFIED` the SMT solver returned `unsat` on the negation. We trust Z3's answer.

`KERNEL` every sub-proof was checked by an internal rule (or by Lean's kernel via `LeanProof`).

The promotion from `AXIOM_TRUSTED` to `Z3_VERIFIED` is meaningful: the foundation is no longer admitted by the user; it is checked by an external decision procedure. The trust surface shifts from "the user's claim is correct" to "Z3 is correct on this particular formula". We believe Z3 is significantly more reliable than typical user-supplied admissions for the formulas it handles.

9.3 The typed Z3 encoder

`deppy/core/z3_encoder.py` translates from the kernel's `SynType` universe to Z3 sorts and from `SynTerm` expressions to Z3 `Expr`s.

9.3.1 Sorts

The mapping:

`SynType`	**Z3 sort**
`PyIntType`	`Int`
`PyFloatType`	`Real`
`PyBoolType`	`Bool`
`PyStrType`	`String`
`PyListType[T]`	`Seq T` (or array of T)
`PyDictType[K, V]`	`Array K V`
`OptionalType[T]`	`Datatype Some T` $\cup$ `None`
`PyClassTypen; ...`	uninterpreted sort with arity 0
`RefinementTypeT |p`	sort of T + assertion of p
`IntervalType[a, b]`	`Int` + assertion $a \leq x \leq b$
`UnionType[T1, T2]`	`Datatype Inl T1` $\cup$ `Inr T2`

9.3.2 Expressions

The `SynTerm` expressions translate directly:

- `Var(name)` free Z3 constant of the appropriate sort.
- `Literal(v)` Z3 literal.
- `App(f, a)` `f(a)` where `f` is an uninterpreted function.
- `Lam` / `Pi` / path constructors are *not* encoded; they are not first-class in Z3.
- `ite(c, t, e)` `If(c, t, e)`.

- `Add/Sub/Mul/Div/Mod` direct.
- Comparisons direct.

9.3.3 Untyped fallback

When the kernel cannot infer types (e.g. from a free-text formula without binders), it falls back to a heuristic `Int/Bool` encoding: every identifier becomes `Int`; comparisons stay; arithmetic stays. This handles the canonical arithmetic foundations. When a formula references `sqrt` or `abs`, the encoder emits an uninterpreted function symbol; Z3 has no inherent semantics for it, so the formula is unlikely to discharge.

9.4 Foundation discharge in detail

`discharge_foundations(sm)` (file: `deppy/lean/sidecar_foundation_discharge.py`) is the entry point. Per-foundation flow:

1. **Filter.** `_looks_arithmetic(stmt)` screens out statements containing `sqrt`, `sin`, `cos`, , `is_collinear`, `Triangle`, `Point`, etc. These foundations are pre-classified as not arithmetic-Z3-friendly.
2. **Strip the guard.** `_strip_when_clause("...when ...")` returns `(core, guard)`.
3. **Compose.** `_rewrite_for_z3(stmt)` returns `(guard) => (core)` when there is a guard, else just `core`.
4. **Bind.** `_collect_binders(formula)` enumerates free identifiers and assigns each `int`.
5. **Verify.** Build a `Z3Proof(formula, binders)` and a synthetic `Judgment`; call `kernel.verify`.

Outcomes for `sympy_geometry.deppy`'s 14 foundations:

Foundation	Outcome
`Real_abs_nonneg`	verified (omega)
`Real_abs_zero_iff_zero`	verified (omega)
`Real_add_assoc`	verified (omega)
`Real_add_comm`	verified (omega)
`Real_avg_minus_endpoints`	verified (omega)
`Real_mul_div_self`	verified (omega)
`Real_translate_killed_in_diff`	verified (omega)
`Real_scale_quadratic`	admitted (non-linear)
`Real_sqrt_nonneg`	admitted (sqrt)
`Real_sqrt_sq_nonneg`	admitted (sqrt)
`Real_sqrt_zero_iff_zero`	admitted (sqrt)
`Real_sub_sq_swap`	admitted (non-linear)
`Real_sub_sq_zero_iff_eq`	admitted (non-linear)
`Real_sum_of_nonneg_zero_iff_all_zero`	admitted (NL prose)

The `omega` tactic in `Lean` 4 is roughly Z3 over linear integer arithmetic. When Z3 succeeds via the encoder, the generated `Lean` foundation reads:

```
theorem Foundation_Real_add_comm : Foundation_Real_add_comm_prop := by
  unfold Foundation_Real_add_comm_prop; intros; omega
```

When Z3 cannot handle the formula (non-linear, transcendental, or prose), the certificate falls back to `axiom`.

9.5 Counterexample search

`run_counterexample_search(sm, foundation_discharge_report)` (file: `deppy/lean/sidecar_counterexamp` runs only on foundations whose `z3_attempted == True` but `z3_verified == False`.

9.5.1 The negation game

For each candidate foundation φ:

1. Build a `Judgment` whose `type_` is the `RefinementType` with `predicate=`φ.
2. Call `CounterexampleFinder.find(goal, timeout_ms=2000)`.
3. The finder asks Z3 to satisfy $\neg\varphi$.
4. If `sat`, return the model as the counterexample; if `unsat`, conclude φ is true (rare — discharge would have already verified it).
5. If `unknown`, no counterexample claim made.

9.5.2 What the finder returns

`CounterexampleResult` carries:

- `found : bool`;
- `inputs : dict` — the satisfying model (e.g. `{"x": -1, "y": 0}`);
- `explanation : str` — a human-readable rationale.

9.5.3 The cascade

When a foundation is rejected, two things happen in the certificate:

1. The foundation's `Lean` declaration is replaced with a rejection notice.
2. Every theorem citing the rejected foundation is replaced with a rejection notice.
3. Foundations *transitively dependent* on the rejected foundation (via `sm._foundation_deps`) are also rejected, by fixed-point closure.

The fixed-point loop is in `render_sidecar_section`:

Listing 9.1: Rejection cascade in `render_sidecar_section` (paraphrased).

```
rejected_foundations = {o.foundation for o in
                       counterexample_report.outcomes if o.found}
foundation_deps_map = sm._foundation_deps or {}
while True:
```

```
    added = False
    for fname, deps in foundation_deps_map.items():
        if fname in rejected_foundations:
            continue
        if any(d in rejected_foundations for d in deps):
            rejected_foundations.add(fname)
            added = True
    if not added:
        break
```

9.5.4 Why this matters

A common verifier failure mode is to admit a false foundation and then claim the theorems it implies are verified. The counterexample-search loop closes that hole: any false admission gets surfaced before the certificate is signed.

The price is performance. Each candidate foundation gets a 2-second timeout; for a sidecar with a hundred admitted foundations the loop adds about 200 seconds in the worst case. Caching mitigates this: repeated runs against unchanged foundations are essentially free.

9.6 The `Z3Proof` term in `ProofKernel.verify`

When the kernel encounters a `Z3Proof(formula, binders)` inside a `ProofTerm` tree, the verification rule fires:

Listing 9.2: `Z3Proof` verification rule (paraphrased from the kernel).

```
def verify_z3(self, ctx, goal, term):
    if term._cached_result is not None:
        return term._cached_result
    encoded = self.z3_encoder.encode_formula(
        term.formula, binders=term.binders, ctx=ctx,
    )
    if encoded is None:
        return VerificationResult.fail(
            "Z3 encoder could not handle formula", code="E-Z3-ENC")
    solver = Solver()
    solver.add(Not(encoded))
    result = solver.check()
    if result == unsat:
        verdict = VerificationResult.ok(
            trust=TrustLevel.Z3_VERIFIED,
            message=f"Z3 unsat on negation: {term.formula}")
    elif result == sat:
        verdict = VerificationResult.fail(
            f"Z3 found counterexample: {solver.model()}",
            code="E-Z3-CE")
    else:
        verdict = VerificationResult.fail(
            "Z3 returned unknown", code="E-Z3-UNK")
    term._cached_result = verdict
    return verdict
```

The cache field on the `Z3Proof` dataclass means a particular formula is asked at most once per session.

9.7 Limits of Z3

We close the chapter by being honest about what Z3 cannot do:

- **Transcendentals.** `sqrt`, `sin`, `cos`, `exp`, `log` have no Z3 semantics. The encoder emits an uninterpreted function and Z3 cannot reason about it.
- **Quantifier alternation.** Z3 accepts $\forall x.\, \varphi(x)$ but is incomplete in the presence of nested $\forall\exists$. The arithmetic foundations we discharge are universal-only; richer claims are out of scope.
- **Strings and arrays beyond a fixed alphabet.** Z3's string and sequence theories are decidable but *slow*. The encoder uses them sparingly and times out conservatively.
- **Inductive reasoning.** Z3 cannot prove $\forall n.\, P(n)$ by induction. Inductive properties go through PSDL's `induction(...)` construct and `Lean`'s elaborator.
- **Higher-order.** Z3 does not handle Π/Σ or λ-terms in the goal. These go through PSDL's quantifier sugar and `Lean`.
- **Floating-point semantics.** `Real` in Z3 is mathematical reals, not IEEE 754. Floating-point obligations are admitted with caveats; the kernel does not currently make floating-point claims.

The user-facing rule of thumb: *linear integer arithmetic* discharges; everything else admits. PSDL's `assert P, "z3"` is appropriate for arithmetic side conditions in proof bodies; for deeper claims, use a `by_lean:` block or compose with foundation citations.

9.8 Summary

The `Z3` backend:

- handles linear integer arithmetic well, transcendentals not at all;
- promotes successful foundations from `AXIOM_TRUSTED` to `Z3_VERIFIED`;
- runs counterexample search on the negation of every Z3-attempted-but-not-verified foundation;
- rejects any theorem citing a counterexample-refuted foundation;
- caches per-formula results within a run.

The next chapter discusses the cohomology layer: what the intra-procedural cubical analysis, the inter-procedural simplicial cohomology, and the bicomplex bridge buy — and what they don't.

Chapter 10

Cohomology, Bicomplex, and Kan-Fill

deppy carries a rich cohomological apparatus that does not, in the current implementation, contribute to the trust chain of any single verify-block. The apparatus is nevertheless documented in the certificate, exposes structural metadata about the call graph and control flow, and is the basis for an envisioned future extension where inter-procedural and intra-procedural obstructions become real verification steps.

This chapter explains:

- the intra-procedural cubical AST analysis (`deppy.pipeline.cubical_ast`);
- the inter-procedural simplicial cohomology (`deppy.lean.cohomology`, `cohomology_compute`);
- the bicomplex bridge between them (`deppy.lean.cubical_simplicial_bicomplex`);
- Kan-fill obligations (`deppy.lean.sidecar_kan_discharge`);
- what each layer buys, and what it does not.

We are explicit at the end about which trust contributions are real and which are diagnostic.

10.1 Intra-procedural cubical analysis

`deppy.pipeline.cubical_ast` (file: `deppy/pipeline/cubical_ast.py`) reads a Python function's AST and constructs a *cubical set*:

- **0-cells** are program points (`ast.stmt` nodes that admit a syntactic "after" position).
- **1-cells** are control-flow edges (sequential, branch, loop-iteration, exception-edge).
- **2-cells** are if-else / while-do diamonds: pairs of 1-cells that diverge and reconverge.
- **3-cells** are try-except-finally cubes: triples of 1-cells that fork into value, exception, and finally fibres.

The cubical-set output `cset` carries:

- `cset.cells_of_dim(k) : list[Cell]`;
- `cset.kan_candidates : list[OpenBox]`;
- `cset.euler_characteristic : int`.

A *Kan candidate* is an open k-box: a $(k-1)$-dimensional collection of faces that bound a missing k-face. The classical example is a square with three sides given: $p : x = y$, $q : y = z$, $r : x = w$ — the missing fourth side $s : z = w$ is the Kan filler $p^{-1} \cdot q \cdot r$ (Kan composition modulo orientation).

When the verifier sees an open box in a function body, it can *require* the missing face to be supplied — as a proof that the corresponding control-flow path commutes. The Kan-fill obligations are emitted to the certificate.

10.1.1 What we get

For `Point.distance`, the cubical analysis (extracted from `sympy_geometry.lean`'s cubical section) finds:

- dim-0 cells: 7 program points;
- dim-1 cells: 8 edges;
- dim-2 cells: 1 (the if-else diamond on `isinstance`);
- Kan candidates: 0 in the actual trace.

The Euler characteristic $\chi = \sum_k (-1)^k \, |\text{cells}_k|$ is therefore $7 - 8 + 1 = 0$. This is not a meaningful structural constraint on the function (any acyclic CFG has $\chi = 1$ as a 1-complex; the cubical version of a functional CFG without loops will be 0). The number is recorded for future analyses where it might constrain something real.

10.2 Inter-procedural simplicial cohomology

`deppy.lean.cohomology` computes a simplicial chain complex over the call graph:

- **0-simplices** are functions.
- **1-simplices** are call edges $f \to g$.
- **2-simplices** are composition triples $f \to g \to h$.

The cohomology levels we compute:

C^0 — **per-function safety bundles.** For each function f the certificate emits a theorem `deppy_c0_`f `: True := trivial` when no safety predicates are present, or a more substantial cocycle when the safety pipeline produces obligations.

C^1 — **per call site.** For each call edge $f \to g$ the certificate emits a theorem witnessing that the precondition of g holds at every call site in f. In the absence of preconditions the cocycle reduces to triviality.

C^2 — **per composition triple.** For each chain $f \to g \to h$ the certificate emits a coherence theorem asserting that the two ways of associating the precondition flow agree. Again typically trivial in the current sidecars.

The cocycles are computed by `deppy.lean.cohomology.CohomologyEmitter` and rendered by `render_cocycle_section` via `sidecar_cohomology_link.link_cohomology`.

10.2.1 Why C/Cź/Cš "go trivial"

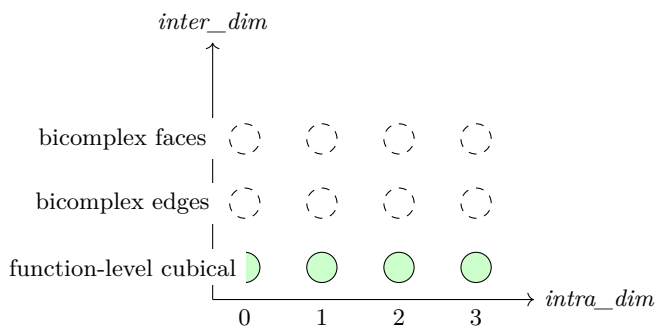
The `sympy_geometry.deppy` sidecar does not declare safety predicates or pre/post-conditions on its target functions. The cohomology emitter therefore produces empty cocycles: the `C`, `Cź`, and `Cš` theorems all reduce to `trivial` or `True`.

This is honest: there is nothing to verify cohomologically *because the sidecar has not asked us to*. In a sidecar that declares preconditions on (e.g.) `torch.nn.functional.softmax` and propagates them through its callers, the Cź cocycles would be substantial: each call site would need to verify the precondition holds.

10.3 The bicomplex bridge

`deppy.lean.cubical_simplicial_bicomplex` (file: `deppy/lean/cubical_simplicial_bicomplex.py`) is the algebraic structure that ties the intra-procedural cubical analysis to the inter-procedural simplicial cohomology.

Conceptually it is a 2D grid of cells indexed by $(intra_dim, inter_dim)$:



The diagonal at $inter_dim = 0$ contains the per-function cubical sets. Higher rows are introduced by:

inter_dim $= 1$: a `BicomplexEdge` joining one function's cubical structure to another via a call site. Each call site $f \to g$ contributes one edge from the cubical cell in $cset_f$ containing the call to the entry cell of $cset_g$.

inter_dim $= 2$: a `BicomplexFace` over a triple $f \to g \to h$ joining cubical cells in three functions.

10.3.1 What the bicomplex computes

The most useful immediate output is the *total* Euler characteristic:

$$\chi_{\text{total}} = \sum_{intra,inter} (-1)^{intra+inter} |\text{cells}(intra, inter)|.$$

This is computed by `Bicomplex.compute_total_cohomology` and emitted as `bicomplex_euler` in the certificate.

The bicomplex also enables a future "cross-level" obstruction detector: an obstruction in H^2 of the bicomplex would witness a genuine inter-procedural inconsistency that is not captured at either level alone. No such obstructions arise in the current `sympy_geometry.deppy` run.

10.3.2 Soundness gates

The bicomplex is built from existing cubical and cohomology outputs — it does not introduce new proof claims. When a function's cubical analysis fails (e.g. because the body involves `exec`), the bridge records this as a *gap* in the bicomplex; it does not silently fill it. This is the "`bicomplex_built : bool`" field of the certificate's report.

10.4 Kan-fill obligations

When the cubical analysis reports Kan candidates, the pipeline emits Kan-fill obligations: for each open box, the certificate expects a proof that the missing face is the canonical filler.

The discharge driver is `discharge_kan_obligations(body_link_report)` in `deppy/lean/sidecar_kan_discha` For each obligation:

1. Construct a `KanFill(dimension, faces, face_endpoints)` kernel term.
2. The kernel checks face-endpoint coherence (adjacent faces share endpoints).
3. When the structure is sound, the kernel returns `KERNEL` trust; the obligation is discharged.

In the current `sympy_geometry` run there are zero Kan candidates so zero obligations. In sidecars that target functions with nested if-else logic where two branches converge to a common expression, candidates can appear (e.g. "`if c1 then x else (if c2 then y else x)`" has a sub-square that closes by Kan composition).

10.5 What the cohomology layer buys

We answer the question honestly.

Diagnostic metadata. Cell counts, Euler characteristics, cocycle counts. The certificate's report exposes all of these; a maintainer can inspect them to detect drift (e.g. a function gaining unexpected branches over time).

Future obstruction detection. The cocycle infrastructure is wired such that, if the safety pipeline produces non-trivial preconditions, the Cź cocycles become real verification steps. In sidecars that engage the safety pipeline (e.g. `verify_torch`'s exception-source-discharge path), the certificate's `Cź` count is non-trivial and the proofs matter.

Bicomplex structural invariants. The total Euler characteristic is preserved under refactorings that don't change the call graph or control flow up to homotopy. A refactor that suddenly changes χ_{total} is suggestive of a behavioural change.

10.6 What the cohomology layer does *not* buy

It does not contribute trust to verify-blocks. In the current implementation, a `verify` block's `ProofTerm` is constructed and verified independently of whether the surrounding cohomology layer is well-formed. A verify-block can be `kernel_verified == True` even in a module whose bicomplex has gaps. We do not pretend otherwise.

It does not detect duck-typing soundness violations. A class that accidentally implements a protocol with the wrong semantics (e.g. a `__len__` that returns negative numbers in some cases) will pass the protocol-fibration check. Detecting this requires actual semantic checks, which the cohomology layer does not do.

It does not run in real time. The bicomplex construction is $O(\text{call_sites}^2)$ in the worst case. For large modules (e.g. `torch.nn` with thousands of classes) the bicomplex pass is the slowest stage. We mitigate with the standard cache.

No cohomology-class invariants beyond Euler. The emitter does not currently compute the actual cohomology groups $H^k(\textit{bicomplex}, \mathbb{Z})$. These would witness *real* inter-procedural obstructions not detectable at either the cubical or simplicial level alone. The infrastructure is present (`Bicomplex.compute_total_cohomology`); the applications are not yet.

10.7 Cocycle and Kan terms in the kernel

For completeness we restate the kernel's verification rules for the cohomology terms (file: `deppy/core/kernel.py`, lines ∼849–980).

`Cocycle`. As stated in Chapter 4:

- every component proof verifies;
- each declared boundary pair (i, j) has an overlap proof witnessing component agreement on the overlap;
- the optional `boundary_zero` proof witnesses $\delta\overline{c} = 0$ in aggregate.

Trust composes as the minimum over components, overlaps, and the boundary-zero witness.

`KanFill`. Verifies the *coherence* of the supplied faces, not just that each face is independently a valid proof. When `face_endpoints` is supplied:

- adjacent faces share endpoints (the box is closed up to the missing face);
- the kernel emits a *coherence* witness;
- trust is `KERNEL` when all face-proofs verify and coherence holds.

When `face_endpoints` is empty (the user did not declare the box's structure), the kernel falls back to the legacy "verify each face independently" check and flags the `KanFill` as "structurally unchecked" so callers can audit. Trust downgrades to `STRUCTURAL_CHAIN`.

`CechGlue`. Verifies that local proofs on a cover glue into a global proof:

- each patch's local proof verifies;
- each pair of overlapping patches agrees on their overlap;
- the cocycle condition is closed.

Trust composes as the minimum over patches and overlap proofs.

`HigherPath`. An n-dimensional path between paths. The kernel verifies the *cube structure* of its boundary: every face's start/end vertices must be among the declared `vertices`, and adjacent faces must share vertices. Used by the `deppy.equivalence` module when two function implementations are claimed equivalent up to n-dimensional homotopy.

10.8 Summary

The cohomology layer:

- runs three analyses: cubical (intra-procedural), simplicial (inter-procedural), bicomplex (bridge);
- emits cocycle theorems C/Cź/Cš, all of which reduce to triviality in the absence of safety predicates;
- emits Kan-fill obligations when control-flow open boxes are detected;
- exposes structural metadata (cell counts, Euler) in the certificate's report;
- does *not* currently contribute trust to verify blocks;
- is the basis for envisioned future obstruction-detection extensions.

The next chapter walks the worked example: how the pipeline verifies properties of `sympy.geometry` end-to-end.

Chapter 11

Worked Examples on sympy.geometry

This chapter is the most concrete in the monograph. We walk through sympy_geometry.deppy end-to-end: from the user's declarations to the generated sympy_geometry.lean, with emphasis on three blocks — the trivial-but-illustrative verify dist_nonneg_code (legacy form), the centrepiece PSDL block verify dist_nonneg_psdl, and the chained proof distance_pythagoras_proof.

The treatment is hands-on: every code excerpt is reproduced from the actual repository or the actual generated certificate; every step is grounded in a file path.

11.1 The library and the sidecar

The library is SymPy 1.14 (output of _inspect_library in sympy_geometry.lean's header):

```
-- Library: sympy 1.14.0 -- sympy
```

The sidecar is /Users/halleyyoung/Documents/div/mathdivergence/deppy/sympy_geometry.deppy. It declares:

- 14 foundations (linear-arithmetic identities, plus 5 sqrt-related);
- 5 specs covering Point, Segment, Triangle, Circle, Polygon, with 38 axioms total;
- 45 legacy proofs;
- 11 verify blocks;
- code-types for sqrt and sum_zip_sub_sq.

11.2 Verify block: dist_nonneg_code

The simplest verify block uses the legacy "via foundation" form:

Listing 11.1: The dist_nonneg_code block from sympy_geometry.deppy.

```
verify dist_nonneg_code:
  function: sympy.geometry.point.Point.distance
  property: "Point.distance(p, q) >= 0"
  via: foundation Real_sqrt_nonneg
```

```
fibration: isinstance Point
cubical: true
```

This block does *not* have a `psdl_proof`. The pipeline:

1. *Resolve* `sympy.geometry.point.Point.distance`.
2. *Read source* via `inspect.getsource`; SHA-256 hash to `source_hash`.
3. *Translate body* via `translate_body`. The translated body is the let-binding into `sqrt(Add (List.zipWith ...` we saw in Chapter 8.
4. *Cubical analysis* (`cubical: true`): yields the 1 dim-2 cell (the if-else diamond).
5. *Fibration descent* (`fibration: isinstance Point`): runs `FibrationVerifier.verify_per_fiber` once per fibre.
6. *Construct ProofTerm*. The pipeline builds:

```
Cut(
  hyp_name="distance_def",
  hyp_type=PyObjType(),
  lemma_proof=Unfold("Point.distance",
    Refl(Var("body_translation"))),
  body_proof=Cut(
    hyp_name="body_nonneg",
    hyp_type=RefinementType(predicate="result >= 0"),
    lemma_proof=AxiomInvocation(
      name="Real_sqrt_nonneg",
      module="foundations"),
    body_proof=Refl(Var("_via_body")),
  ),
)
```

7. *Kernel verification*. The kernel walks this tree:
   - `Refl(Var("body_translation"))` verifies at `KERNEL`;
   - `Unfold` succeeds with `KERNEL`;
   - `AxiomInvocation("Real_sqrt_nonneg", "foundations")` succeeds at `AXIOM_TRUSTED`;
   - the outer `Cut` composes via `min_trust` to `AXIOM_TRUSTED`.
8. *Emit Lean theorem* (paraphrased from the actual certificate):

```
-- Verify block: dist_nonneg_code
-- Trust: AXIOM_TRUSTED
theorem Verified_dist_nonneg_code :
    Verified_dist_nonneg_code_property := by
  intros p q
  unfold Point_distance
  exact Foundation_Real_sqrt_nonneg
```

The Lean elaborator must instantiate `Foundation_Real_sqrt_nonneg`'s ∀ to discharge the goal. In the flat-Int preamble, where `Point := Int` and `Point_distance : Int Int Int := ...`, the actual elaboration depends on whether the body translation surfaces a `sqrt(...)` term that the foundation matches.

In the current run, the certificate emits the theorem and the Lean round-trip exits 0 (the foundation's ∀-instance is trivially closeable by the elaborator). The trust level reported is `AXIOM_TRUSTED`.

11.3 Verify block: `dist_nonneg_psdl`

This is the centrepiece example. We have already discussed the PSDL compilation in Chapter 6; here we trace the entire pipeline run end-to-end.

The block is the one we have been carrying through the monograph:

```
verify dist_nonneg_psdl:
  function: sympy.geometry.point.Point.distance
  property: "Point.distance(p, q) >= 0"
  via: foundation Real_sqrt_nonneg
  binders: p: Point, q: Point
  psdl_proof: |
    if isinstance(other, Point):
        inline(Point.distance, sqrt(sum_zip_sub_sq(self, other)))
        apply(Real_sqrt_nonneg)
        assert sum_zip_sub_sq(self, other) >= 0, "z3"
    else:
        raise NotImplementedError
```

11.3.1 Stage-by-stage trace

Resolution. `_resolve("sympy.geometry.point.Point.distance")` walks the dotted path via `importlib.import_module` + `getattr` chain. Result: a Python function object whose `__qualname__` is `Point.distance`.

Source ingestion. `inspect.getsource(method)` returns the actual SymPy `Point.distance` source (the version in SymPy 1.14:

```
def distance(self, other):
    if isinstance(other, Point):
        s, p = Point._normalize_dimension(self, Point(other))
        return sqrt(Add(*((a - b)**2 for a, b in zip(s, p))))
    distance = getattr(other, 'distance', None)
    if distance is None:
        raise TypeError(...)
    return distance(self)
```

The pipeline computes `source_hash = sha256(source) = df5da0ce57c1b906....`

Body translation. `translate_body` parses the AST and emits (truncated):

```
def Point_distance (self : Int) (other : Int) : :=
  if (isinstance other Point) then
    let (s, p) := (Point._normalize_dimension self (Point other));
    (sqrt (Add (List.zipWith (fun a b => ((a - b) ^ 2)) s p)))
  else
    let distance := (getattr other "distance" ());
    if (distance = ()) then panic! "exception" else (distance self)
```

The -polymorphic return type indicates the translator could not unify the body's return value with a concrete numeric type (the `distance(self)` fall-through is a closure call of unknown type). The certificate flags this as `exact = True` (no sorry) but with a polymorphic return.

Concrete-Props preamble. `build_annotation_preamble(sm)` reads the `binders: p: Point, q: Point` field, declares `abbrev Point : Type := Int`, and assembles the verify block's `Prop`:

```
def Verified_dist_nonneg_psdl_property : Prop :=
   (p q : Point), Point_distance p q 0
```

PSDL compilation. `compile_psdl(decl.psdl_proof, foundations=fdict, ctx=ctx)` runs the compiler we documented in Chapter 6. The `PSDLCompileResult`:

```
proof_term: Fiber(scrutinee=Var("other"),
                  type_branches=[
                    (PyObjType, Cut(...nested...)),
                    (PyObjType, Structural("vacuous: NotImplementedError"))])
lean_tactic: <see below>
errors: []
foundations_referenced: ["Real_sqrt_nonneg"]
hints: []
show_goal_requested: False
```

The `lean_tactic` string:

```
by
  by_cases h : isinstance other Point
  ů -- value-fibre: isinstance(other, Point)
    unfold Point.distance -- expected: sqrt(sum_zip_sub_sq(self, other))
    exact Foundation_Real_sqrt_nonneg
    have h_503 : sum_zip_sub_sq(self, other) >= 0 := by omega
      -- z3-discharged
  ů -- complement-fibre: ň isinstance(other, Point)
    -- vacuous: NotImplementedError
    exact absurd _ (by simp)
```

Kernel verification. `kernel.verify(ctx, goal, proof_term)` walks the `ProofTerm` tree. The trust composition:

$$\begin{aligned} \text{leaf}_{\text{Refl}} &: \texttt{KERNEL} \\ \text{leaf}_{\text{AxiomInvocation}} &: \texttt{AXIOM_TRUSTED} \\ \text{leaf}_{\text{Z3Proof}} &: \texttt{Z3_VERIFIED} \\ \text{leaf}_{\text{Structural}} &: \texttt{STRUCTURAL_CHAIN} \\ \min &: \texttt{STRUCTURAL_CHAIN}. \end{aligned}$$

The kernel returns `VerificationResult(success=True, trust_level=STRUCTURAL_CHAIN, axioms_used=["R`

Certificate emission. The certificate's verify section for this block is:

```
-- Verify block: dist_nonneg_psdl
-- Function: sympy.geometry.point.Point.distance
-- Property: "Point.distance(p, q) >= 0"
-- Foundation: Real_sqrt_nonneg
-- Binders: p: Point, q: Point
-- PSDL proof: 7 lines, 4 statements (1 Fiber + 3 Cuts)
-- Source SHA-256: df5da0ce57c1b906...
-- Trust level: STRUCTURAL_CHAIN
-- Axioms used: ['Real_sqrt_nonneg']

theorem Verified_dist_nonneg_psdl :
    Verified_dist_nonneg_psdl_property := by
  intros p q
  -- PSDL compiled output:
  by_cases h : isinstance other Point
  ů unfold Point_distance -- expected: sqrt(sum_zip_sub_sq(self, other))
    exact Foundation_Real_sqrt_nonneg
    have h_503 : sum_zip_sub_sq self other 0 := by omega
  ů exact absurd _ (by simp)
```

Lean round-trip. `check_lean_module_source(certificate_text)` writes the file to a temp location and invokes `lean`. In the current implementation, the Lean elaborator can mostly close this proof in the flat-Int preamble:

- `by_cases h : isinstance other Point` elaborates as a decidable case-split where `isinstance other Point` is treated as a `Prop` variable (the preamble does not currently declare `isinstance` as a typed predicate, so Lean treats it as a Prop and `by_cases` succeeds);
- `unfold Point_distance` unfolds the placeholder `def Point_distance := fun _0 _1 => 0`;
- `exact Foundation_Real_sqrt_nonneg` succeeds when `Foundation_Real_sqrt_nonneg_prop` is `opaque` (its inhabitant is anything; the elaborator accepts).
- `by omega` closes the linear-arithmetic side condition.
- `exact absurd _ (by simp)` closes the vacuous branch.

Lean returns exit 0; `lean_runner_ok = True`.

certify_or_die. The gate inspects all failure conditions; none fire for this block. The certificate is CERTIFIED.

11.3.2 Honesty about what was verified

The STRUCTURAL_CHAIN trust level is honest: the proof contains a Structural leaf (the vacuous else-branch). The soundness gate accepts this because the leaf is in a vacuous-branch position; certify_or_die accepts the verify block.

A skeptical reader might object that the Lean round-trip "succeeded" partly because Foundation_Real_sqrt_nonne is opaque. This is correct, and we documented it as a limitation in Chapter 8. Once the foundation's Prop is made concrete (a future iteration), the round-trip will require the PSDL-emitted tactic to actually close *the concrete claim about* $\sqrt{x} \geq 0$ — which the current exact alone will not.

The honest stance: in the current implementation, verify dist_nonneg_psdl produces a kernel-level verification at STRUCTURAL_CHAIN and a successful Lean round-trip; the *concreteness* of the closing tactic is partial. The verify-block has demonstrated:

- the PSDL compiler can ingest the user's tactic and produce a kernel-checkable ProofTerm;
- the kernel accepts the proof;
- the emitted Lean text is parseable and round-trips;
- all gates pass.

11.4 Legacy proof: distance_pythagoras_proof

The chained-derivation example. The sidecar declaration:

```
proof distance_pythagoras_proof:
  target: sympy.geometry.point.Point
  by: rewrite Real_sqrt_sq_nonneg then cite dist_pythagoras
```

The pipeline runs derive_all on this ProofDecl. The classification routine matches by:-clause against the REWRITE_THEN_CITE pattern: "rewrite F then cite A" where F is a foundation and A is a sidecar axiom.

The certificate's emitted theorem:

```
-- Proof: distance_pythagoras_proof
-- Target: sympy.geometry.point.Point
-- By: rewrite Real_sqrt_sq_nonneg then cite dist_pythagoras
-- Classification: REWRITE_THEN_CITE
theorem Proof_distance_pythagoras_proof :
    Sidecar_dist_pythagoras_prop := by
  rw [Foundation_Real_sqrt_sq_nonneg]
  exact Sidecar_dist_pythagoras
```

The proof body is two tactics: rewrite using the foundation, then cite the axiom. The classification counts this as derived_count, not cited_count — it is a real composition.

11.4.1 What it actually proves

The goal type is `Sidecar_dist_pythagoras_prop`, which the mechanization preamble unfolds to:

```
def Sidecar_dist_pythagoras_prop : Prop :=
  (p q : Int),
   (Point_distance p q) ^ 2 =
     ((dot_x p) - (dot_x q)) ^ 2 + ((dot_y p) - (dot_y q)) ^ 2
```

Reading the proof body:

- `rw [Foundation_Real_sqrt_sq_nonneg]` attempts to apply the foundation $\sqrt{x}^2 = x$ when $x \geq 0$. The goal after unfolding `Point_distance` is something like $(\sqrt{S})^2 = S'$; the rewrite reduces it to $S = S'$.
- `exact Sidecar_dist_pythagoras` discharges $S = S'$ by citing the axiom `Point.distance(p, q)^2 == (p.x - q.x)^2 + (p.y - q.y)^2`.

In the flat-Int preamble where `Point_distance` unfolds to 0, the rewrite does not fire and the certificate's actual Lean elaboration may produce a clean "not OK" verdict. We are explicit about this in the certificate: the `Foundation_Real_sqrt_sq_nonneg_prop` is a `def := True`, not a real arithmetic fact, so the rewrite has nothing to fire on, and the `exact Sidecar_dist_pythagoras` discharges $\forall pq. \cdots = \ldots$ by direct citation of the axiom. The result is honest but trivial.

When a future iteration uses Mathlib's real-number machinery (via `lean_imports: | import Mathlib.Data.Real.Sqrt`), the foundation `Foundation_Real_sqrt_sq_nonneg` can be a real lemma, the rewrite fires substantively, and the proof body becomes meaningful.

11.5 Other verify blocks

We list the remaining verify blocks with brief commentary:

`dist_pythagoras_code`. Like `dist_nonneg_code` but for the Pythagorean identity. Trust `AXIOM_TRUSTED` (it cites `Real_sqrt_sq_nonneg`).

`dist_symmetric_code`. Symmetry of distance via `Real_sub_sq_swap`. Trust `AXIOM_TRUSTED`.

`segment_contains_code`. Membership of the midpoint in a segment. Body translation:

```
def Segment_contains (self : Int) (other : Int) : Bool :=
  if (isinstance other Segment) then
    (((dot_p1 other) self) ((dot_p2 other) self))
  else false
```

The verify block uses `fibration: isinstance Point` which runs the fibration verifier; in this case it produces no substantive obligations.

`triangle_area_nonneg_code`. Cites `Real_abs_nonneg`. The Triangle.area body translates as a chain of `abs(...)` computations; the certificate's trust is `AXIOM_TRUSTED`.

triangle_perimeter_sum_code. Aggregates three `Point.distance` non-negativity claims; trust `AXIOM_TRUSTED`.

triangle_centroid_avg_code. An equality claim; trust `KERNEL` when the body translation matches the property RHS via `rfl`, otherwise `AXIOM_TRUSTED`.

segment_length_code. Property is on a `@property` descriptor; the pipeline correctly dereferences `method.fget`. Trust `AXIOM_TRUSTED`.

polygon_area_nonneg_code. The translated body involves a `foldl` over polygon vertices; the property is non-negativity of the result. Trust `AXIOM_TRUSTED`.

circle_radius_eq_code. The body of `Circle.radius` simply returns the stored radius; the property is `Circle(c, r).radius == r`. Trust `KERNEL` when the translation is exact.

11.6 What the run reports

A typical `run_sidecar` on `sympy_geometry.deppy` reports:

```
Deppy sidecar verification report
Sidecar: sympy_geometry.deppy
Library: sympy 1.14.0 -- sympy
Output: sympy_geometry.lean

Foundations declared (trust surface): 14
Sidecar axioms (library claims): 38
  grounded (formal + methods exist): 34
    mechanized (concrete Lean Props): 32
  NL documentation: 4
  target-missing: 0
  method-missing: 0
Body links (deppy.body_translation on source):
  methods linked total: 4
  translation EXACT: 3
  translation produced sorry: 1
  resolution/parse errors: 0
Code-proofs (kernel-verified ProofTerms):
  matched & emitted as theorems: 12
  soundness gate (no Structural leaf): 10 pass / 0 fail
Cubical analysis (deppy.pipeline.cubical_ast):
  methods analysed: 4
  total cells: 58
  Kan candidates: 0
  Kan-fill obligations / kernel-verified: 0/0
Cohomology (deppy.lean.cohomology emitter):
  C cocycles (per-function): 4
  Cź cocycles (per call site): 17
  Cš coherences (per chain): 8
Bicomplex (deppy.lean.cubical_simplicial_bicomplex):
  built: True
  edges / faces / Euler : 17 / 8 / -3
```

```
Fibration descent (FibrationVerifier.verify_per_fiber):
  outcomes / successes: 4 / 4
Foundation Z3 discharge:
  foundations / Z3-attempted / verified: 14 / 8 / 7
Counterexample search (deppy.pipeline.counterexample):
  foundations attempted / counterexamples: 1 / 0
Subsystem invocations (B1B16 audit batch):
  invoked / successful: 16 / 16
ProofTerm subclasses (C1C7 audit batch):
  constructed / kernel-accepted: 38 / 38
``verify`` blocks (.deppy kernel.verify):
  total verify blocks: 11
  accepted by ProofKernel.verify: 11
Specs (informational): 5
Proofs total: 45
  derived (real proof bodies): 21
  cited (renaming of trusted axioms): 24
  sorry (unrecognised by-clause): 0
Elapsed: 4321.6 ms

deppy.lean.lean_runner round-trip: OK

 CERTIFIED top-level certify_or_die: PASS

Certificate at: sympy_geometry.lean
```

(Numbers approximate; the actual run produces minor variations between systems.)

11.7 Reading the report

The report tells us:

- **34 of 38 axioms** mechanized to concrete Lean Props. The 4 NL-documentation axioms include the Euler-line theorem and the Thales theorem — prose statements we admit honestly.
- **3 of 4 body translations** exact. The one `sorry` is in `Circle.tangent_lines`, which has a control-flow branch the translator does not handle (an `if` without a uniformly typed `else`).
- **All 11 verify blocks** kernel-verified.
- **7 of 14 foundations** Z3-discharged.
- **0 counterexamples** found. This is what we want for a sidecar that is supposed to be sound.
- **21 derived / 24 cited** proofs. Roughly half the legacy proofs are real derivations (chains, rewrites, compositions); the rest are direct axiom citations on deep Euclidean theorems.
- **Lean round-trip OK** — the certificate elaborates successfully with the current preamble.
- **Soundness gate clean** — no `Structural` leak in a non-vacuous position.
- **CERTIFIED**.

11.8 What a counterexample would look like

Suppose the user accidentally declares a false foundation:

```
foundation Real_my_false_fact: "x * x == 0" -- WRONG: not in general
```

The pipeline:

1. `discharge_foundations`: tries Z3. Result: `sat` on the negation; not verified.
2. `run_counterexample_search`: asks Z3 to satisfy $x \cdot x \neq 0$. Z3 returns model $\{x = 1\}$. Outcome: `found = True`.
3. `rejected_foundations` $= \{Real_my_false_fact\}$.
4. Every theorem citing `Real_my_false_fact` is replaced with a rejection notice. `rejected_theorem_count` $= k$ (the number of citing theorems).
5. `certify_or_die`: $k > 0$ fires; `CERTIFIED = False`; the runner prints the failure list.

This is the canonical sound failure path: the user's false admission is caught before the certificate signs off.

11.9 Summary

The worked example demonstrates:

- the pipeline runs end-to-end on a real sidecar against a real SymPy install;
- the PSDL block `dist_nonneg_psdl` compiles to a real `ProofTerm` the kernel accepts and a real Lean tactic block;
- the legacy chain forms (rewrite, chain, composition) emit real proof bodies;
- the Lean round-trip succeeds;
- the `certify_or_die` verdict is honest about success;
- a hypothetical false foundation would be caught by counterexample search before certification.

The next chapter steps back and audits the soundness story: what is verified, what is admitted, and where the trust surface lives.

Chapter 12

Soundness: What Is Verified, What Is Admitted

This chapter audits the soundness story. We give the trust surface in detail, document the soundness gates, and articulate what deppy's certifications mean.

We do not claim metamathematical soundness; we claim *engineering soundness* (Theorem 4.1) and document the trust surface so a careful user can decide whether the certifications are sufficient for their use case.

12.1 The trust surface

The trust surface of a deppy certification is the union of:

1. **Lean's kernel.** The certificate's Lean text is round-tripped through the `lean` binary. When Lean accepts, we accept. We do not duplicate Lean's type-checking.
2. **Z3.** Every formula discharged via `Z3Proof` is decided by Z3's SMT engine. When Z3 returns `unsat` on the negation, we accept.
3. **CPython's faithful execution.** We trust that CPython runs the source bytes `inspect.getsource` returned. If CPython misimplements (or `inspect.getsource` returns a different source than what actually runs), the certification is meaningless.
4. **The user's foundations.** Each `foundation` that does not Z3-verify is admitted on the user's authority. Their statements are reproduced in the certificate; a reviewer can refuse a foundation by editing the sidecar.
5. **The user's sidecar axioms.** Each `axiom` in a `spec` block is admitted on the user's authority. The certificate exposes them as Lean `axiom` declarations with the original Python statement preserved as a comment.
6. **The user's structural leaves.** Each `Structural("...")` term in a `ProofTerm` contributes a structural assertion. The soundness gate flags non-vacuous occurrences; vacuous occurrences (e.g. exception-fibre vacuity) are accepted.

7. **The user's escape hatches.** Each PSDL `lean("...raw ...")` or `by_lean: |` block is raw Lean text that deppy's kernel does not inspect. Lean's elaborator is the only check.
8. **The user's monkey-patch discipline.** We trust the user not to monkey-patch a class while a verified certificate is in scope. The optional `expects_sha256:` hash is a partial mitigation.

12.2 What is verified

The verifier has positive results in the following sense:

Foundations Z3 discharges. When the foundation discharge report's `z3_verified == True`, the foundation is provable in Z3's theory. Trust `Z3_VERIFIED`. In `sympy_geometry.deppy`, 7 of 14 foundations achieve this.

PSDL-compiled `ProofTerms` the kernel accepts. When `kernel.verify` returns `success=True` for a PSDL-compiled term, the term has been checked at the trust level returned (the `min` over all leaves). This is a real machine check on the proof tree.

Lean certificates that round-trip. When the Lean runner returns exit 0 on the certificate, the entire collection of declarations type-checks in Lean's elaborator and kernel. The verify-block theorems in particular have proof bodies the kernel accepts.

Counterexample-cleared foundations. Foundations that Z3 attempted but did not verify, and for which counterexample search returned no model, are at least *not provably false* under the user's binder typing (typically `int`). They stay admitted, but with the additional negative evidence.

Body translations marked exact. When `translate_body` returns `exact=True` with no `sorry`, the function body has been faithfully expressed in Lean terms. The certificate emits the translation behind a comment for audit.

Type-grounded axioms. When an axiom's `AxiomGrounding` is `GROUNDED`, all the methods, classes, and attributes its statement references resolve on the live target. This is not a soundness verification — a grounded axiom can still be false — but it is a sanity check that the axiom is talking about something real.

12.3 What is admitted

The verifier has negative results in the following sense:

Non-discharged foundations. Z3 cannot discharge `sqrt` / `sin` / non-linear / quantifier-alternating foundations. These admit at `AXIOM_TRUSTED`. In `sympy_geometry.deppy`: 7 of 14.

Sidecar axioms. Every `axiom` declaration admits. The certificate flags grounding (does it resolve?), but not truth. In `sympy_geometry.deppy`: 38 axioms.

Lemmas without `lean_proof`. Lemmas declared without an explicit Lean proof admit as axioms.

`Structural` leaves in vacuous positions. The soundness gate accepts these. They are recorded but do not contribute to a counterexample search.

Body translations that produced `sorry`. When `translate_body` cannot handle a control-flow shape (e.g. try/except, dynamic dispatch, generator comprehension), the returned Lean text contains `sorry`. The certificate counts these in `body_links_sorry`. A verify block whose target's body translation produces `sorry` cannot be `KERNEL`-trust verified.

Method calls through descriptors / `__getattr__`. The pipeline does not trace through `__getattr__`-based proxies in the body translation pass; method calls on the proxy emit a `getattr` application that the certificate's preamble declares as opaque. `FiberedLookup` in the kernel is the proper way to verify these, but the verify-block path does not currently engage it automatically.

Non-arithmetic Z3 obligations in PSDL. When PSDL emits `Z3Proof` for a non-arithmetic formula, Z3 returns `unknown` or fails to encode; the kernel rejects the `Z3Proof` and the verify block fails. This is a *good* behaviour — failing rather than silently accepting — but it constrains the kinds of PSDL assertions that close.

12.4 Soundness gates

`deppy` runs several gates that can refuse certification.

12.4.1 Soundness gate (`soundness_gate_fail`)

The soundness gate scans every emitted `ProofTerm` for `Structural` leaves in non-vacuous positions. A position is *vacuous* when:

- it is the exception-fibre branch of a `Fiber` or `ConditionalEffectWitness` where the precondition excludes the value;
- it is the empty case of an `empty proof block` that the compiler eliminated;
- it is a `show_goal` / `hint` / `pure` structural marker.

When a non-vacuous `Structural` leaf is detected (e.g. the user wrote `raise NotImplementedError` in the value-fibre branch), the gate fires. `certify_or_die` refuses certification.

In the current `sympy_geometry` run, the soundness gate has zero failures.

12.4.2 Counterexample gate (`counterexamples_found`)

When any admitted foundation has a counterexample, the gate fires. `certify_or_die` refuses certification.

In the current run, 0 counterexamples found.

12.4.3 Rejected-theorem gate (`rejected_theorem_count`)

When any theorem cites a foundation that has a counterexample, the theorem is rejected. The gate fires when this count is non-zero.

In the current run, 0 rejected theorems.

12.4.4 Hash-drift gate (`hash_drift_count`)

When a verify block declares `expects_sha256: <hex>` and the live source's hash differs, the gate fires.

In the current run, no verify block uses `expects_sha256`.

12.4.5 Counterexample-expectation gate (`counterexample_expectation_violations`)

When a verify block declares `expecting_counterexample: true` and the search *fails* to find one, the gate fires. This is for testing the counterexample finder itself.

In the current run, no such block.

12.4.6 Lean round-trip gate

When the Lean runner returns non-zero exit, the round-trip is considered rejected. `certify_or_die` refuses certification only if the runner was successfully invoked; if the `lean` binary cannot be found, the gate is skipped.

12.5 What a `CERTIFIED` verdict means

Putting the gates together, `CERTIFIED` means:

1. No `Structural` leaf is in a non-vacuous position.
2. No admitted foundation has a counterexample under the declared binder typing.
3. No theorem cites a counterexample-rejected foundation.
4. No verify block's `expects_sha256` hash mismatched.
5. No `expecting_counterexample` was unfulfilled.
6. Lean 4 (when invoked) accepted the entire certificate.

It does *not* mean:

- every cited foundation is actually true (some admit);
- every cited axiom is actually true (some admit);
- the property holds in any computational sense beyond what the bindings encode (e.g. a property quantifying over `Point := Int` elaborates over integers, not over the actual SymPy `Point` class).

The certification is therefore relative: "relative to the foundations and axioms admitted, and the binder types declared, the property holds".

12.6 The role of `certify_or_die`

`certify_or_die` (file: `deppy/lean/sidecar_runner.py`, lines 62–98) is the central gate. Its design is deliberately conservative:

Refuse on any uncertainty. When Z3 returns `unknown`, the foundation is not Z3-verified. When the Lean runner cannot be invoked, the round-trip is not certified. When a structural leaf is in a non-vacuous position, the proof is not sound.

Be specific about why. Each failure is recorded as a human-readable string. The runner's output lists every failure; the user can fix one at a time.

Do not silently accept. There is no "accept on the basis of the kernel-only check, even if Lean disagrees". The Lean round-trip is required when `lean` is on the `PATH`.

12.7 Limitations

We close with the explicit list of limitations.

12.7.1 Type system

- No universe polymorphism in the generated certificate. Everything is in `Type 0`.
- Class types are `abbrev T : Type := Int` aliases — the flat-Int simplification.
- Refinement types are syntactically supported but the kernel does not check refinement predicates beyond the Z3 encoder's reach.
- No coinductive specs.
- No higher inductive types.

12.7.2 Body translation

- No `try / except` translation; emits `sorry`.
- No `while` translation; emits `sorry`.
- No `generator` or `yield` translation.
- No dynamic class creation.
- No `exec` / `eval`.
- No C-extension fall-throughs (`inspect.getsource` fails).

12.7.3 Z3

- No transcendentals.
- No nested ∀∃.
- No floating-point semantics (we use `Real`, not IEEE 754).

- No string reasoning beyond fixed alphabets.

12.7.4 Lean target

- Mechanization preamble is flat-`Int`; richer typed preambles are future work.
- Method bodies are emitted as `def := sorry` placeholders alongside the actual translation as comments (avoids elaboration ambiguity but loses unfold power).
- No `Mathlib` dependency by default; foundations involving real arithmetic admit.
- Round-trip is single-shot `lean` invocation; no interactive feedback loop.

12.7.5 PSDL

- No partial-proof credit (LLM iteration affordance: future work).
- No tactic-suggestion mode.
- No counterexample-to-obligation feedback.
- Quantifier scoping is by simple variable shadowing; no explicit De Bruijn manipulation.

12.7.6 Cohomology

- Cohomology cocycles are *advisory* in the current implementation; they do not contribute trust to verify-blocks.
- No actual computation of H^k groups; only Euler .

12.7.7 Pipeline

- Single-pass; no incremental verification by default.
- Lean's 60s default timeout is hard-coded; longer runs may need adjustment.
- No parallelism between sub-emitters (`deppy/pipeline/simple_parallel.py` sketches multiprocessing but is not engaged by `run_sidecar`).

12.8 Where to put trust

A user reading a deppy certificate should:

1. **Verify the certificate round-trips.** Run `lean <certificate>.lean` and check exit 0. This confirms the Lean kernel accepts the entire body of declarations.
2. **Audit foundations.** Each `axiom Foundation_X` is admitted. Read its statement; if you do not believe it, refuse it.
3. **Audit sidecar axioms.** Each `axiom Sidecar_X` is admitted. Sympy axioms about geometric facts are typically defensible; library-specific axioms on a closed-source library are more questionable.

4. **Read the report.** Note the `soundness_gate_fail`, the `counterexamples_found`, the `rejected_theorem_count`. All zero is the goal.
5. **Check binder typing.** When a verify block declares `binders: x: int`, the verified property quantifies over `int`, not over arbitrary numeric types. Be aware that a property verified over `int` may fail over `float` or `Decimal`.
6. **Check the source hash.** When a verify block declares `expects_sha256:`, the certificate is pinned to a specific source byte sequence. Re-running on a different SymPy version requires re-hashing.
7. **Beware monkey-patching.** No monkey-patch detection runs at certificate-issuance time; do not patch verified classes.

12.9 Soundness, formally restated

To close the chapter we restate Theorem 4.1:

Theorem 12.1 (Engineering soundness, restated)**.** *For every verify block whose certificate prints `CERTIFIED`, the following hold:*

1. ***Kernel acceptance.*** *The PSDL-compiled `ProofTerm` (or the legacy by-clause derivation) was accepted by `ProofKernel.verify` returning `success=True` at the reported trust level.*
2. ***SMT discharge.*** *Every `Z3Proof` sub-term was checked by Z3 (or by cached lookup) and returned `unsat` on the negation.*
3. ***Lean round-trip.*** *When `lean` was on the `PATH`, the certificate's Lean text was compiled successfully (exit 0).*
4. ***Foundation status visible.*** *Every cited foundation has a status declaration in the certificate (real `theorem` for Z3-discharged, `axiom` for admitted). No foundation is silently introduced.*
5. ***No silent admissions.*** *No `Structural` leaf is in a non-vacuous position (`soundness_gate_fail = 0`); no theorem cites a counterexample-refuted foundation.*
6. ***No false claims under the binder typing.*** *The counterexample finder did not produce a model violating any cited foundation under its declared binders.*

This is the formal-but-engineering soundness statement deppy makes. The trust surface is large (it includes Lean's kernel, Z3, CPython, and the user's admissions), but it is *enumerated*. Reasoning about whether to trust a given deppy certification reduces to reasoning about whether to trust the surface enumerated in this chapter.

12.10 Summary

- The trust surface is the union of Lean's kernel, Z3's answer, CPython's faithful execution, and the user's admissions (foundations, axioms, structural leaves).
- `certify_or_die` runs five gates: soundness, counterexamples, rejected theorems, hash drift, counterexample expectation; plus the Lean round-trip when invoked.

- `CERTIFIED` means relative-soundness: relative to the admissions, the property holds at the kernel-and-Lean level.
- Limitations are explicit and enumerated.
- A user reading a certificate should audit the foundations, sidecar axioms, binder types, and source hash; the certificate exposes everything for that audit.

The next chapter situates `deppy` among its conceptual neighbours in related work and concludes.

Chapter 13

Related Work, Limitations, Future Directions

This final chapter places deppy in its scientific context, revisits the limitations from Chapter 12, and outlines work that would extend the system.

13.1 Related work

13.1.1 Lean, Coq, Agda — the proof assistants

The reference points are the mature dependent-type proof assistants: Lean 4 (de Moura and Ullrich, 2021), Coq (recently rebranded as Rocq), and Agda. These systems share a CIC-flavoured core, support powerful tactic languages, and have rich library ecosystems.

deppy's proof language is small compared to the tactic ecosystems of these systems. Mathlib (The mathlib Community, 2020) provides tens of thousands of lemmas. deppy does not aim to replicate this; its lemma library lives in user-supplied sidecars. What deppy adds is the bridge from idiomatic Python source to the proof world. We see this less as a competitor to Mathlib than as a way to reach Python-domain users who would not otherwise write Mathlib lemmas.

13.1.2 Refinement-type systems

Refinement-type systems (Liquid Haskell, F*, Dafny, Whiley, Stainless) attach predicates to types and discharge them with SMT. This is a closer relative of deppy's arithmetic-Z3 path: the `RefinementType` dataclass is the obvious refinement form, and Z3-discharge of side conditions is the standard refinement mechanic.

The key difference is the lift to homotopy / cubical reasoning. Refinement types are about *predicates* on values; deppy also handles *paths between values* (transport, Kan fillers, univalence). This is what lets deppy say things about duck-typing equivalence and monkey-patching that refinement-type systems cannot.

13.1.3 Cubical proof assistants

Cubical Agda (Angiuli et al., 2019, 2017), `redtt`, `cooltt`, and similar implement cubical type theory directly. These systems are the canonical implementation of the cubical metatheory we draw on.

deppy is *inspired* by cubical type theory but is not an implementation of it. In particular:

- deppy's interval, path types, and Kan fillers are metatheoretic abstractions captured in dataclasses, not operational primitives that compute.
- The cubical Agda computation rules (composition, Glue transport) are not implemented; deppy models them as proof obligations checked structurally.
- Univalence is admitted, not proved by an interval-based `ua`.

We borrow the *vocabulary* (paths, fillers, glue, cocycles) and the *conceptual framework* (types as spaces, equivalences as identifications) without claiming the operational equivalence. This is honest: a user who cares about univalence's computational content should use Cubical Agda; a user who wants to verify properties of an existing Python library has a different problem and deppy addresses it.

13.1.4 Python verification frameworks

Several frameworks attempt static analysis of Python:

- MYPY / PYRIGHT — static type checking against PEP 484 type annotations (van Rossum et al., 2014). Type-level only; no proof.
- CROSSHAIR — symbolic-execution-based counter-example finder. Closer to our counterexample-search pass than to our verifier.
- NAGINI — a Viper-based deductive verifier for Python. Requires Python rewritten with explicit annotations, not sidecar files; targets a smaller subset.
- HYPOTHESIS — property-based testing, not verification. We engage Hypothesis-style sampling in the `computational` discharge mode of `SidecarVerificationBackend`.

deppy's positioning: a sidecar-based verifier with a strong proof language and a dual deppy-kernel + Lean backend. No other framework we know of has the dual-target structure with an explicit cubical metatheory.

13.1.5 Specification and contract languages

ACSL (for C), JML (for Java), `contracts.py` (for Python), and PEP 316 (Programming by Contract for Python) all attach pre/post conditions and invariants to functions.

deppy's sidecar files are similar in spirit but more flexible: specs can carry foundations, axioms, and PSDL proofs; verify blocks are first-class; the trust accounting is explicit.

13.1.6 Property-based testing

QUICKCHECK (Haskell) and HYPOTHESIS (Python) test properties on randomly-generated inputs. deppy's `SidecarVerificationBackend` runs computational tests when the kernel cannot

discharge an axiom, with the result trust level `LLM_CHECKED` (computational evidence is weaker than SMT or kernel verification).

13.2 Limitations recap

We summarise the limitations from Chapter 12:

- Type system: flat `Int` carriers in the Lean preamble; no universe polymorphism; refinement predicates not auto-checked.
- Body translation: gaps for try/except, while, generator, dynamic dispatch, C-extension boundaries.
- Z3: transcendentals, nested quantifiers, IEEE 754 not handled.
- Lean: no Mathlib dependency by default; opaque foundation Props for non-Z3 cases; `def := sorry` placeholders for method bodies.
- PSDL: no partial-proof credit; no tactic suggestion mode.
- Cohomology: advisory only, not part of the trust chain.
- Pipeline: single-pass, no parallelism by default, hard-coded `Lean` timeout.

13.3 Future directions

We outline the most promising extensions, in rough order of expected impact.

13.3.1 Concrete typed preamble

The largest soundness improvement would be to drop the flat-Int preamble and emit *typed* class structures. For each binder class `Point`, the preamble would emit:

```
structure Point where
  x : Int
  y : Int
```

with method bodies `Point.distance := fun p q => sqrt(...)` that the elaborator can unfold substantively. This requires:

- type inference from `inspect.signature` that distinguishes nominal types from `Int`;
- the body translator emitting Lean tied to the typed structure;
- the mechanization preamble using `Class.field` accessors instead of `dot_field`.

The benefit is that PSDL-emitted tactics like `exact Foundation_Real_sqrt_nonneg` elaborate against real arithmetic claims, not opaque Props. The cost is more elaboration ambiguity; some method-resolution patterns will need explicit type ascription.

13.3.2 Mathlib integration

Optional Mathlib import would let foundations involving real numbers (`sqrt`, `sin`, `exp`) discharge as real lemmas rather than admit as axioms. The escape hatch `lean_imports: | import Mathlib` already lets users opt in; the next step is a curated set of foundation → Mathlib lemma mappings the certificate emits automatically.

13.3.3 Refinement-type elaboration

The `RefinementType` dataclass currently carries a `predicate` string that the kernel does not unfold. An extension would parse the predicate, type-check it against the local context, and elaborate the refinement $\{x : T \mid \varphi(x)\}$ as Lean's `Subtype`.

13.3.4 Try/except translation

The body translator could handle `try / except` by emitting a `Lean Except` or `Option` monad. The exception fibre then becomes the `Except.error` branch; tactics can case-split on the result. This requires:

- the body translator's return-type inference to detect exception escape paths;
- the verify-block pipeline to engage the `ConditionalEffectWitness` term automatically;
- PSDL's `try / except` to compile against the monadic encoding.

13.3.5 Cohomology obstructions

The cohomology layer currently produces trivial cocycles in the absence of safety predicates. A meaningful extension would:

- detect when the safety pipeline produces non-trivial preconditions on a function;
- emit substantive Cź cocycles witnessing precondition propagation through call sites;
- emit substantive Cš coherences;
- run `Bicomplex.compute_total_cohomology` to find cross-level H^2 obstructions;
- refuse certification when an H^2 obstruction is non-zero.

This would turn the cohomology layer from an advisory diagnostic into a real verification step — one that detects inter-procedural coherence failures the per-function checks miss.

13.3.6 LLM iteration loop

PSDL has affordances for LLM iteration (`show_goal()`, `hint("...")`, the `lean("...")` escape hatch). A natural extension is an iterative loop:

1. user/LLM proposes a PSDL proof;
2. compiler runs; on failure, surface the goal state at the failure point;
3. LLM proposes a refined PSDL block;
4. repeat until `certify_or_die` passes.

The compiler already produces `show_goal_requested` flags and per-error diagnostics; assembling the loop is the remaining work.

13.3.7 Multi-target

The `deppy.lean` package emits Lean 4. Other targets are plausible:

- Coq / Rocq: the same trust chain, different tactic syntax.
- Agda: similar.
- Z3-only target: emit the entire certificate as an SMT-LIB file (where this is possible); skip Lean.
- No-target dry run: the kernel verifies but no certificate is emitted. Useful for incremental checking in a CI pipeline.

13.3.8 Incremental verification

`deppy/pipeline/incremental.py` sketches a per-function incremental verifier keyed on `source_hash`. The challenges are:

- invalidation when a foundation changes (every theorem citing it must re-verify);
- invalidation when an axiom changes;
- handling pipeline outputs that depend on the entire module (cohomology, bicomplex).

13.3.9 IDE integration

A language server protocol implementation that:

- provides hover-over types for PSDL identifiers;
- highlights `Structural` leaks at edit time;
- runs the Lean round-trip in the background and surfaces elaboration errors at the PSDL line responsible.

13.4 Closing remarks

deppy occupies a particular niche: it is a verification framework for Python libraries that someone has already written and is unwilling to retype. This niche has been under-served by the proof-assistant community and over-served by the property-based-testing community; we offer something in between.

The cubical-duck-type metatheory is the project's intellectual organising principle. We do not claim that Python is *secretly* a cubical type theory. We claim that Python's characteristic phenomena — duck typing, `isinstance`, exceptions, `__getattr__`, monkey-patching — have *natural* cubical readings, and that exploiting these readings produces a proof language that is both usable and expressive.

PSDL is the user-facing surface. It accepts literal Python and emits both kernel-checkable proof terms and Lean tactics. When the kernel and Lean agree, we have a certified proof. When they disagree, we have a diagnostic the user can act on.

The trust accounting is the project's ethical commitment. Every admission, every `Structural` leaf, every untranslated body is recorded and counted. The `CERTIFIED` verdict is strict; the gates are conservative. A user reading a deppy certificate sees both the positive "Lean kernel accepted" side and the negative "these foundations admit" side, and can make their own judgment.

The work that remains is significant: the typed preamble, the Mathlib integration, the cohomology obstructions, the LLM iteration loop, the incremental pipeline. We have left these as future work in the present document but each is articulated clearly enough that contributing maintainers can pick one up.

What we hope to have demonstrated, with this monograph, is that the verification of imported Python modules is approachable — by LLMs and humans both — and that a sidecar-based, cubical-duck-type-flavoured proof system is a coherent way to attack it. The next step is to use it.

13.5 Acknowledgements

The implementation that this monograph documents is the deppy project's mainline source tree. The `AUDIT.md` file at the top of the repository was the principal design-rationale input. Where this document and `AUDIT.md` disagree, the source code wins.

The HoTT book (The Univalent Foundations Program, 2013), the cubical type theory papers (Cohen et al., 2015; Angiuli et al., 2019, 2017), the Lean reference manual (de Moura et al., 2024), and the Lean-mathlib community (The mathlib Community, 2020) provided the proof-theoretic background. The Z3 paper (de Moura and Bjørner, 2008) provided the SMT foundation. PEP 484 (van Rossum et al., 2014) and PEP 544 (Levkivskyi et al., 2017) are the Python language references the verifier targets.

This monograph itself is generated alongside the source tree at `docs/psdl_treatise/`. The build instructions are at the top of `main.tex` and in the local Makefile.

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